Physical Inventory Counting

Front Royal, VA cold storage · Zebra MC9000-series handhelds

This is the working manual. It assumes you have never opened the app before and takes you from an empty deployment through to a finished count sent to the ERP. Read Part 1 and Part 2 once, when the app is first set up. Parts 3 to 5 are what you do for every count. Part 6 is the cycle-count program, Part 7 is what to do when something goes wrong.

> **About the screenshots.** Every picture below is the real application, photographed against a count part-way through a morning: the actual Front Royal bin list, three teams, four scanners, two aisles finished and three being counted. The names, pallet IDs and lot codes are made up. If a screen changes, `npm run docs` rebuilds every picture in this manual from scratch and re-prints **SOP.pdf** — the same manual typeset for paper.

> **Step-by-step guides.** For the jobs that have the most steps, docs/guides has a short guide each, with its own PDF: **second and third counts**, **the ERP upload, an aisle at a time**, **employees, teams and gear sizes**, and **LIVE and TEST**.

Printed October 20, 2026

Contents

Part 0 — What the system is 3

Part 1 — First-time setup: getting in 6

Part 2 — First-time setup: the warehouse 16

Part 3 — Setting up a count 34

Part 4 — Counting day 47

Part 5 — Finishing a count 69

Part 6 — The cycle-count program 81

Part 7 — When something goes wrong 83

Part 8 — Capacity: what the system will take 89

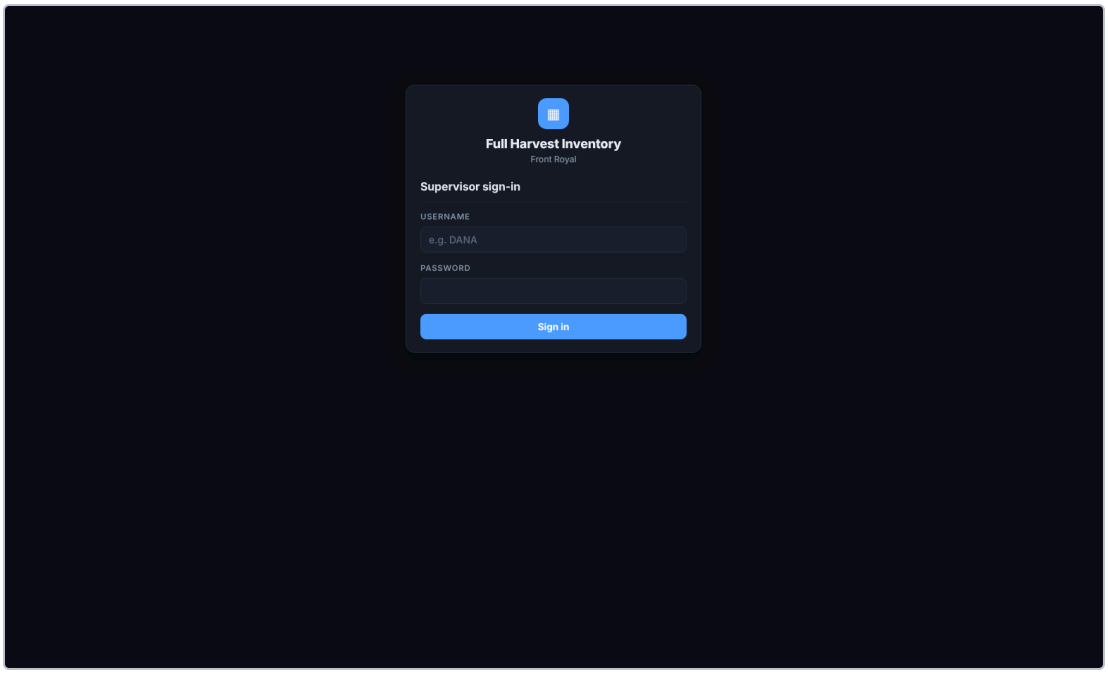
Appendix A — File column reference 90

Appendix B — Server settings reference 93

Part 0 — What the system is

One server, four screens. Everything below happens in a browser; nothing is installed on a PC.

SCREEN	ADDRESS	WHO USES IT	SIGN-IN
Scanner	its own link, `/?d=…`	Counters, on the handhelds	Each scanner has its own link; the bare address opens the supervisor sign-in
Dashboard	`/admin`	Supervisor running the count — the live view of the count in the picker	Supervisor login
Full Counts	`/full`	Every wall-to-wall on the site: open and closed, how far each got, which one the scanners land on; start a new one; what a count still needs	Supervisor login (the Dashboard's access)
Inventory Control	`/inventory`	The dashboard's Second counts, Adjustments and Reports on a page of their own, with an **Overview** in front: how much is counted, what is still being counted again, what the count adds and takes off, how much is in the ERP, and whether the final report can be filed. With cycle counts on, the open cycle counts are there too.	
Settings	`/settings`	Whoever sets the count up	Supervisor login
Office board	`/board`	Anyone — put it on the office TV	**None.** Read-only
Practice data	`/practice`	An admin: every login's runs in the Testing Suite	Admin login
Cycle counts	`/cycle`	Whoever runs the daily program	Supervisor login
User guide	`/guide`	Anyone with a login — new starters first	Supervisor login, any access



Every supervisor page opens on the same sign-in. Signing in on one carries to all four.

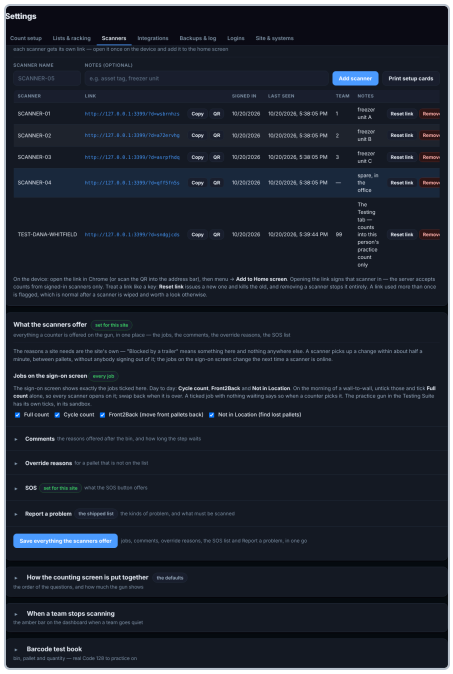
What the system does. It holds a list of every bin in the warehouse and a list of what the ERP thinks is in them. Counters scan pallets into bins on the handhelds. The app compares the two, tells you where they disagree, sends people back to look again where it matters, and produces an adjustment file for the ERP.

What a "count session" is. One count — a wall-to-wall in October, a spot check of the freezer, the year's cycle-count program. Everything (bin list, inventory report, team assignments, every scanned line) belongs to a session. You can have several open at once; scanners choose which one they are working on at sign-on.

Two words you need.

- **Aisle** — a rack run, e.g. `F01`. Teams are assigned whole aisles.
- **Block** — aisles that back onto each other share a **block**, and only one team may be active in a block at a time, so two crews never work opposite faces of the same racking. Pairing aisles into blocks is a one-time job (Part 2, step 7).

Folded cards. The cards a site sets once and then leaves — the counting-screen layout, the barcode test book, the site name and logo, the SOS and comment lists, OneDrive, *Find a lot* and the like — show their heading alone, with a ▸ in front of it. Click the heading to open one; click it again to fold it away. Each computer remembers which ones you keep open. The heading still carries the card's status chip (*the defaults*, *connected*, *not set up*), so a folded page still says what is set. A search hit, a *Ready for count day* link or the Testing Suite's coach opens the card it sends you to.



Settings → Scanners with the set-once cards folded: what the scanners offer stays open, the rest is one click away.

Bin codes. A code like `F01A001` reads as zone `F`, aisle `01`, level `A`, position `001`. Four positions per bay; odd positions are the front face, even are the back. The app derives the aisle from the code when your file has no aisle column.

Part 1 — First-time setup: getting in

> Do this once, with whoever administers the server. Allow 20 minutes.

Step 1 — Deploy the server

On Railway (how this site runs). Deploy the repository; `railway.json` builds the Dockerfile, and the app listens on the port Railway gives it. Attach a **persistent volume** and point `DB_PATH` at it — see step 2 — or the database is wiped on every redeploy.

Your own address (e.g. `fullharvest-inventory.app`). Railway gives the service a `*.up.railway.app` address; a name of your own goes on top of it. In Railway: the service → **Settings → Networking → Custom Domain**, type the name, and Railway shows the DNS record to add at the registrar. Then at the registrar:

- a **subdomain** such as `count.fullharvest-inventory.app` takes a plain **CNAME** to the target Railway shows — the simplest, and what we suggest for the scanners;
- the **bare name** `fullharvest-inventory.app` cannot carry a CNAME under ordinary DNS, so it needs a registrar or DNS host that offers *CNAME flattening* / an *ALIAS* record (Cloudflare, for one), or point the bare name at the subdomain with a redirect.

Railway issues the HTTPS certificate itself within a few minutes. A `.app` name is HTTPS-only by design (the whole top-level domain is on the browsers' preload list), which the app needs anyway for the scanners' offline mode. Once the new address answers, open **Settings → Scanners** on it: the scanner links shown there carry the new address, and each handheld is opened from its new link and installed again — an app installed from the old address is a different app to the device. Supervisors just use the new address.

On a PC inside the warehouse (no internet needed).

```
SUPERADMIN_PASSWORD='pick-something' docker compose up -d
```

Then give that PC a static IP or a DHCP reservation, because the handhelds are pointed at its address and that address must never change. Scanners reach it at `http://<that-ip>:3000/`.

Step 2 — Set the server settings

On Railway these are **Variables**; on a PC they go in `docker-compose.yml`. Set these before anybody signs in:

SETTING	SET IT TO	WHY
`DB_PATH`	A path on the persistent volume, e.g. `/data/inventory.db`	Otherwise a redeploy loses the count
`SITE_TIMEZONE`	`America/New_York`	Dates on reports and cycle-count due dates
`SUPERADMIN_USER`	e.g. `sitelead` (defaults to `ADMIN`)	The one admin login the site starts with, made at start-up
`SUPERADMIN_NAME`	That person's name, as it should appear in the log	Shown against everything they do
`SUPERADMIN_PASSWORD`	A strong password, at least 8 characters	Its password; `ADMIN_PASSWORD` is read as the same thing

The superadmin account is created when the server starts and cannot be deleted from inside the app, so you can never lock yourself out. The app refuses to create it if the password would be the default `changeme` or is shorter than 8 characters — it will say so in the start-up log and carry on without it.

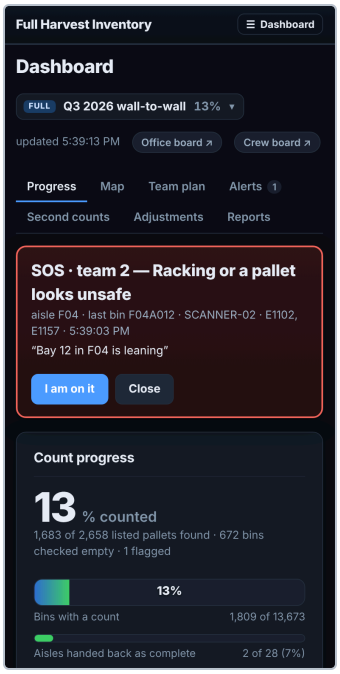
> **Never put a password in the repository.** These are server settings for a reason: anything committed to git is readable by anyone with access to the code, for ever.

Step 3 — Sign in for the first time

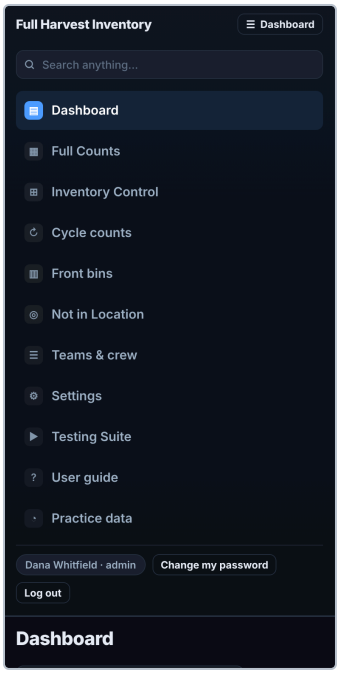
Open `/admin`. The page is a plain sign-in until you are in. Sign in as the superadmin — its username and the password you set in Railway (a password typed with no username is taken as the superadmin's). There is no shared password.

Wrong passwords. Five wrong passwords for one login from one place lock that login out *from that place* for 15 minutes — the right password included — and the sign-in says how long is left. Twenty wrong from one place, whatever the logins, lock the place. A lock never shuts the office out because of somebody guessing from outside: the same login from another address still works. An admin sees a lock on the person's row under **Settings → Logins → Supervisor logins** (a red *locked* tag) and presses **Unlock**; resetting the password lifts it too. Every lock is in the log.

On a phone. The office side works on a phone and installs like an app: open the site's address in the phone's browser, sign in, then **Add to Home Screen** (Safari: Share → *Add to Home Screen*; Chrome: ⋮ → *Install app*). It opens on its own, without the browser's bars, named after the site (*FH Office*). On a phone the side panel folds into a bar along the top: the page's name and a **Menu** button that opens the pages, the search box and *Log out*.



The dashboard on a phone, installed from the home screen.

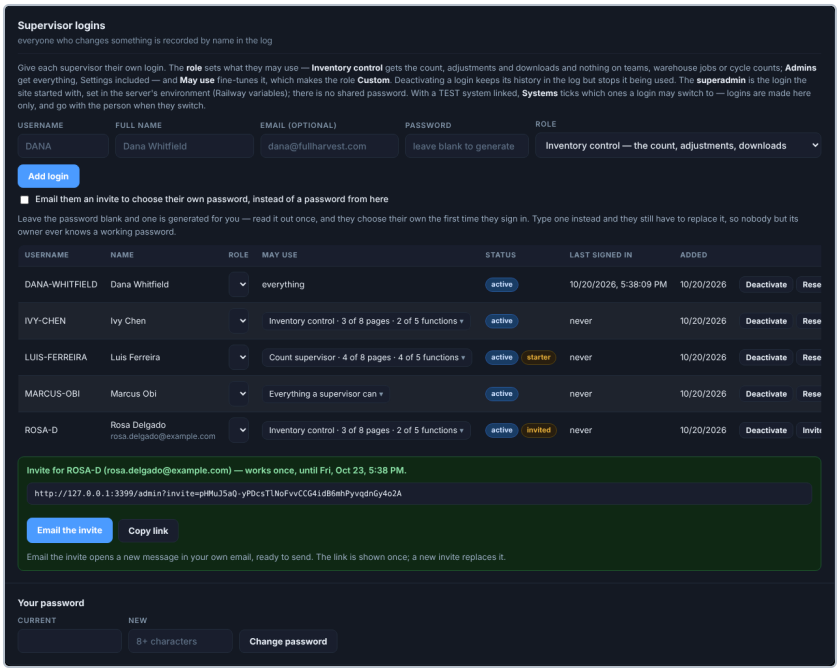


Menu opens the pages; picking one folds it away again.

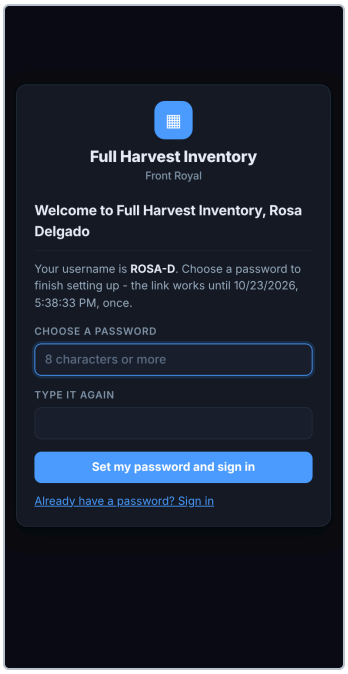
Step 4 — Create a login for each supervisor

Settings → Logins → Supervisor logins.

1. Type the person's **name**, a **username** and choose their **role**: downloads. Nothing on teams, warehouse jobs or cycle counts. aisles, downloads, Testing Suite. The role can be changed on the person's row at any time, and **May use** fine-tunes it (which shows the role as **Custom**).
   - **Admin** — everything, Settings and logins included.
   - **Supervisor** — everything a supervisor can: every page and function except Settings.
   - **Inventory control** — the count only: Dashboard, Testing Suite, approving adjustments,
   - **Count supervisor** — runs the floor: Dashboard, Teams & crew, messages, SOS, queueing
   - **Warehouse jobs** — Front bins, Not in Location, Cycle counts, downloads, Testing Suite.
   - **Cycle counter** — Cycle counts and the Testing Suite only.
2. Either **invite them by email** or give them a **starter password**: login**. The app makes a link that works** once, for three days**, and shows it with** Email the invite **(a ready-made message in your own email — it names the site, their username and when the link runs out) and** Copy link** (for a text or Teams). They open the link, choose a password, and are signed straight in. No password ever goes into an email. Their row shows *invited* until they do; **Invite again** makes a new link and the old one stops working. A password reset, or switching the login off, ends a waiting invite. login**. The app shows a** starter password** once — something like `winter-4k2p`. Write it down and hand it to them; it is not shown again. The first time they sign in they are made to choose their own password before they can do anything.
   - **Invite** — type their **Email** (the *invite* box ticks itself) and press **Add
   - **Starter password** — leave the email blank (or untick *invite*) and press **Add
3. Nobody but them knows their password after that.



An invite: the link, shown once, with a ready-made email to send it from your own mail.

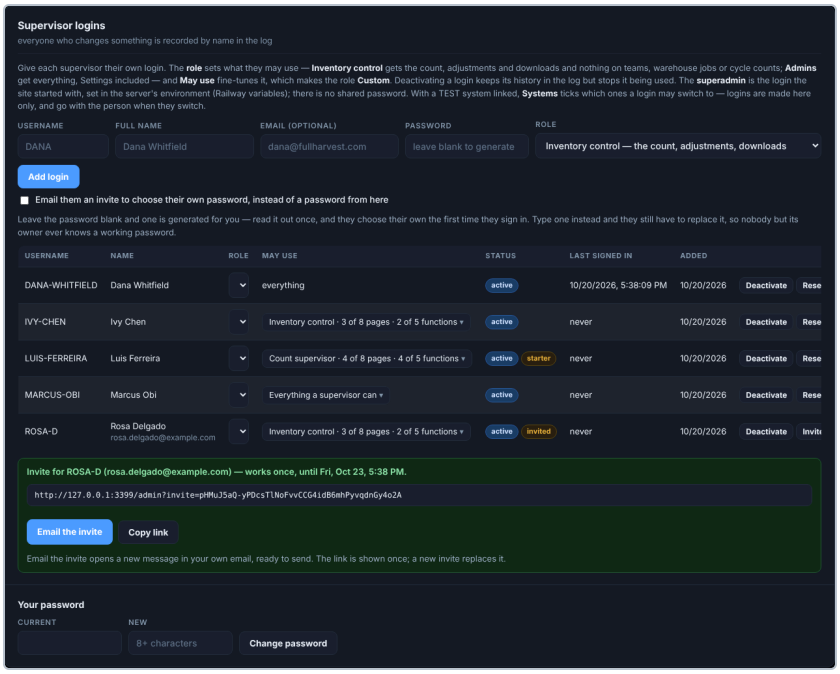


What they see when they open it: their username, and a password to choose.

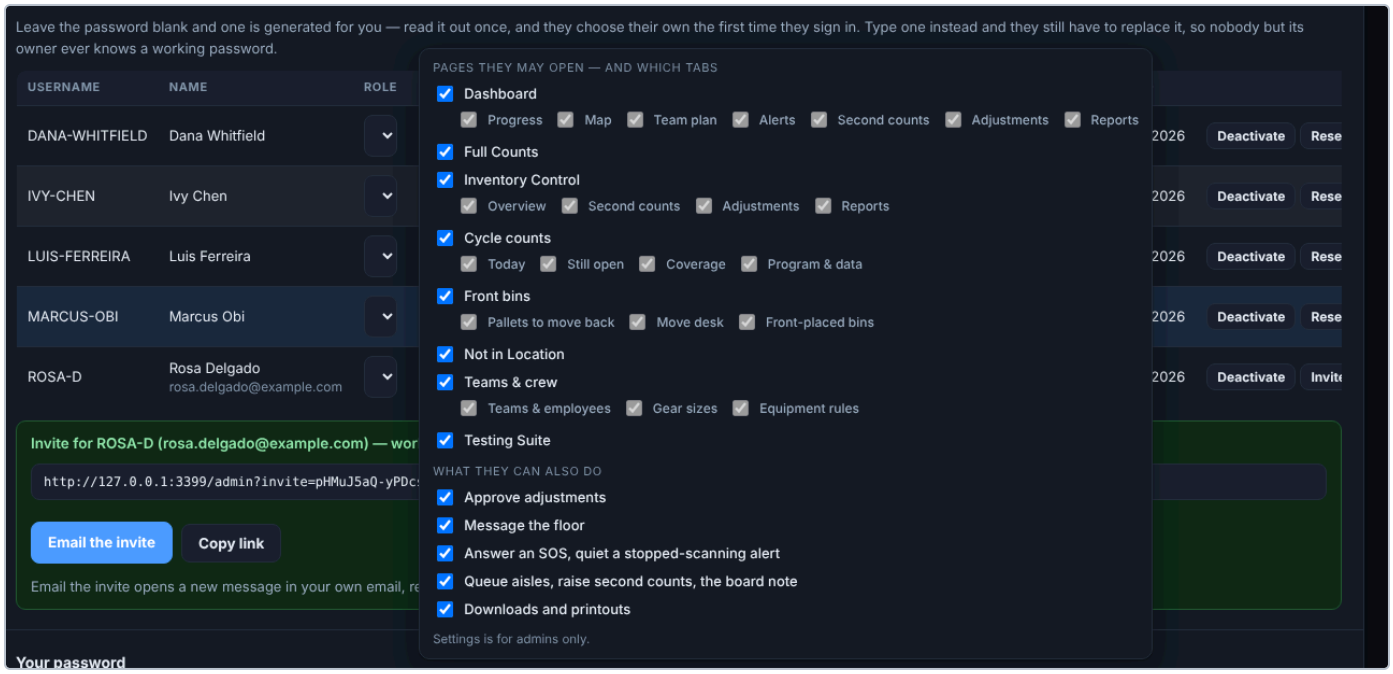
If somebody forgets their password, an admin presses **Reset password** on their row and gives them the new starter password. The same forced change happens again.

Roles. Each row has a **Role**: *Admin* (everything, Settings included), *Supervisor* (every page and function but Settings), *Inventory control* (the Dashboard, the Testing Suite, approving adjustments and downloads — nothing to do with teams, warehouse jobs or cycle counts), *Floor lead* (the Dashboard and Teams & crew, messages, SOS and aisles), *Warehouse jobs* (Front bins and Not in Location), *Cycle counts*, and *Custom* for a list of your own. Picking a role sets the list; changing the list makes it Custom.

What each login may use. Every supervisor row has a **May use** column: tick the **pages** they may open (Dashboard, Cycle counts, Front bins, Not in Location, Teams & crew, Testing Suite) and what they **can also** do (approve adjustments, message the floor, answer an SOS, queue aisles and raise second counts, downloads and printouts). A new login may use everything a supervisor can until you untick something; a tick takes effect on their very next click, with no sign-out needed. The sidebar shows them only the pages on their list, and the buttons they may not press are grayed out and say why. **Settings** — every section of it — is for admins only, whatever the list says; supervisors change their own password from **Change my password** at the foot of the sidebar.



Settings → Site & systems. Each person has their own login, a role and their own password; the starter password is shown once.



May use, opened on one login: the pages, each page's tabs, and the functions — a tick takes effect on their next click.

Which count a page opens on. The picker in the header opens on an open full count — the one the scanners land on, else the newest — and, when no full count is open, on an open cycle count. It lists only the kinds of count the login may work and the site has switched on (Step 4a): with cycle counts switched off, no cycle count appears in it.

Practice counts are not in the picker. Nobody's practice count from the Testing Suite is listed in a dashboard's count picker — not even an admin's — so no page ever opens on one by mistake. An admin reads every login's practice under **Learn → Practice data** (Step 8c-1).

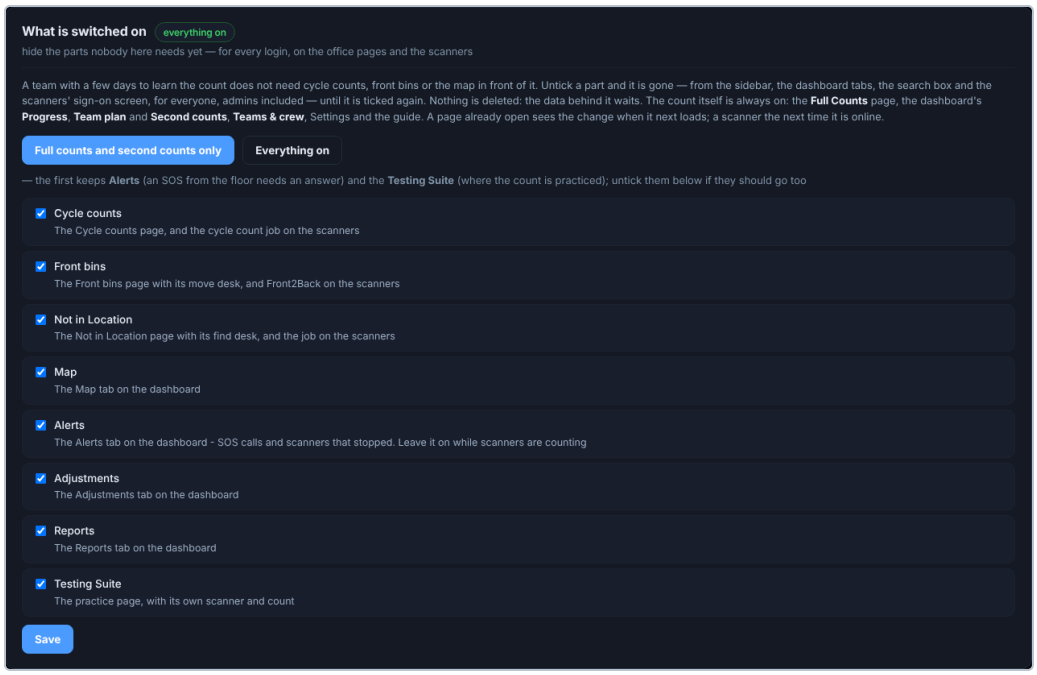
A procedure per role. Each profile has a short SOP of its own under `docs/roles/` (and as a PDF beside it): Admin, Supervisor, Inventory control, Count supervisor, Warehouse jobs, Cycle counter, and one for the counters on the handhelds. Hand a new login the right one; this manual is the reference behind them.

Step 4a — Switch off what nobody needs yet

Settings → Site & systems → What is switched on. Admins only.

A team with a few days to learn the count does not need cycle counts, front bins or the map in front of it. An admin can switch the parts of the app the site is not using off — for everyone, on the office pages and the scanners — and back on when the time comes.

1. Press **Full counts and second counts only**. Cycle counts, Front bins, Not in Location, the Map, Adjustments and Reports go: from the sidebar, the dashboard tabs, the search box and the scanners' sign-on screen, for every login, admins included. The **Alerts** tab stays, because an SOS from the floor needs an answer, and so does the **Testing Suite**, because that is where the count is practiced; untick either in the list if it should go too.
2. Or tick the parts one by one and press **Save**.
3. **Everything on** brings it all back, exactly as it was.



What it is and is not:

- **Hidden, not forbidden.** A login's role still says what it may use (Step 4). Switching a part off takes it out of everyone's way; nothing is deleted, and the data behind a hidden page waits for it to come back.
- **Always on:** the Dashboard's Progress, Team plan and Second counts, the Full Counts page, Inventory Control's Overview and Second counts, Teams & crew, Settings and the User guide — the count itself.
- **A bookmark** to a hidden page says *"Cycle counts is switched off"* with the way back for an admin, rather than wandering somewhere else without a word.
- **The scanners** stop offering the job of a part that is off — Cycle count, Front2Back, Not in Location — whatever is ticked under Settings → Scanners → Jobs; those ticks are kept (grayed, with a note saying why) for when the part comes back. The practice gun and the Testing Suite's other jobs follow the same switches.
- **When it takes effect:** a page already open sees the change when it next loads; a scanner the next time it is online. The log records who switched what off.

Step 4b — Systems: LIVE and TEST, from one sign-in and one scanner link

A **TEST** system is the app deployed a second time, with its own database: the place to try new code, new settings or a practice count on real guns without touching the live stock. Each system is at its own address; people and scanners switch between them from inside the app, so nobody needs a second password and no scanner needs a second link.

LIVE is the home. It keeps the logins and decides who may use which system. TEST is *linked* to it: it takes the people and the scanners LIVE hands over, and nothing else.

Setting TEST up (once):

1. In Railway, deploy the app again as its own service with its **own volume**, from the branch TEST should run.
2. Give that service three variables: `SYSTEM_NAME` = **TEST**, `SYSTEM_HOME_URL` = LIVE's address (for example `https://full-harvest.up.railway.app`), and `SYSTEM_LINK_KEY` = LIVE's link key (on LIVE: **Settings → Site & systems → Systems → The link key → Copy**). Or sign in on TEST with its own superadmin and paste the same two into its Systems card.
3. On LIVE, **Systems → Add a system**: name **TEST**, TEST's Railway address, a color. Under **On the scanners**, choose where the guns count: **LIVE and TEST** (switched on the gun), **LIVE only**, or **TEST only**. **Save**.
4. Say which logins may use TEST: tick **TEST** in the **Systems** column of **Supervisor logins** (or in **Who may use each system** on the Systems card — the two are the same ticks). A new login can be given it as it is made: *may also switch to TEST* under **Add login**. Admins always may.

Logins are made on LIVE only. Nobody needs a login on TEST: when someone switches, LIVE hands over who they are, their role and the pages they may use, and TEST makes or updates their

login there to match. Change someone's role on LIVE and it applies on TEST the next time they switch. Signing in on TEST directly is only for its own superadmin (its Railway variables), for setting it up.

Logins someone already made on TEST. If people were set up on TEST before it was linked, bring them over instead of making them again: on LIVE, **Settings → Logins → Supervisor logins → Copy logins from TEST**. LIVE asks TEST (signed with the link key, as a switch is) and lists TEST's own logins with what would happen to each:

- *copied here, same password* — ticked; they sign in on LIVE with the password they use on TEST, keep their name, email, role and pages, and are given TEST in the Systems column;
- *already a login here — left as it is* — the LIVE login is never changed;
- *never chose a password* — an invite not yet taken up on TEST: add them on LIVE and invite them;
- TEST's own superadmin is listed but not ticked — tick it only if that login should be on LIVE too.

Untick anyone who should not come over, then **Copy the ticked logins**. People who only ever switched over from LIVE are not listed: they are LIVE's already. The log records who was copied.

Switching:

- **In the office.** A login with TEST ticked sees **Switch to…** at the bottom of the sidebar, and **Sign in to** on LIVE's sign-in box. TEST opens on the same page, signed in as the same person with the same role and pages. Every TEST page has an orange **TEST — not the live system** bar with **Back to LIVE**. With more than one system, the logo and site name at the top of the sidebar (and on the sign-in box) sit in a ring of the system's color with its name on it — green **LIVE**, orange **TEST** — so which one you are on shows at a glance.
- **On a scanner.** *On the scanners* (LIVE's Systems card) decides. **LIVE and TEST**: the sign-on screen shows **System: LIVE | TEST**. **LIVE only**: no System row, and a gun left on TEST comes back by itself. **TEST only**: every gun goes to TEST by itself (once its counts have sent) and is offered no way back until the choice changes. With both, one tap moves the gun to TEST under its own name, and **Back to LIVE** brings it home. The header always says which system the gun is on. A gun with counts still waiting to send is not allowed to switch until they have gone — they belong to the system they were counted in. Each system keeps its own storage on the gun, so nothing mixes. With more than one system the gun's header has a band of the system's color across the top, the site name in a ring of it, and the current system's button in that color — green LIVE, orange TEST.

- **Scanners are registered on LIVE only.** Register each gun, print its setup card and open its link under **LIVE → Settings → Scanners**, as always. When it switches, LIVE hands the scanner over and TEST makes it there under the same name and the same link; it is never registered on TEST by hand (a gun registered on TEST would be a second scanner with a second link, and LIVE would not know it). So on TEST, **Settings → Scanners → Scanner setup** has no *Add scanner*, *Print setup cards*, *Copy* or *Reset link* — just the guns that have come over, and a note linking to LIVE's scanner page. Reset a scanner on LIVE. *Ready for count day* on TEST counts the guns that have come over instead of asking for scanners to be registered. The Testing Suite's own practice gun works on TEST as it does anywhere.

What to know:

- TEST has its own everything: counts, bin lists, crew, settings, backups. A count made on TEST is never on LIVE.
- A switch is a hand-off signed with the link key, good for two minutes and once. TEST believes LIVE; LIVE believes nobody, so TEST can never sign anyone in to LIVE. A login handed to TEST lasts twelve hours there, so taking TEST away from someone takes effect within a shift.
- **Make a new link key** on LIVE stops every switch until each other system is given the new key.
- **Check the link** (a button on each system's row, on LIVE) asks TEST, server to server, whether a switch would work, and says what to change if not: TEST cannot be reached; it runs an older version; it thinks it is a home itself (set `SYSTEM_HOME_URL`); its home is set to another address; it calls itself something other than it is listed as (match `SYSTEM_NAME` and the name here); or it refuses the key (copy the link key into its `SYSTEM_LINK_KEY`, with no spaces or quotes, and redeploy). Run it after setting TEST up, and whenever a switch fails.
- **Back to LIVE goes nowhere, or says "this system's own address". Back to LIVE** and *Switch to... LIVE* use one thing only: `SYSTEM_HOME_URL` on the TEST service. If it holds TEST's own address the button would just reload TEST, so it says so instead, and so does TEST's Systems card. Set it to LIVE's address (`https://…-prod.up.railway.app`, nothing after it) and redeploy TEST. LIVE refuses, the same way, a system listed with LIVE's own address.

Step 5 — Change the superadmin's password

Sign in as the superadmin, press **Change my password** at the foot of the sidebar, and pick one only you know. The environment variable no longer opens the account after that; a restart never puts the old password back.

Part 2 — First-time setup: the warehouse

> Once, before the first count: the bin list, the racking, the scanners, the counting screen and the reasons the guns offer. A count of its own comes in Part 3.

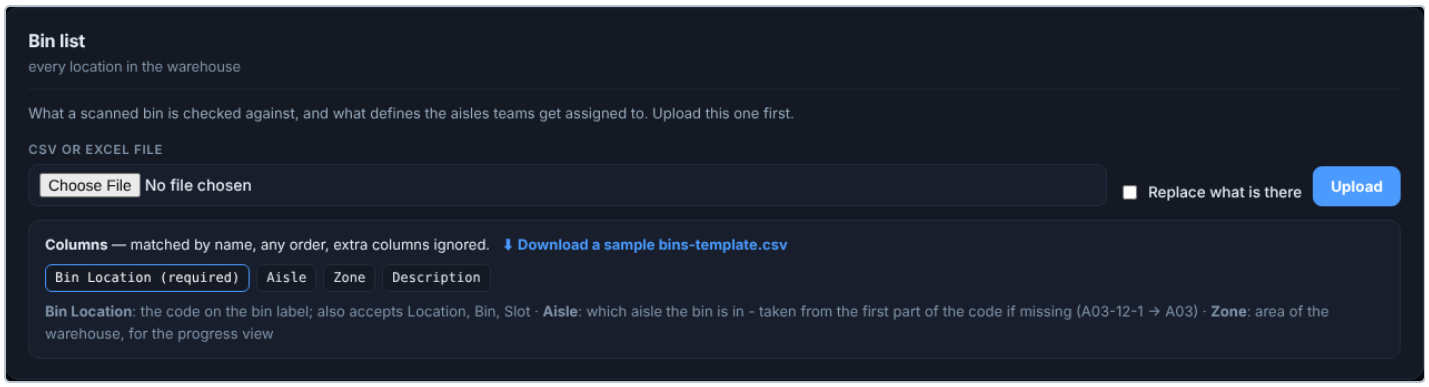
Step 6 — Upload the bin list

Settings → Lists & racking → Bin list.

This is every location in the warehouse. It is what a scanned bin is checked against, and it is what defines the aisles teams get assigned to. **Upload this before anything else.**

- **No bin list ships with the app.** A new installation starts empty, and the site uploads its own: the ERP's bin export, as it comes. For Front Royal that is racks F01–F24 and A01–A04 plus WIP, the NIL bin and other areas; staging lanes and dock doors are counted manually and are left out on upload.
- Choose the CSV or Excel file. Column names are matched loosely — see Appendix A. The only column you must have is the bin location.
- **Replace what is there** wipes the existing list first. Leave it unticked to add to it.

The card reports what it read: how many bins, how many aisles, and anything it skipped.



Settings → Lists & racking. Each list has its own card, and each reports exactly what it read.

Step 7 — Pair the aisles into racking blocks

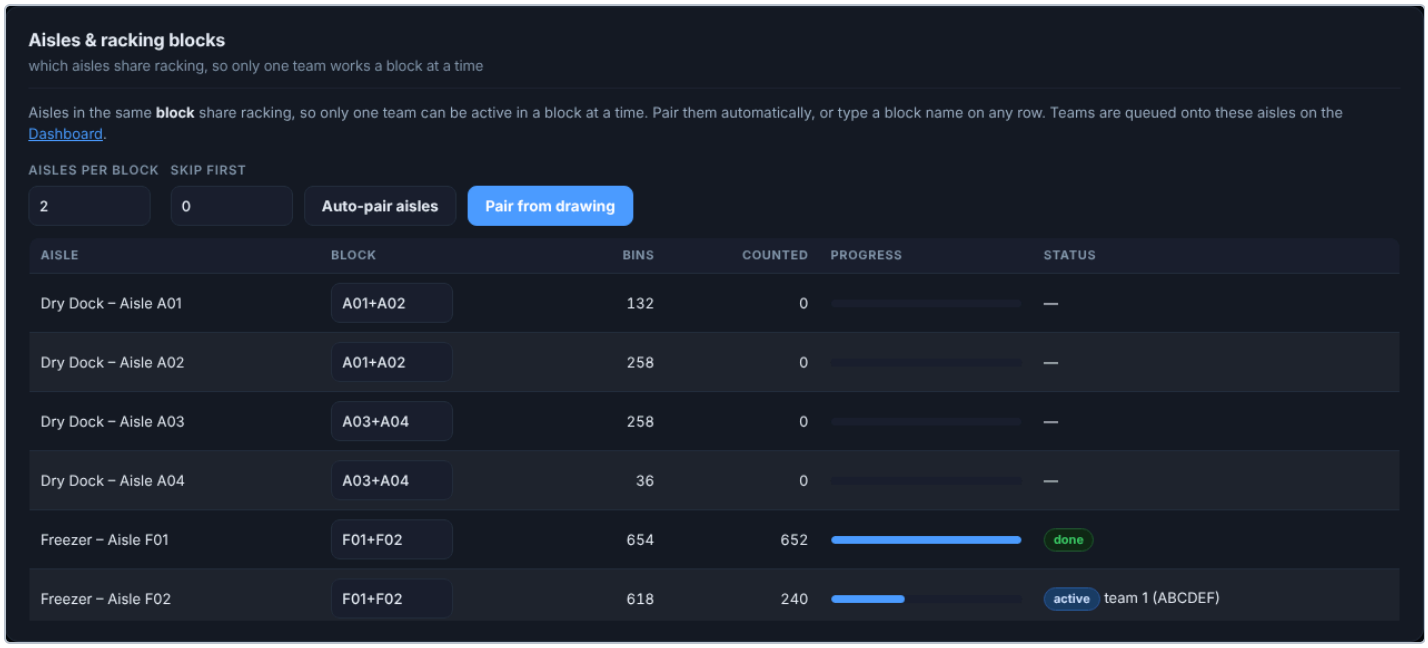
Settings → Lists & racking → Aisles & racking blocks.

Aisles that back onto each other must share a block so two teams are never working the same racking from both sides.

1. Set **Aisles per block** to 2 (most double-deep racking).

2. **Skip first** is 1 if the first aisle has a wall behind it rather than another aisle.
3. Press **Auto-pair aisles**, then read the table and correct anything that is wrong by typing a block name straight onto a row.

If you have uploaded a rack drawing, **Pair from drawing** uses it instead — more reliable than counting aisles off by hand.



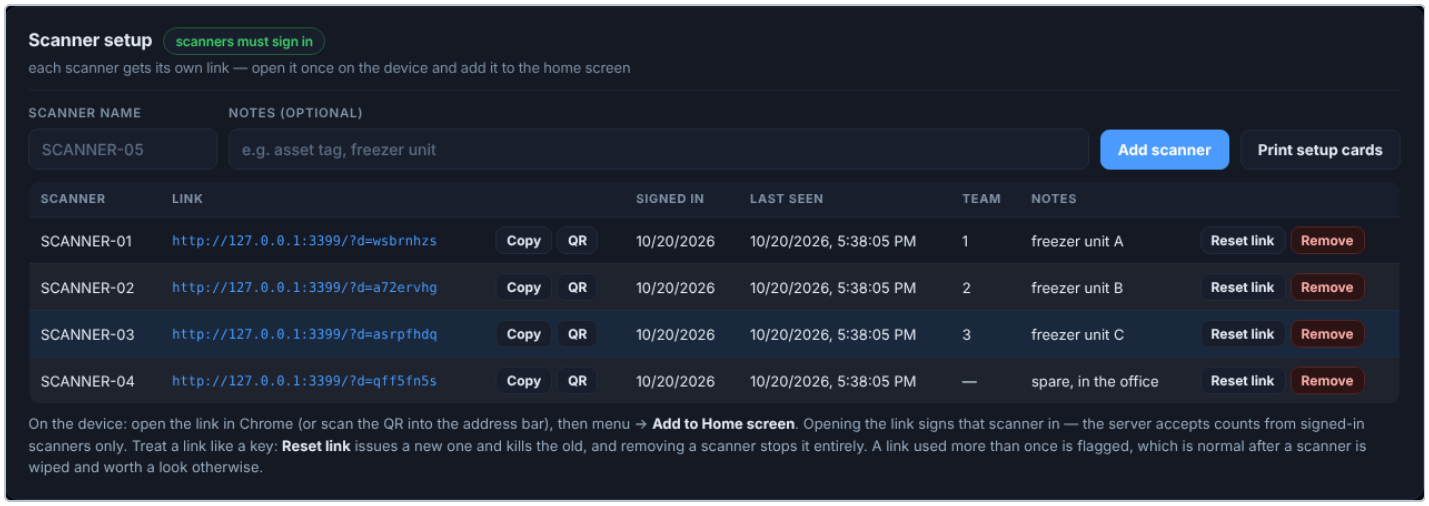
Aisles paired into racking blocks. F01+F02 share racking, so only one team works that block at a time.

Step 8 — Register the scanners

Settings → Scanners → Scanner setup.

Each handheld gets its own link. Opening that link once is what signs the scanner in; after that the server accepts counts from it and stamps its name on every line.

1. Type a **scanner name** — use what is written on the device, e.g. `SCANNER-05`. Add notes (asset tag, "freezer unit") if it helps.
2. Press **Add scanner**. Repeat for every handheld.
3. Press **Print setup cards**. You get one card per scanner with its own QR code. Cut them up and tape one inside each cradle.



Settings → Scanners. One row per handheld: its link, when it was last seen, Reset link and Remove.

On each handheld, once:

1. Open Chrome, scan the QR code from its card into the address bar (or type the link).
2. **Install it. Do not leave it running as a browser tab.** Chrome menu → **Install app** / **Add to Home screen**, or tap **Install on this scanner** on the app's own sign-on screen. Installed, it runs in its own window — no address bar at all — and starts even with no signal.

A browser tab is the one setup that causes trouble: Chrome keeps its address bar above the page, brings it back whenever the page moves, and if the keyboard focus ever leaves the page a scan is typed into that bar instead of into the count. The app warns you when it is running in a tab and offers **Hide the browser bar for this shift** as a stopgap.

Still want a locked-down device? See **Kiosk mode** at the end of this step.

1. Set up **DataWedge** so scans arrive as keystrokes: either as the end of a scan, and neither moves the cursor off the box)*
   - the profile associated with Chrome → **Keystroke output: enabled**
   - **Basic data formatting** → send a suffix: **ENTER** *(or TAB — the app takes
   - **Barcode input** → enable the symbologies your labels use.

Kiosk mode. Three levels, strongest last:

	WHAT IT TAKES	WHAT THE COUNTER SEES
Installed app (do this)	Chrome menu → Install app, once per device	Its own window. No address bar, no tabs, no back button
Full screen for a shift	The button on the sign-on screen, one tap	The bar goes away until the app is closed. The browser says so once, with a banner carrying the address

	WHAT IT TAKES	WHAT THE COUNTER SEES
Locked device	Zebra **Enterprise Home Screen** (built into the device) or your MDM, set to launch this app and nothing else	The handheld does nothing but count. Survives a reboot

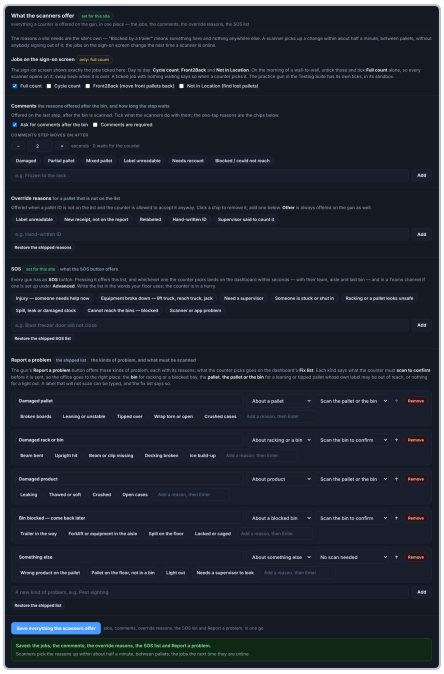
The first two are inside the app. The third is a device setting — it is configured on the handheld or through your MDM, not here, and it is the only one that stops somebody leaving the app altogether.

Treat a scanner link like a key. Reset link issues a new one and kills the old — use it if a device is lost or a link gets out. Removing a scanner stops it entirely. A link used on more than one device is flagged in the table; that is normal after a scanner is wiped and worth asking about otherwise.



Print setup cards: one per scanner, each with its own QR code and the three steps to do on the device.

Which jobs the scanners offer. *Settings → Scanners → What the scanners offer → Jobs on the sign-on screen.* A gun's sign-on screen shows exactly the jobs ticked here — **Full count**, **Cycle count**, **Front2Back**, **Not in Location (NIL)** — whether or not there is work waiting for each; a ticked job with nothing to do says so when it is picked. Day to day a site ticks the cycle count, Front2Back and Not in Location. On the morning of a wall-to-wall, untick those and tick the full count alone, so every scanner opens on it and nobody signs on to anything else by mistake; tick the others back on when it is over. At least one job has to stay on. Scanners pick the change up the next time they are online. The practice gun in the Testing Suite has its own ticks instead, under *Scanner settings in this sandbox → Jobs on the sign-on screen* — all four to start with.



Settings → Scanners → What the scanners offer: the full count alone ticked, for count day.

What Report a problem offers. *Settings → Scanners → What the scanners offer → Report a problem.* One box per kind of problem the gun offers: its name on the gun, what it is about (it sorts the fix list), **what the counter must scan to confirm it** — *Scan the bin*, *Scan the pallet*, *Scan the pallet or the bin*, or *No scan needed* — and its reasons (× takes one off; type a new one and press Enter). **Add** a kind of your own (a pest sighting, a door left open), ↑ moves one up the gun's list, **Remove** takes it off, **Restore the shipped list** puts the five shipped kinds back. Up to ten kinds, each with at least one reason. Saved with **Save everything the scanners offer**, or its own **Save**; a gun picks it up the next time Report a problem is opened with signal. Admins only.

Step 8a — A barcode test book, to practice on

Settings → Scanners → Barcode test book. A page of real **Code 128** labels to print, cut up and scan — the same symbology the racking uses, so a gun that reads the book reads the rack. Use it to train a crew at a desk, to check a new scanner reads properly, or to dry-run a count before the real one.

The page is a table, one row per line to count: **Bin**, **Pallet**, **Qty**. Work down it the way a counter works down an aisle — scan the pallet, key the quantity, scan the bin. The quantity is printed as a number to type; tick **Quantity as a barcode too** and it is scannable as well, for demonstrating a scan-everything workflow.

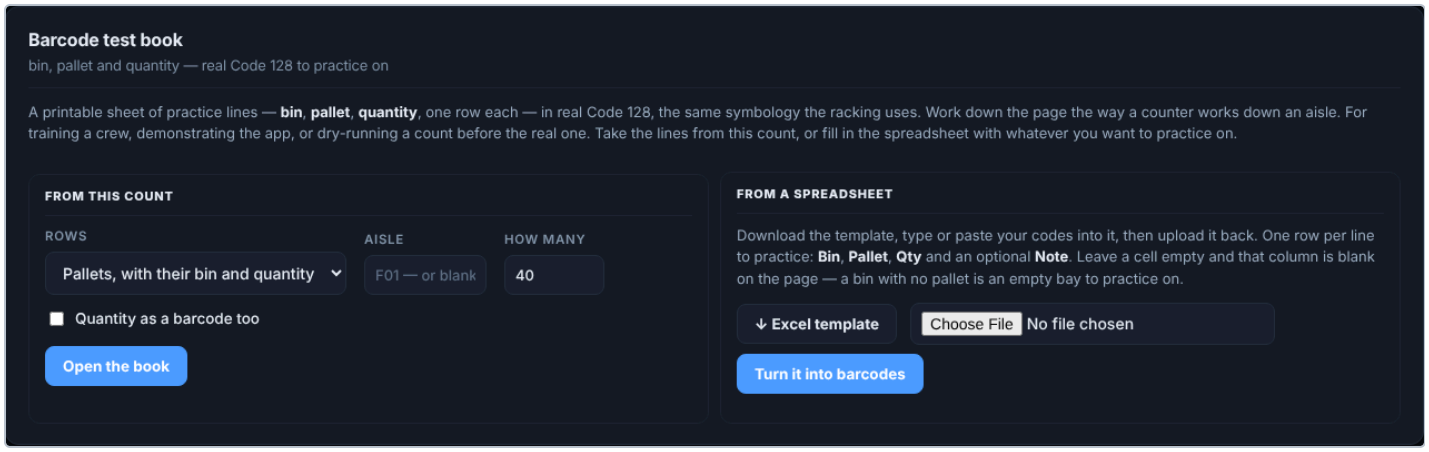
Two ways to fill it:

- **From this count — Pallets, with their bin and quantity** gives a full practice line per row, straight from the uploaded lists, so what you scan off the paper is what the gun expects. **Bins on their own** prints rack labels with the other two columns empty, which is what you want for practicing empty bays. Narrow it to one aisle, and say how many rows.
- **From a spreadsheet — ↓ Excel template** downloads a workbook with four columns:

COLUMN	WHAT GOES IN IT
Bin	The bin barcode for that row — leave it out and the column is blank
Pallet	The pallet barcode
Qty	The quantity to key in (or to scan, if you ticked the box)
Note	A small line under the pallet — the item, or what the row is for

Fill it in, upload it back, and the book opens ready to print. The codes need not exist in any count — that is the point of practicing.

> **Print at 100%.** “Fit to page” shrinks the bars and a scanner will refuse them. Plain white paper only: glossy or colored stock scatters the beam.



The test book card: a practice line per row from the count itself, or from a spreadsheet of whatever you want to practice on.

Step 8b — The SOS list, and the Teams channel

Settings → Scanners → SOS from a scanner.

Every gun has an SOS button. What it offers when pressed is this list — write it in the words your floor uses, because the person pressing it is in a hurry. It ships with what usually goes wrong in a cold store, most urgent first: an injury, **equipment broken down**, **needing a supervisor**,

somebody shut in, racking that looks unsafe, a spill, blocked bins, a scanner problem. Each SOS shows on the dashboard (a red bar above every tab, with a chime) and, if the channel is set up, in Teams.

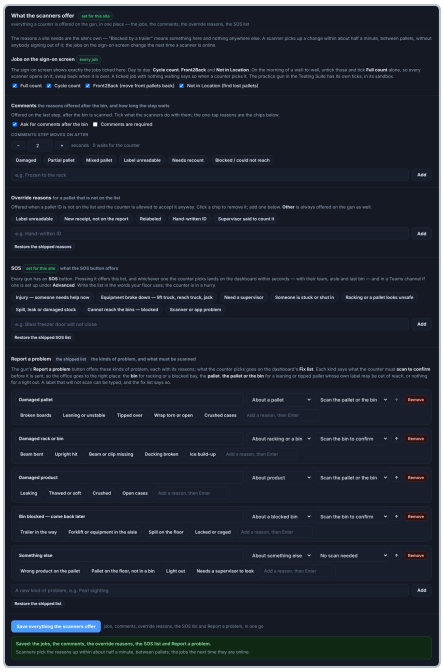
To send alerts to a Teams channel as well (Settings → Integrations → Microsoft Teams channel):

1. In Teams, click **⋯** beside the channel → **Workflows** → *Post to a channel when a webhook request is received*.
2. Follow it through; Teams gives you a web address.
3. Paste that into **Webhook address**, tick **Send alerts to Teams**, and **Save**.
4. **Send a test card** — a card should appear in the channel within a second or two. Do this now, not during an emergency.

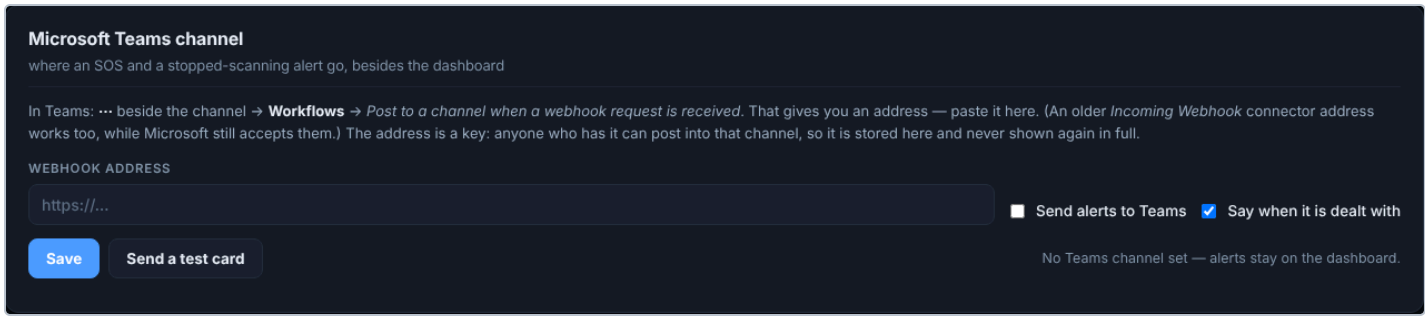
An older *Incoming Webhook* connector address also works while Microsoft still accepts them. The address is a key — anybody holding it can post into that channel — so it is stored once and never shown again in full; the page only shows which server it points at.

> **A Teams outage never costs you an alert.** The SOS is on the dashboard either way, and the row says whether the channel took it (`teams`, or `teams failed: …`).

When a team stops scanning. On the same card: a team that is signed on and still has an aisle, but has not scanned for this many minutes (10 to start with), gets an amber bar on the dashboard — and a card in the Teams channel if **Send it to the Teams channel too** is ticked. **0** turns it off.



The SOS list, under Scanners.



The Teams channel it goes to, under Integrations: the address is kept, never shown again in full.

Step 8c — The Testing Suite: try the scanner on test data

Testing Suite in the sidebar. The fastest way to show somebody the gun, or to try a setting before a count, without a handheld and without touching a real count.

On the left is the **real scanner app** — the same one the Zebras run — on a picture of an MC9300. On the right is a sheet of test data: two short aisles of real Front Royal bins (F01 and F02, levels A–B), what is on each shelf, and a made-up inventory report laid over them.

Click any bin, pallet or quantity on the sheet and it is scanned into the gun, exactly as the trigger would send it. The **SCAN** box under the screen sends anything you type — a wrong ID, a typo — to see what the gun does with it. You can also click on the gun's screen and use it directly.

1. **Sign on.** Click team **99** and a clock-in number, then **Sign on & load list** on the gun. Team 99 already has F01, then F02. Your own clock-in number works; **T1001** and **T1002** are made up, so the gun flags them as not on the crew list — which is what it does with a typo.
2. **Start counting**, and work down the sheet. For each pallet: pallet, quantity, bin.
3. **Things to try** ticks itself off from what actually reached the server. The shelves are wrong in places on purpose, each with a **Try this** line saying what to do:

BIN	WHAT IT TEACHES
F01A002, F01B004	Two and three pallets in one bin
F01A003	An empty bin — **Bin is EMPTY — scan the bin**
F01A004	A pallet short of the report (report 36, shelf 32)
F01A005 / F01A006	A pallet in the next bay, and one the report never heard of
F01B001	A torn pallet label — **Label will not scan**, then type it
F01B002	Best-before already passed
F01B003	A missing rack label — **Bin label will not scan**

BIN	WHAT IT TEACHES
F01B006	A pallet on the report that has gone — record the bin empty
F02B001	A quantity big enough that the gun asks for it twice

Then **Aisle complete — next aisle**, press **SOS** once, and **Open the dashboard on this count** to see it all from the supervisor's side: progress, the pallet report, the SOS bar, adjustments.

It is walled off from the real thing.

- It counts into a **Practice count** of its own. A gun on the floor is never offered it, and the test gun is never offered anything else.
- The office board never shows it, and no dashboard's count picker lists it. The dashboard at the top of the Testing Suite is the one place it is on screen (and **Own tab** opens that dashboard on it, marked **PRACTICE**).
- **It is yours.** Each login has its own practice count and its own test scanner (**TEST-** and their name, under Scanners), kept from one sign-in to the next. Nobody else sees it, two people practicing at once never touch each other's, and a **new user** opens the tab to a fresh one. The test gun keeps its own storage in the browser, separate from a real scanner's and from anyone else's.

First time here? The tab opens with a short guide at the top — what the two halves are, and the six steps from signing on to opening the dashboard — and a **tip bubble** that points at the next thing to click and follows the gun: *click 99*, *now a clock-in number*, *tap Sign on on the gun*, *click this pallet*, *now the quantity*, *now the bin*. Both fold away (**Hide**, **I've got it**) and stay folded for anyone who has counted here before; **Show tips** brings the bubble back.

Before you start at the top of the tab is the checklist: signed in with your own login, test data loaded, the test scanner registered and signed in, team 99 with aisles, the gun running — each with what to do if not. It reads **ready to test** when there is nothing to do.

Your own test pallets. Upload a sheet with **Bin**, **Pallet** and **Qty** (and, if you have them, **Note**, **Lot**, **Best Before**) and the practice run is built from it instead. It is the same sheet as the barcode test book (step 8a): print the book, upload the same file here, and scan the paper. A row with a bin and no pallet is an empty bin. The file is checked row by row and told back plainly (*row 7: "ten" is not a quantity*). **Back to the built-in test data** returns to the shipped aisles.

Try the features that ship turned off. Lot codes, best-before dates, adjustments needing approval (with its thresholds), the pallet check (allow override / no overrides / accept any), automatic second counts, the comments step — each can be switched on for **your practice count only**; nothing else changes. Each one says whether the practice data has what it needs (*✓ 25 of 25 pallets have a lot code*) and, when it does not, what to add to your file. Settings that apply to every

scanner (the Teams channel, the stopped-scanning limit, the scanner screen) are not changed from here.

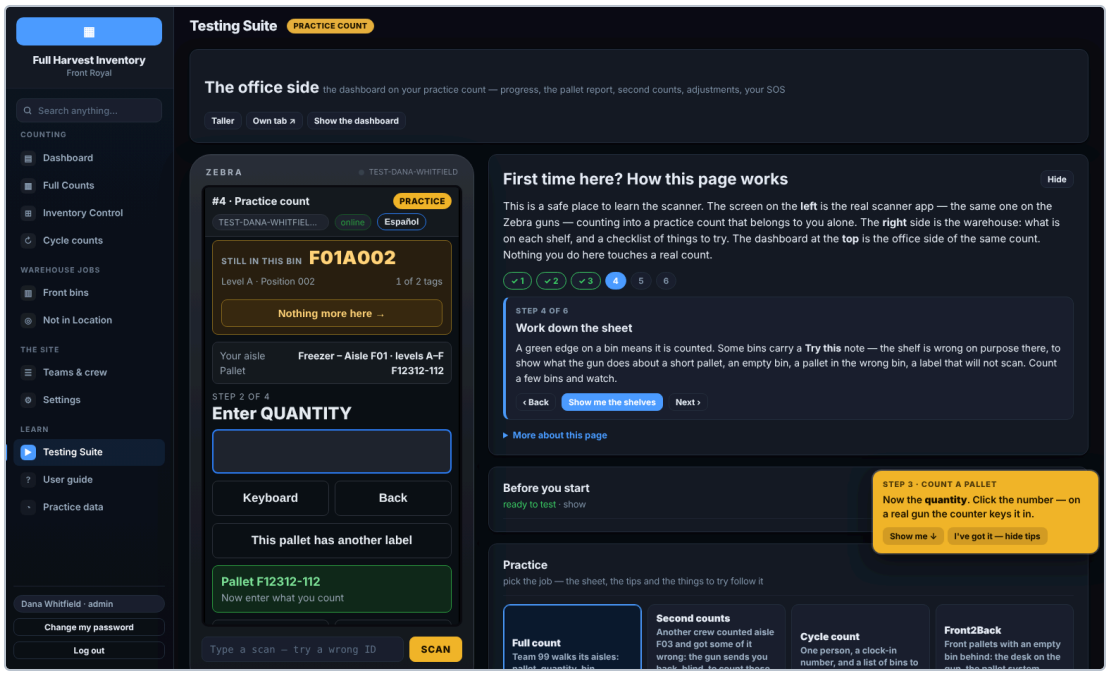
Start over begins a new practice run. The one you were on is kept under **Your earlier runs** — when it started, how many of the twelve things you tried, how much you counted — so you can see your own progress. The last ten are kept. Nothing on a real count is touched.

Practice — pick the job. The first card after *Before you start* lists the jobs a scanner does, and the page follows the one picked: the full count keeps its sheet and sign-on chips; the others bring their own pane in their place, and *Things to try* lists only that job's. Each is set up on your practice the first time you pick it — and only the jobs your login may do for real are offered (a login without Cycle counts is not offered the cycle count; the card says which are missing and why, whether they are not on the login or switched off for the site):

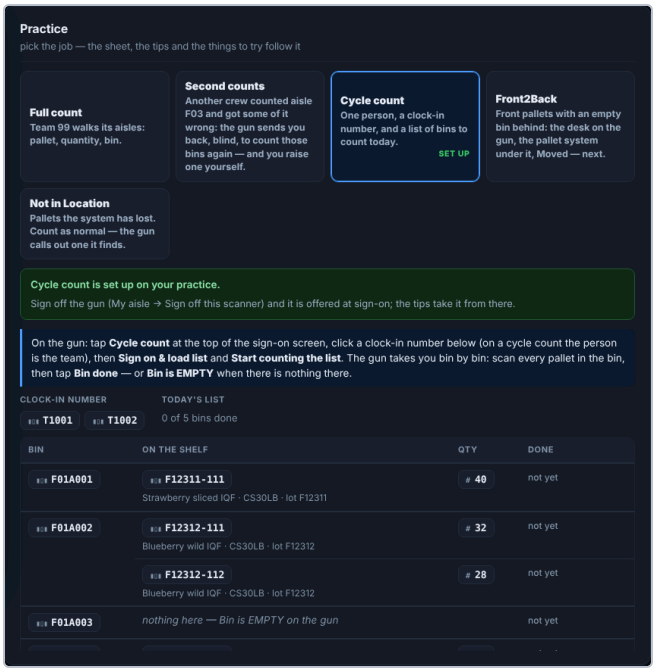
PICK	WHAT IS SET UP	WHAT THE GUN DOES
Second counts	Another crew — team 98 — has counted a third aisle, F03, and got some of it wrong: a pallet keyed short, one missed, one found in the wrong bin, one not on the report. Four bins come back as second counts; a fifth, which they recorded empty, is left for you to raise by hand. The crew that made a first count is never offered the second, which is why it has to be somebody else's aisle	Sign on to the **full count** as team 99 as usual. The My aisle screen says **Second counts available** — tap **Start second counts**. The gun names the bin and why it came back, never the numbers: scan every pallet in it, then **Bin done — nothing more here**; it moves to the next. Then raise one yourself: in the dashboard at the top, **Second counts → Bin F03A004 → Request second count**, tap **Refresh** on the gun, and do it with **Bin is EMPTY — scan the bin**. Watch each line settle under the dashboard's Second counts tab
Cycle count	A practice cycle count beside your run, with a five-bin list: a plain bin, two pallets, an empty bin, a short pallet, one whose pallet has gone	**Cycle count** at the top of the sign-on screen; sign on with a clock-in number alone; **Start counting the list**; every pallet in the bin, then **Bin done**
Front2Back	Three front pallets to put back in the empty bin behind — one of them into a bin that turns out not to be empty	**Front2Back** at sign-on with a clock-in number alone; pick the aisle; the gun shows the move desk — pallet, from → to, the pallet system under it; put it back, tap **Moved — next**; the one that will not go is left and skipped from the office with the reason
Not in Location	Three lost pallets on the list, for you alone: two are on the shelves, one is nowhere	Count the full count as normal; the gun says **found!** the moment it scans one, and the office is told where

Each job adds its own lines to **Things to try**, the tips follow it on the gun, and the pane under the card shows the list — bins, pallets, moves, lost pallets — with what is done. The planted lost pallets

never reach the real Not in Location page or a scanner on the floor. **Start over** clears the lot.



The Testing tab: the real gun on the left, the shelves and what to try on the right.



Practice, as inventory control sees it: the full count and its second counts, and a line saying which jobs are not on this login.



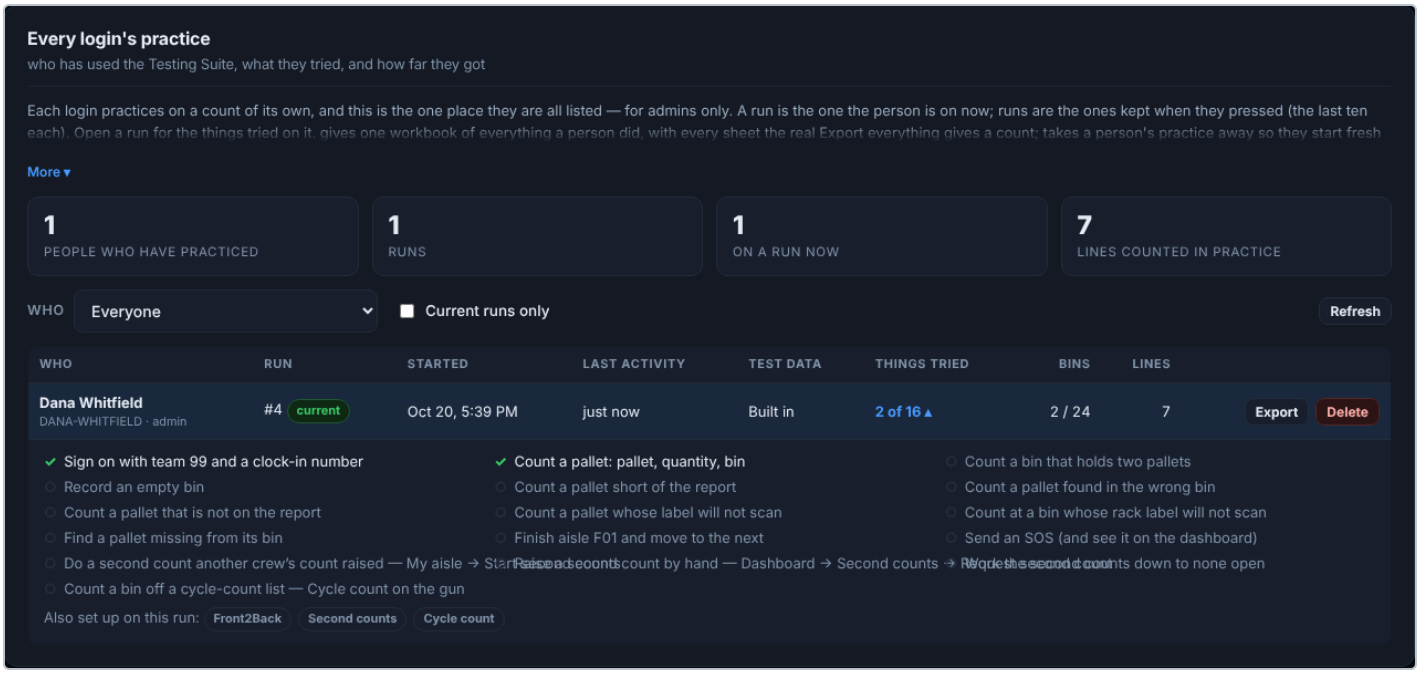
Second counts picked: the other crew's aisle, what they keyed beside what is on the shelf, and the state of each bin.

Step 8c-1 — Practice data: every login's practice, for an admin

Learn → Practice data (`/practice`) is in the sidebar for admins only. It lists every login's practice from the Testing Suite in one place — the practice itself stays each person's own, and this is the one page that sees all of it:

- **Who** has practiced, named with their login, the most recent first; the tiles count the people, the runs, who is on a run now, and the lines counted in practice.
- **A row per run**: *current* (the one they are on) or *earlier* (kept when they pressed Start over — the last ten each), when it started, when they last did anything on it, whether it was the built-in data or their own file, **things tried** (open it for the list, done ones ticked, and which other jobs — second counts, a cycle count, Front2Back, Not in Location — were set up on the run), bins and lines counted.
- **Who** and **Current runs only** narrow the list. **Export** gives one workbook of everything a person did (the same one the Testing Suite gives them). **Delete** takes a person's practice away — every run — so the Testing Suite starts them fresh next time; it is written to the log.

Nothing here touches a real count, and a supervisor opening the address is sent on to a page they may use.



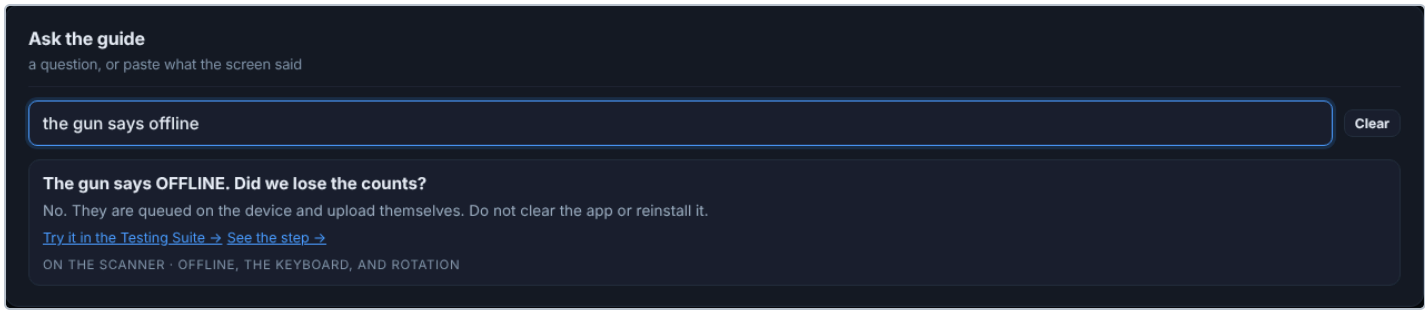
Learn → Practice data: every login's runs, who tried what, and how far they got.

Step 8d — The user guide: this document, inside the app

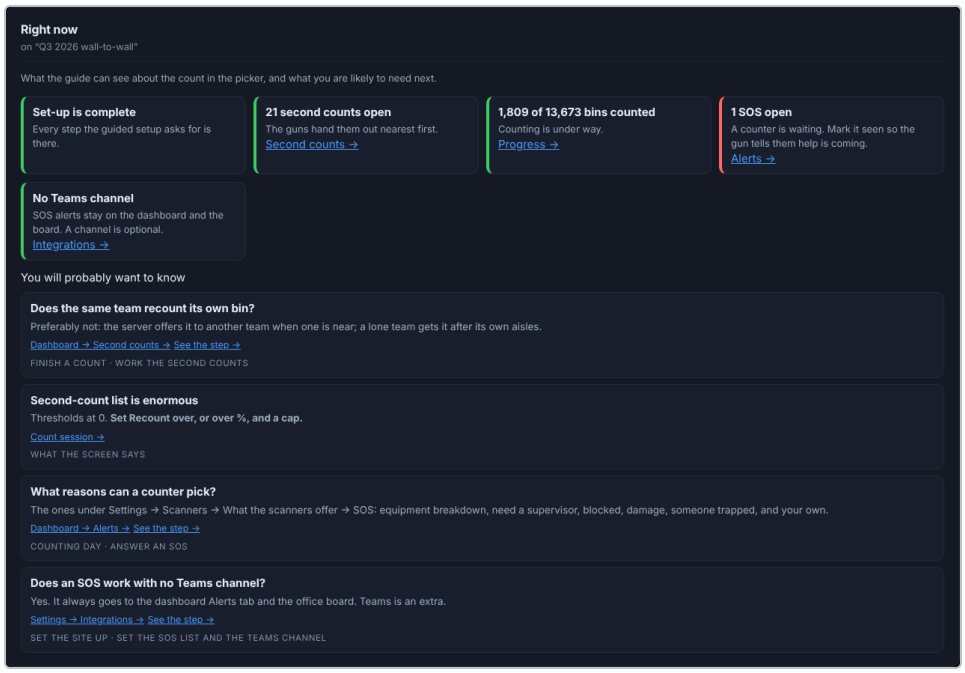
User guide in the sidebar, under *Learn*. Every login can open it, whatever else it may or may not do. It follows this SOP job by job and is the first place to send a new starter.

- **Journeys.** Your first day, Set the site up, Set up a count, Counting day, Finish a count, Cycle counts, Front bins and pallet moves, Not in Location, and On the scanner (the counter's loop, for training). Each step says what to do, links to the card it is about, and then lists **what tends to go wrong here** — the message you will see, what it means, what to do — and **what people ask at this point**. Tick a step when it is done; the ticks stay in that browser, so each person keeps their own place.
- **How-to guides.** Step-by-step guides with pictures — second and third counts, the ERP upload an aisle at a time, employees teams and gear sizes, LIVE and TEST. **Read** shows one on the page (its contents list jumps within it, and it stays open on a reload); **⬇ PDF** downloads it to print. Every login gets the guides for users; an admin also gets the admin editions, with the setting up in, under *For admins*. They are the same files as docs/guides.
- **Ask the guide.** Type a question in your own words (*"the gun says offline — did we lose the counts?"*, *"how do I approve adjustments"*) or paste a message straight off the gun or the dashboard. It searches every step, every question, every section of the how-to guides, and every message the gun and the server can show, and answers with the fix and a link. *Gun*, *handheld* and *Zebra* all mean the scanner; *Wi-Fi* means signal; *variance* means adjustment.

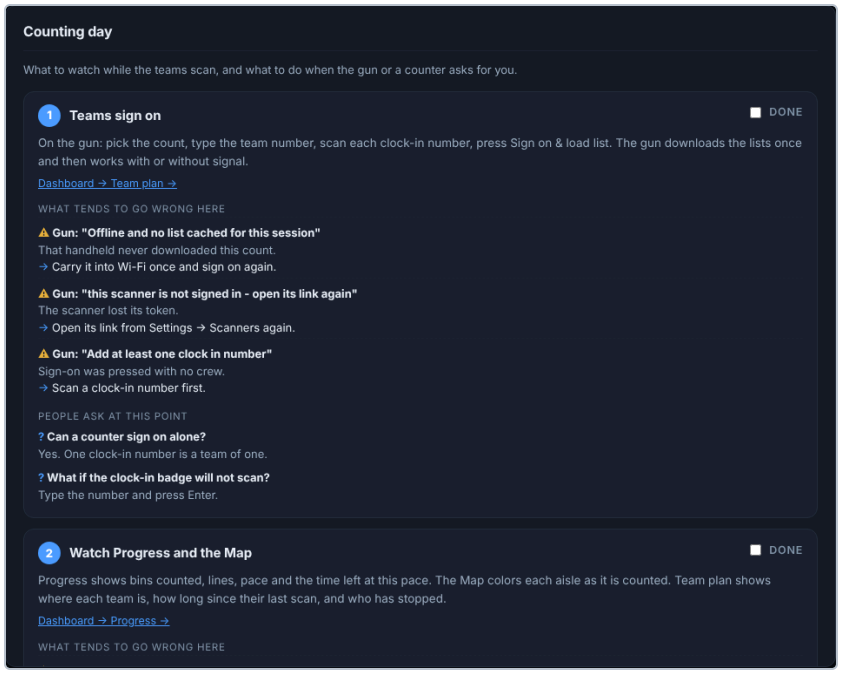
- **Right now.** The guide reads the count in the picker — the guided setup, open SOS, second counts, the thresholds, the scanners, the Teams channel — and lists what is missing, red for required and amber for wanted, each with a link. Under it, **You will probably want to know** offers the questions those gaps usually raise. It refreshes every fifteen seconds.
- **Only doors you can open.** A login without Settings sees *"ask an admin"* in place of a Settings link, never a link that bounces.



Ask the guide: a question in plain words, the answer, and where to go.



Right now: what this count is still missing, worked out from the data.



A journey open: the step, what goes wrong there, and what people ask.

Step 9 — Set up the counting screen

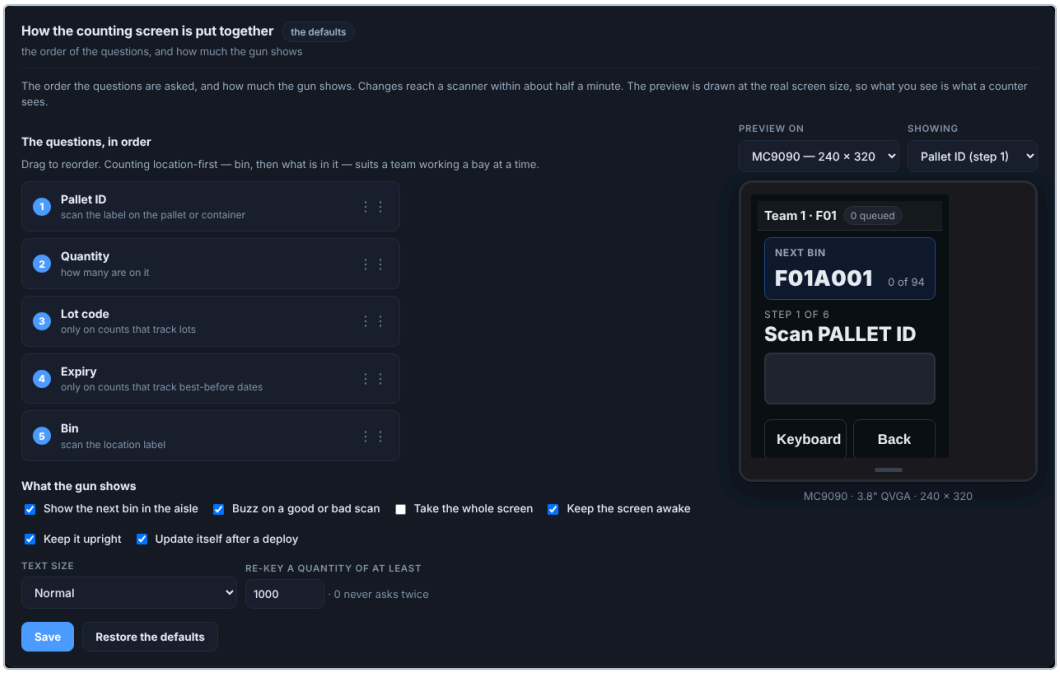
Settings → Scanners.

This is what a counter actually sees. The preview on the right is drawn at the real pixel size of the device (MC9090 is 240 × 320; MC9200 is 480 × 640) using the scanner's own stylesheet, so it cannot drift from reality.

- **The questions, in order.** Drag to reorder. The default asks for the pallet, then the quantity, then the bin. Counting **location-first** — bin, then what is in it — suits a team working a bay at a time. Lot code and expiry sit in the order too, and only appear on counts that ask for them (Part 3). A question can never be dragged off the list entirely.
- **Show contents after a pallet scan** — shows the SKU and description the ERP has for that pallet, so the counter can see they are at the right one.
- **Show the next bin in the aisle** — once a team starts a section, the gun tells them which bin comes next (001 → 002 → 003), so nothing is skipped. A bin holding several pallets keeps the guide until every tag in it is accounted for — see step 15.
- **The on-screen keyboard never appears by itself**, on any screen — not for the quantity, not for a clock-in number. The scanner is the keyboard. The **Keyboard** button on the counting screen, and **Type it instead** on the sign-on screen, bring one up for whoever has to type.

- **Update itself after a deploy** — on unless you turn it off. A scanner left running keeps the version it started with: these are installed apps that sit on a cradle overnight and are the same page in the morning. With this on, the gun compares what it is running against what the server is serving and reloads itself when it has fallen behind — **never mid-pallet, and never with counts still queued on the device**. If it is in the middle of a line it shows a green **Update ready** bar and waits for the line to finish; tapping the bar takes it at once. It reloads once for any one version: a gun that comes back still behind says *"update did not take"* rather than reloading in a loop halfway down an aisle.
- **Keep it upright** — on unless you turn it off. These are portrait handhelds held one-handed at a rack face, and a screen that flips halfway down an aisle is unusable. There is **one** way up: the installed app asks the device for it by name, and a gun flipped end over end is not "still portrait" as far as this app is concerned. Only an app that owns the screen (installed, or full screen) is allowed to ask at all — a browser tab is always refused — so when the handheld turns anyway, the app turns *itself* back: sideways gets a quarter turn, end over end gets a half turn, and either way the counting screen reads upright in the hand. It is only a picture being turned — scanning, tapping and the comments countdown all carry on exactly as before. Turn it off and the gun is left however the device turned it, with a yellow **Hold the scanner upright** strip across the top. Turning auto-rotate off on the handheld (an Android setting, not an app one) is still the tidiest fix of all.
- **Take the whole screen** — off unless you turn it on. It makes the app fill the display at sign-on, but the browser announces that with a bar carrying the site's address, right over the counting screen. **Installing the app** on the handheld (step 8) is the quiet way to lose the browser, and the one to prefer.
- **Keep the screen awake** — holds the display on while a team is counting, instead of it sleeping between bays.
- **Buzz on a good or bad scan** — useful with ear defenders.
- **Text size: Large** — for gloves and a freezer.
- **Re-key a quantity of at least** — anything this big has to be typed twice. 0 never asks twice.
- **Comments step moves on after N seconds** — how long the comments step waits before it goes to the next bin by itself. It ships at **2 seconds**: long enough to tap a reason, short enough that a counter with nothing to say is not standing there. Use **−** and **+** to change it, or type a number. **0** turns the clock off and waits for the counter. Typing or tapping a chip always stops the countdown, whatever it is set to.

Press **Save**. Scanners pick the change up within about half a minute, between pallets — nobody has to sign out.



Settings → Scanners. Drag the questions into the order your site counts in; the preview is the real MC9090 screen size.

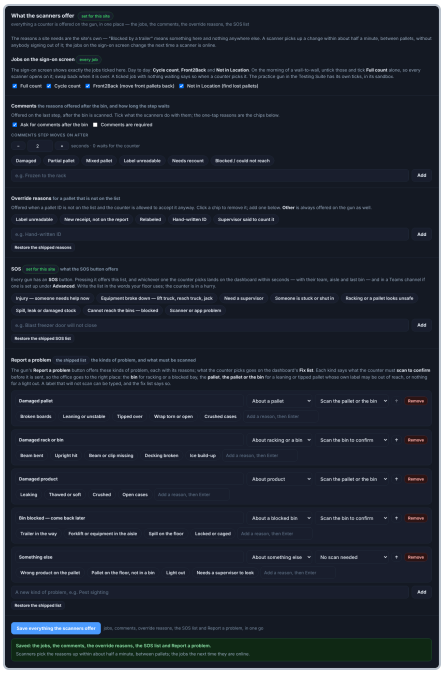
Step 10 — Set the one-tap reasons

Settings → Scanners → What the scanners offer. One section for everything a counter is offered on the gun, with one **Save everything the scanners offer** button at the bottom. The jobs are open; **Comments**, **Override reasons**, **SOS** and **Report a problem** are folded under their headings — click one to open it:

- **Jobs on the sign-on screen** — the full count, the cycle count, Front2Back, Not in Location (see step 8).
- **Comments** — offered on the last step, after the bin is scanned. Two tick boxes: **Ask for comments after the bin** (untick it and no scanner asks, on any count, whatever the count's own setting says) and **Comments are required** (the counter must tap a reason or type a note: no Skip, no blank, nothing moving on by itself). Under them, how many **seconds** the step waits before moving on when comments are optional (0 waits for the counter), and the one-tap reasons.
- **Override reasons** — offered when a pallet ID is not on the list and the counter is allowed to accept it anyway. Click a chip to remove one, type and **Add** for a new one. **Other** is always offered on the gun as well, whatever you configure.
- **SOS** — the list a counter picks from when they press SOS.
- **Report a problem** — the kinds of problem on the gun's Report a problem button, and what must be scanned to confirm each (see step 8).

Counters wearing gloves in a freezer will not type. These are the buttons they tap instead, and the reasons a site needs are its own — "Blocked by a trailer" means something here and nothing

anywhere else. A scanner picks a change up within about half a minute, **between pallets**: a reason added while a counter is halfway through a line appears after that line is finished. The jobs on the sign-on screen change the next time a scanner is online.



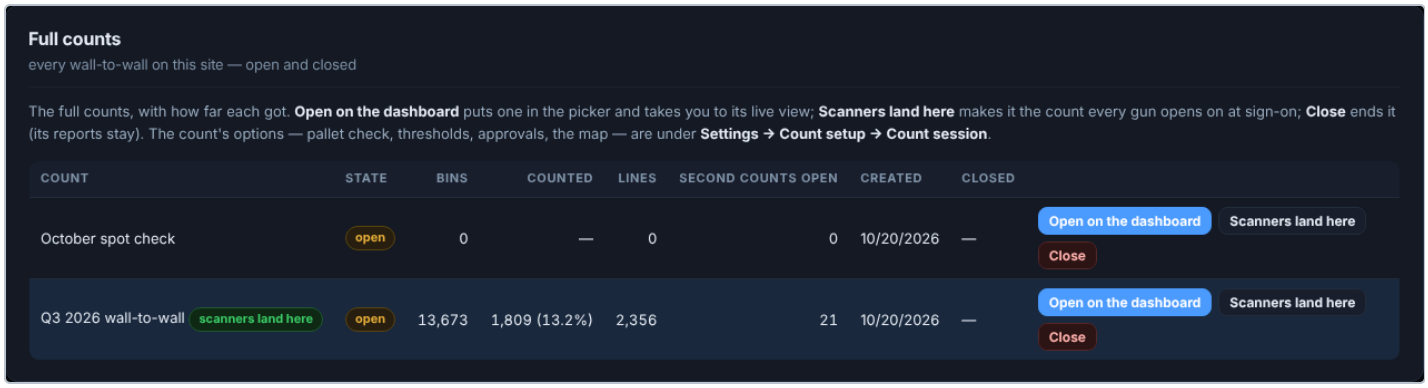
Settings → Scanners → What the scanners offer. These are the buttons a counter taps instead of typing.

Part 3 — Setting up a count

> Do this the day before, not on the morning of the count.

Step 11 — Create the count

Full Counts in the sidebar is the home of the wall-to-walls: a table of every full count, open and closed, with bins counted, lines and second counts open, **Open on the dashboard**, **Scanners land here** (the count every gun opens on at sign-on) and **Close** / **Reopen**; a **New count** tab that creates one; and a **Set-up** tab that lists what the count in the picker still needs, each step linking to where it is done. The count's options stay under *Settings → Count setup → Count session*, and the Dashboard stays the live view of whichever count the picker holds. Cycle counts have their own page, *Cycle counts*, the same way.



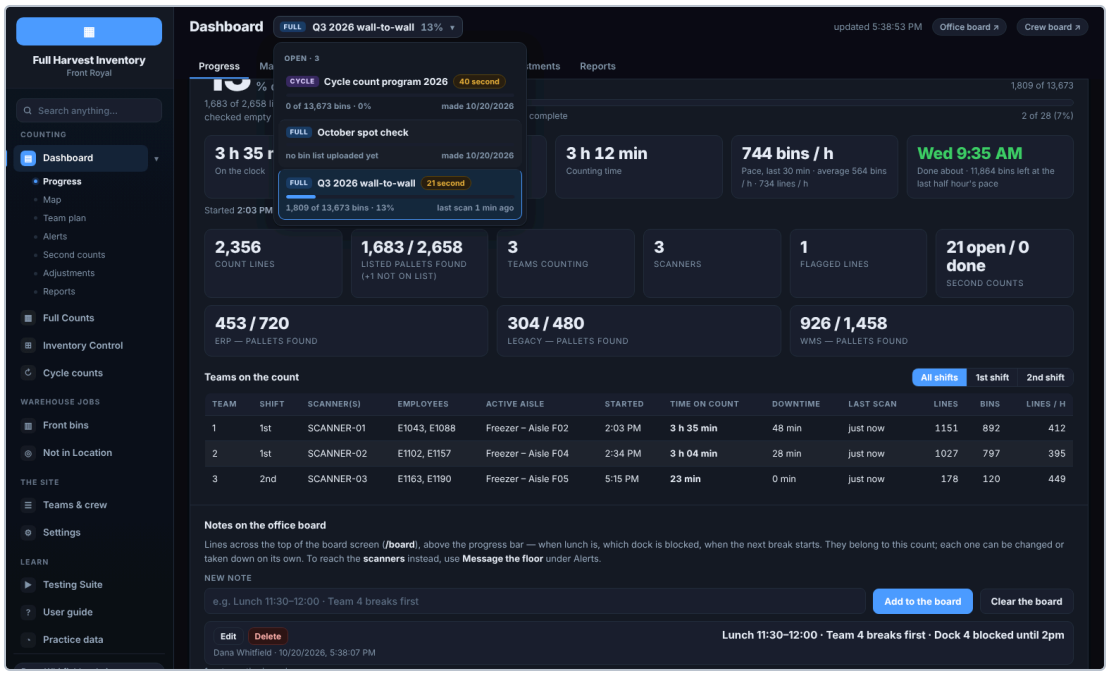
Full Counts: every wall-to-wall, how far it got, and what to do with it.

The picker shows what the login may work. A login without the *Cycle counts* page sees no cycle count in the picker at the top of any page, and a cycle-counter login sees no full count.

Settings → Count setup → Count session → Start a new count.

1. **Name** it something you will recognize later: `Q3 2026 wall-to-wall`.
2. **Type**:
   - **Full count** — teams work whole aisles. This is a wall-to-wall.
   - **Cycle count** — a batch of bins per day or week; see Part 6.
3. **Bin list** — attach it here, or upload it later in Settings.
4. **Inventory report** — what the ERP thinks is on hand. **Attach it.** With it you get variances, second counts and an ERP adjustment file. Without it the count still records what is there, but nothing is compared against anything.
5. Press **Create count**.

The new count inherits the rack drawing the last one used, so the map works immediately.



The count is picked in the header, on every supervisor page. Each row carries its own progress, so two counts are never confused.

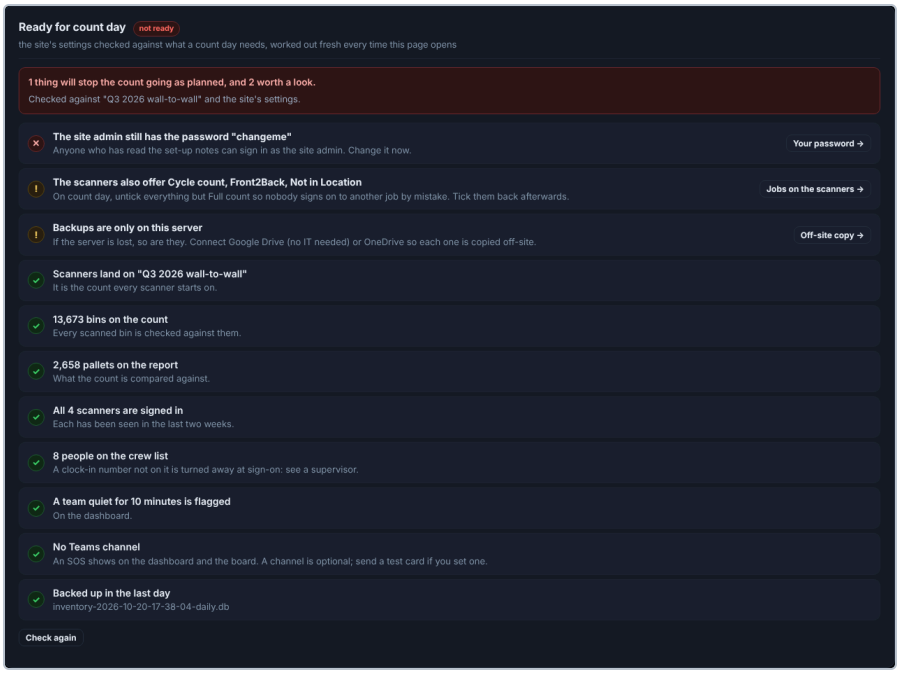
Step 11a — Ready for count day

Settings → Count setup → Ready for count day.

The first card on Settings checks the *site* against what a count day needs, worked out fresh every time the page opens — the count's own lists are the checklist below it. Each line is green, amber (*works, but you would want to know*) or red (*the count will not go as planned*), the worst first, and each amber or red line has a button that opens the card that puts it right. It looks at:

- the count the scanners land on — open, not still a trial run, with a bin list and a report;
- **the jobs the scanners offer** — on count day the full count alone; anything else ticked is amber, the full count unticked is red;
- the scanners — registered, each one's link opened on its device, seen lately;
- the crew list — without one any clock-in number can sign on;
- the stopped-scanning alert, the override reasons, the Teams channel;
- the backups — taken in the last day, and copied off-site (Google Drive or OneDrive);
- the site admin's password — still *changeme* is red.

Run it the afternoon before, and again first thing. **Check again** re-reads it after a change.

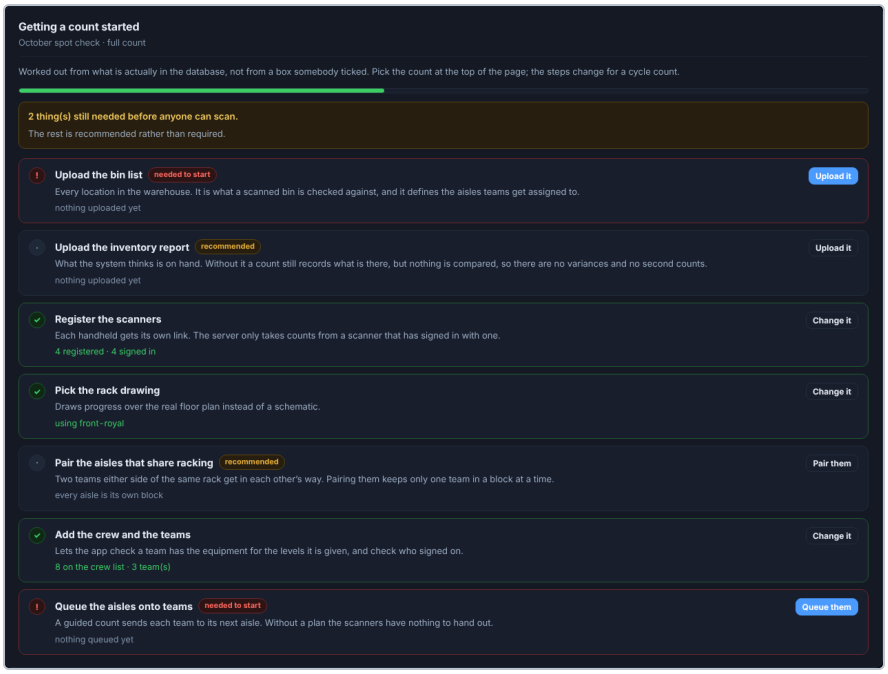


Ready for count day: red first, then amber, then what is already right — each with a way to the fix.

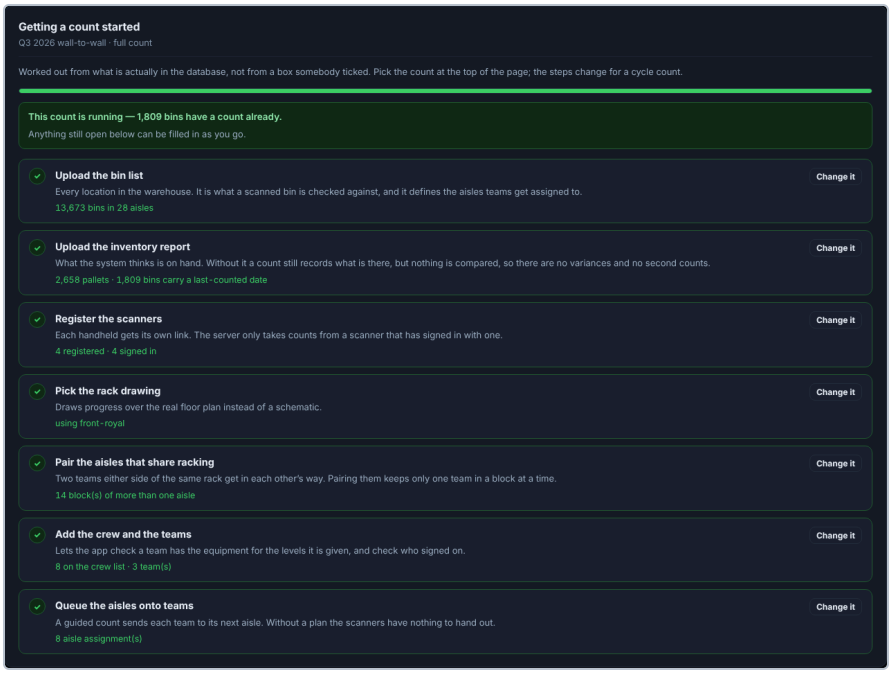
Step 12 — Check the guided setup

Settings → Count setup.

A checklist worked out from what is actually in the database, not from a box somebody ticked. It tells you what is still missing and takes you straight to the screen that fixes it. For a full count it wants: the bin list, the inventory report, registered scanners, the scanner screen, racking blocks, a crew roster and a team plan.



A new count: two things stop anybody scanning, the rest are recommended. Every step links to the screen that fixes it.



The same checklist once that count is ready to run: every step green, and the count already running.

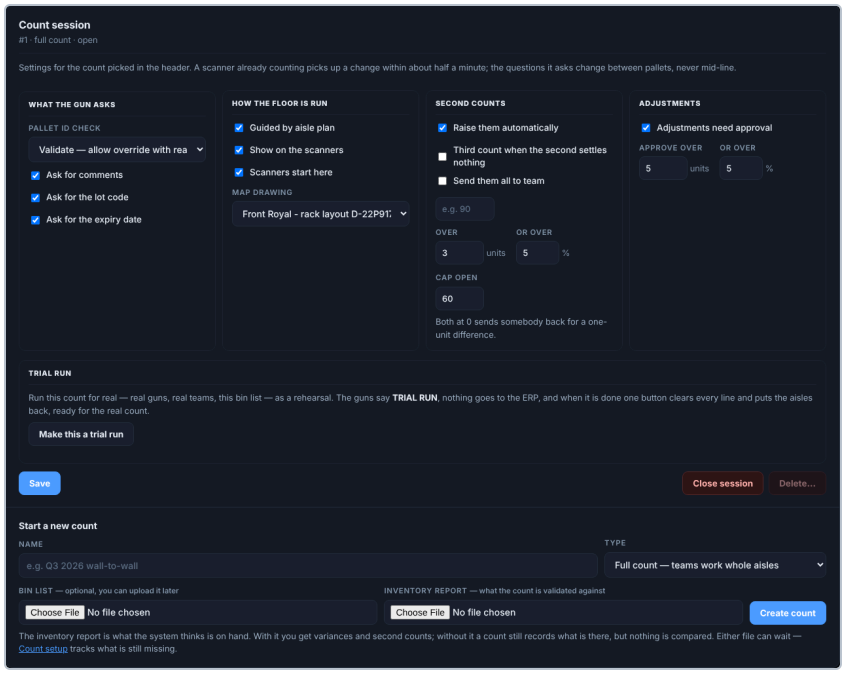
Step 13 — Set the count's options

Settings → Count setup → Count session. These apply to the count picked at the top of the page, and a scanner that is already counting picks up a change within about half a minute. The questions it asks change between pallets, never mid-line.

OPTION	WHAT IT DOES	SUGGESTED
Pallet ID check	*Validate — allow override*: a pallet not on the report asks the counter **Count it anyway? YES / NO**; YES counts it and it shows as a positive adjustment. *Validate — no overrides*: it cannot be counted. *Accept any ID*: no checking, though duplicates are still blocked.	Validate with override
Guided by aisle plan	Teams are sent to their assigned aisle and warned when they scan a bin outside it. Off means anyone can count anything.	On for a wall-to-wall
Ask for comments	Adds the optional comments step at the end of each pallet.	On
Auto second counts	Raises a "go back and look again" task automatically when a line disagrees with the report.	On
Show on the scanners	Untick to keep this count off the scanners' list — while it is being set up, or once the floor is done with it. The dashboards still have it, marked *not on scanners*.	On, once the lists are uploaded
Scanners start here	Every scanner lands on this count at sign-on. They can still pick another.	On, on count day
Ask for the lot code	Adds a LOT CODE question, checked against the report. Wrong lot is called out at the pallet.	On only if you track lots
Ask for the expiry date	Adds an EXPIRY question and flags anything already out of date.	On for frozen food
Recount over N units	A difference smaller than this raises no second count.	2–5 units
or over N %	...or at least this share of the expected quantity. Either threshold is enough.	5–10 %
Cap open	Stop raising automatic second counts once this many are open, so the list stays walkable.	50–100
Adjustments need approval	Every difference from the report has to be approved, with a reason code, before it reaches the ERP file. Off by default.	On for a wall-to-wall that is audited
Approve over N units	A difference smaller than this goes through without a signature.	5–10 units

OPTION	WHAT IT DOES	SUGGESTED
or over N %	…or at least this share of what the report expected. Either threshold is enough.	5–10 %
Map drawing	Which rack drawing the map uses. *Schematic* builds one from the bin codes.	Your drawing

> **Why the thresholds matter.** With both at 0 the app sends somebody back for a one-unit difference, and by mid-morning the second-count list is longer than the count. Set them once and the list stays short enough that people actually walk it.



Settings → Count setup → Count session: every option above, on one card, for the count picked at the top of the page.

Step 13a — A trial run, before the real count

Settings → Count setup → Count session → Trial run.

The way to find out that a crew does not know the empty-bin button, or that an aisle has no rack labels, is to run the count for real a shift or a day before it matters. **Make this a trial run** turns the count into a rehearsal on the same bin list, report and team plan:

- the guns show **TRIAL RUN** in the header, and so do the dashboard and the office board;
- nothing can be sent to the ERP, and nothing counted moves a bin's *last counted* date;
- everything else works exactly as on the day — teams, aisles, SOS, second counts, reports.

When it is done, **End the trial — clear it for the real count**, and type **CLEAR**. Every line, sign-on, SOS and second count goes; the bin list, the report, the team plan and every setting stay, and each team's aisles go back to the start. The guns are told the next time they check in (within a minute, between pallets): they forget what they counted in the trial, so no pallet comes back as "already counted", and a trial line a gun sends late is dropped.

Step 14 — Give the teams their aisles

Dashboard → Team plan.

Either upload a **counting plan** (Settings → Lists & racking → Counting plan: one row per aisle, in the order each team counts it, with the levels their equipment reaches), or queue aisles by hand here.

- A team's next aisle starts automatically as soon as the block it belongs to is clear. That is what staggers the crews apart.
- A team is not sent where its equipment cannot reach: if the roster says a crew is on foot, they will not be queued onto level D. A supervisor can override this.
- Two teams **can** share one aisle as long as their levels do not overlap — a forklift crew high and a crew on foot low.
- **Two teams on the same aisle and levels, from opposite ends.** Queue one team as usual (*Start from: the 001 end*) and the other with **Start from: the far end**. Both start at once; the far-end team's gun walks the bins backwards (F01A010, F01A009 …) and says *start at the far end · team 1 from the other end*. Each gun sees the bins the other has counted, so they meet in the middle. At most one team each way. The aisle is checked for missing pallets only when the second of the two hands it back. **Auto-assign the aisles → Two teams per aisle, from opposite ends** pairs the teams (1 with 2, 3 with 4 …) and plans it for every aisle; an odd team out works alone. A counting-plan file takes the same thing in an optional **Start** column: *far end* or *001 end*.

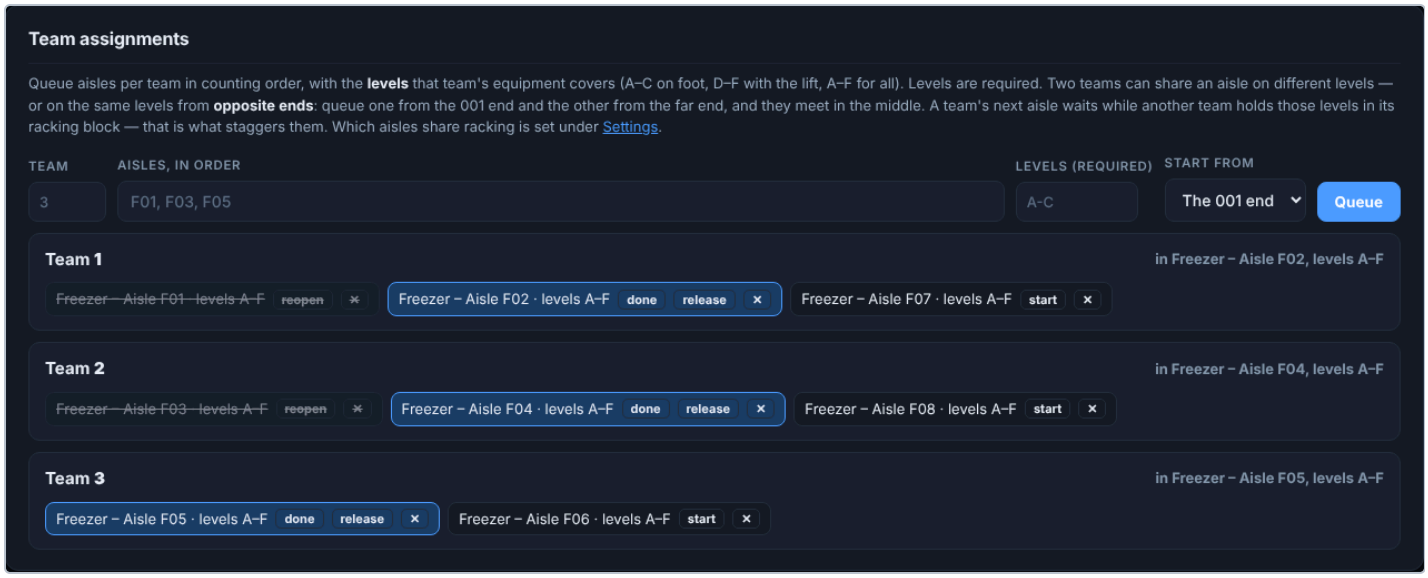
The buttons on an aisle. Each aisle in a team's row has what fits its state: **start** on one waiting (if its racking block is free); **done** on the one being counted (the same as the gun's *Aisle complete*); **release** sends the one being counted to the **back** of the team's queue and the next one starts — with nothing else queued it simply starts again, and the card says so; **reopen** on a finished one queues it again; × takes the aisle off the plan whatever its state — the aisle's counts stay, and if the team was counting it, their next aisle starts. On an aisle being counted or finished, × asks first.

Auto-assign the aisles. The card above the queue does the whole plan at once, staggered:

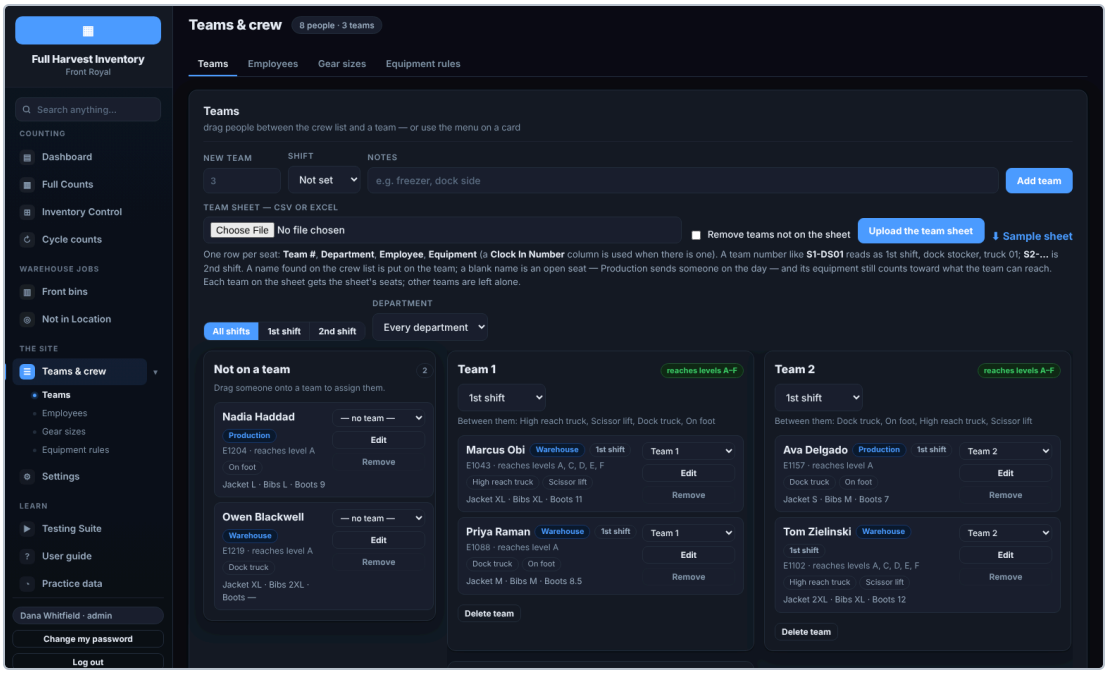
1. Pick the **teams** (blank is every team on Teams & crew, or one **shift**), the **levels** (blank is all), and which way to **walk the aisles**.

2. **Preview.** The aisles are split into one continuous stretch per team, about the same racking each, so the teams start spread across the building and never bunch up. A team gets only the levels its equipment reaches; what it cannot reach goes to the nearest team that can, after its own stretch (marked `*`). The preview lists each team's aisles in order, with warnings — two teams starting in the same racking block, a level nobody can reach — and what it left alone.
3. **Queue this plan.** Each team's first aisle starts at once. **Replace what is queued** re-plans the queued aisles and leaves anything being counted, or done, alone.

The rules it follows are shown on the card, with what to fix if one is missing: **racking blocks** (step 7 — without them two teams can be released into back-to-back racking), **equipment** on Teams & crew (a team with nobody on it is given every level), and the **shift**.



Dashboard → Team plan: each team's aisles in counting order — what they are on now, and what is queued behind it.

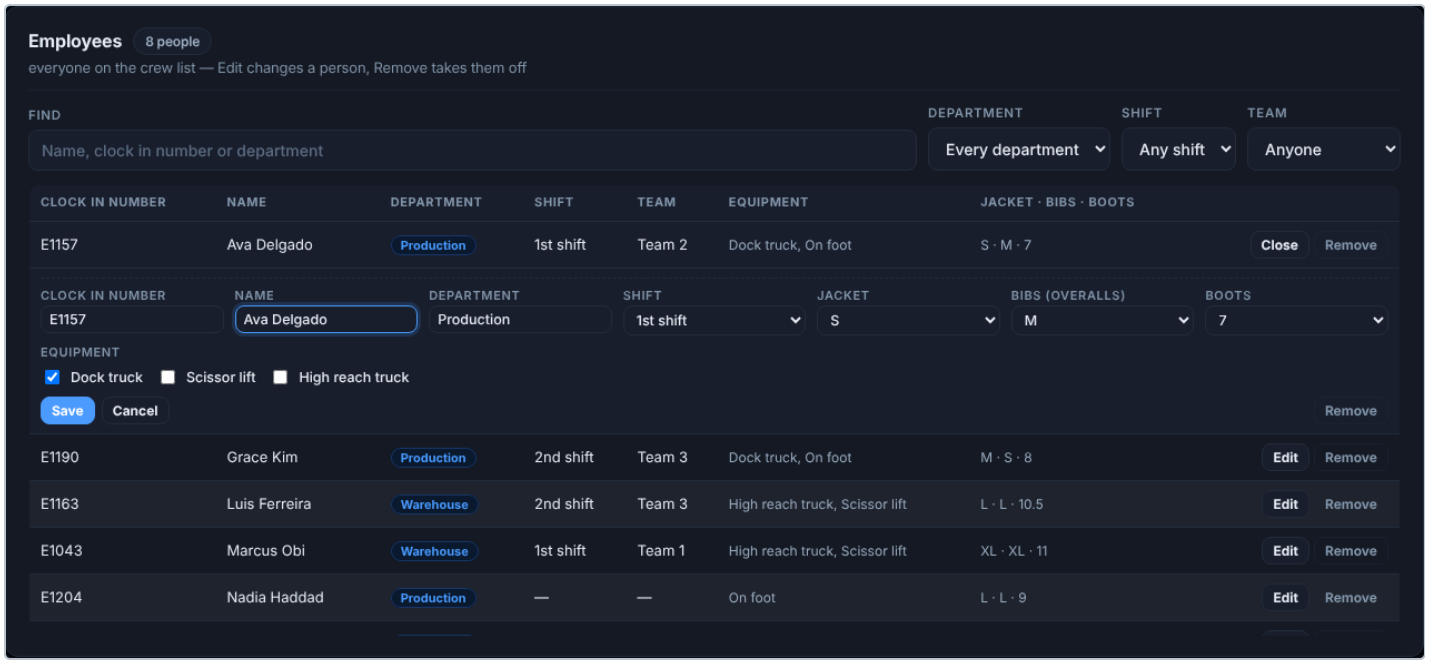


Teams & crew (`/teams`) → Teams: who is on which team. Equipment is what decides which levels a team can be sent to.

Employees: add, edit, remove. *Teams & crew → Employees* is the crew list itself.

- **Add employee** — type the **clock-in number** and **name**; the department, **shift** (1st or 2nd), equipment and jacket, bibs and boots sizes are optional. Press **Add employee** (or Enter in the name box) and the form clears for the next person. A clock-in number that is already on the list is refused, with the name it belongs to — use **Edit** on that person instead.
- **The whole list from a file** — the same card uploads a CSV or Excel crew list (Appendix A); **Replace the whole list** clears it first. **Export current list** downloads it.
- **Edit** — every row of the **Employees** table has **Edit**: it opens the person's clock-in number, name, department, shift, equipment and sizes under the row; **Save** keeps all of it. A new clock-in number moves their team and their team-sheet seat with them (it asks first); counts already scanned keep the number they were scanned under. Their card on the **Teams** tab has the same **Edit**.
- **Remove** — on the row, on their card, or inside the Edit form. It asks first, takes them off the crew list and off their team; the counts they scanned stay as they are. A removed clock-in number can no longer sign on to a scanner.
- **Find**, **Department** (or *No department*), **Shift** (1st, 2nd or *No shift*) and **Team** (or *Not on a team*) narrow the table. The list has a **Shift** column, and so do the crew-list export and the gear-sizes workbook. A search for a person's name in the search box lands on their row.

Every add, edit and removal is in the log (Settings → Backups & log) with who made it and what changed.



Teams & crew → Employees: everyone on the crew list, with Edit and Remove on every row — here one row open for editing.

The team sheet. *Teams & crew → Teams → Upload the team sheet* takes the sheet the floor already keeps, one row per seat on each team: **Team #**, **Department**, **Employee**, **Equipment** (and a **Clock In Number** column, if the sheet has one).

TEAM #	DEPARTMENT	EMPLOYEE	EQUIPMENT
S1-DS01	Warehouse	Michael Roman Travera	Dock Stocker
S1-DS01	Production		On Foot with Dock Stockers
S1-RT05	Warehouse	Jason Davies	High Reach
S1-RT05	Warehouse	Donovan Lopez	Scissor Lift - Counting
S1-RT05	Production		Scissor Lift - Driving

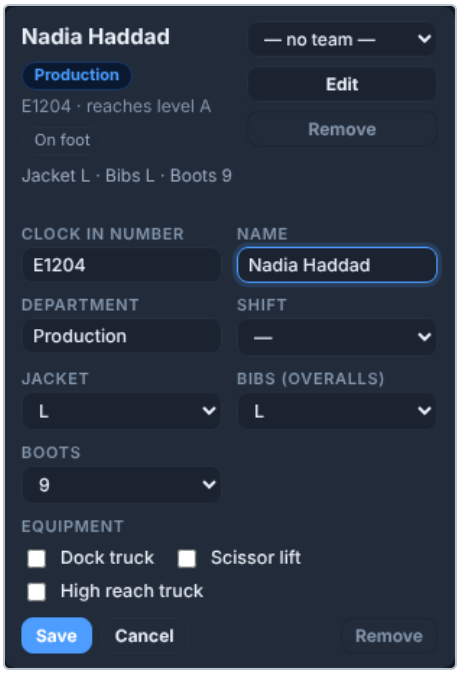
- **The team number says the shift and the truck.** *S1-DS01* is 1st shift, dock stocker, truck 01; *S1-RT05* is a reach truck, truck 05; *S2-…* is 2nd shift; *S1-T0L* is the team lead. Each team on the sheet is made (or brought up to date) with that shift and a note saying what it is.
- **A named person on the crew list goes on the team** — matched by clock-in number when the sheet has one, else by name. A name that is not on the crew list yet stays on the card, marked *not on the crew list*: add them under Employees and upload the sheet again.
- **A blank name is an open seat** — Production sends someone on the day. Its equipment still counts toward what the team can reach: the truck is the team's either way.

- **The equipment words are read as the floor writes them**: *Dock Stocker* is the dock truck, *High Reach* the high reach, *Scissor Lift - Counting* and *- Driving* both the scissor lift, *On Foot with Dock Stockers* on foot, *Supervisor* a job rather than a machine.
- Uploading again replaces each listed team's seats, never doubles them; teams not on the sheet are left alone unless **Remove teams not on the sheet** is ticked (their people go back to the crew list). The upload says how many teams, how many people were found on the crew list, the open seats, and any equipment word it did not know.
- On the gun, the team box takes the number as scanned or typed — **Type it instead** brings up the full keyboard, so *S1-DS01* can be keyed by hand.

> **Check the level rules against these teams.** With the shipped rules (*What each level needs*), a dock stocker team with someone on foot reaches **level A** only, because levels B–C need a dock truck *and* a scissor lift; a reach-truck team (high reach and scissor lift) reaches **A and C–F**. If a dock stocker reaches B–C on its own at your site, change the rule under *Teams & crew → Equipment rules* before you plan the aisles.

1st and 2nd shift. Each team on Teams & crew has a **shift** — set it when you add the team, or on the team's card (a team from the team sheet takes it from the S1 / S2 in its number). The page can show one shift at a time, and so can the team table on the dashboard, so the 2nd-shift lead sees only their crews.

Departments and gear sizes. Every crew card shows the person's **department** as a blue chip beside their name (*No department* when it is blank), and their **jacket**, **bibs (overalls)** and **boots** sizes under their equipment. **Edit** on the card opens the same form as the Employees table; **Save** keeps it. The department filter above the teams uses the same field.

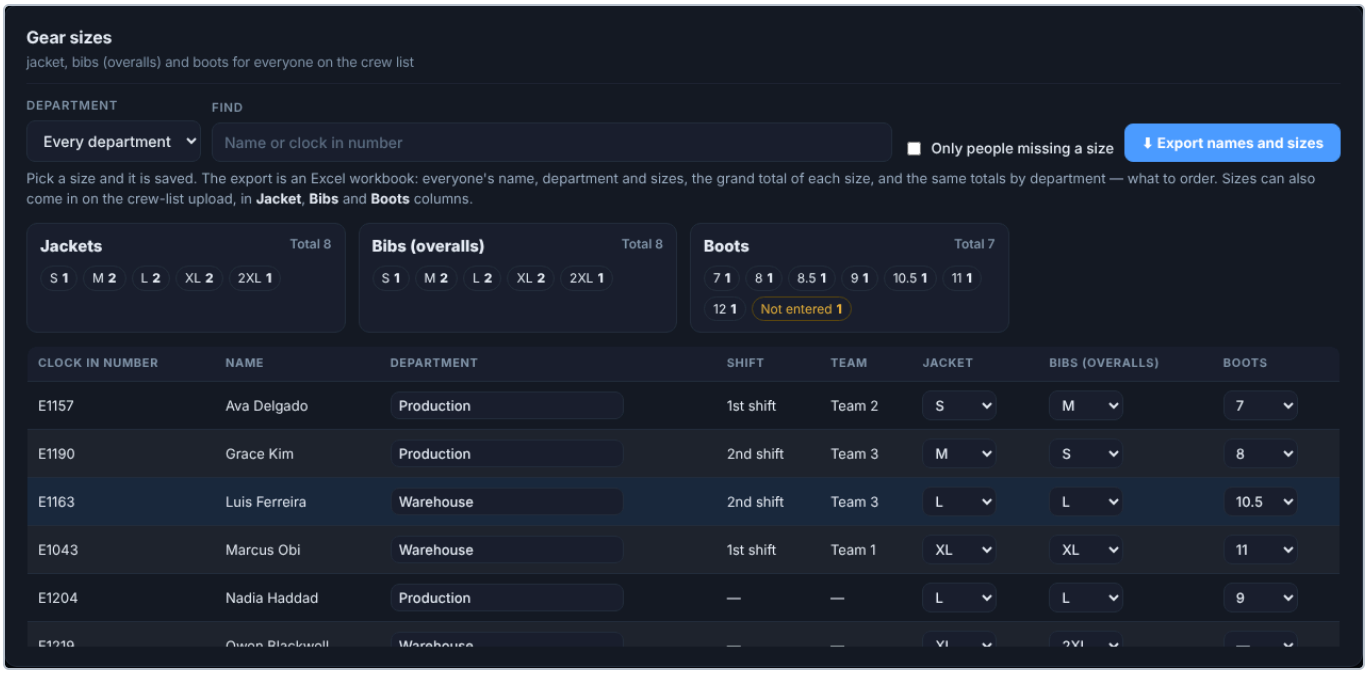


A crew card with Edit open: clock-in number, name, department, sizes and equipment, saved in one go.

Teams & crew → Gear sizes is everyone at once, for ordering:

1. Pick a size in the table and it is saved — the cell flashes green. Jackets and bibs run XS–5XL; boots 5–16 in half sizes.
2. The three boxes above the table are the **totals**: how many of each size, the **total** of each item, and how many people have no size entered yet (amber).
3. Narrow it with **Department**, **Find** (name or clock-in number) or **Only people missing a size** — the list to chase before the order goes in.
4. **Export names and sizes** downloads an Excel workbook: *Names and sizes* (everyone, by department), *Grand totals* (each size of each item, the not-entered count, a grand total per item and the head count) and *By department*. With a department picked, the export is that department alone.

Sizes can also come in on the crew-list upload, in **Jacket**, **Bibs** (or *Overalls*) and **Boots** columns, written any common way — *xx-large* and *XXL* read as **2XL**, *10 1/2* as **10.5**, a wide boot keeps its **W**. A crew list with no size columns leaves the sizes already entered alone. **Gear sizes** is its own tick under a login's *May use*, so the person who orders the gear can have it without the rest of the page.



Teams & crew → Gear sizes: totals per size above, everyone's sizes below, and the Excel export.

Part 4 — Counting day

Step 15 — The counter's procedure (on the handheld)

> **There is a film of this.** `docs/gun-demo.webm` (about four minutes) is a recording of the real app on real data, working through everything below: signing on, counting a pallet, a bay with three tags, two labels on one pallet, a label that will not scan, an empty bin, a pallet nobody expected, a message from the office, voiding a line, losing the wifi, and handing the aisle back. Open it in any browser. It is worth showing a new crew before their first shift.

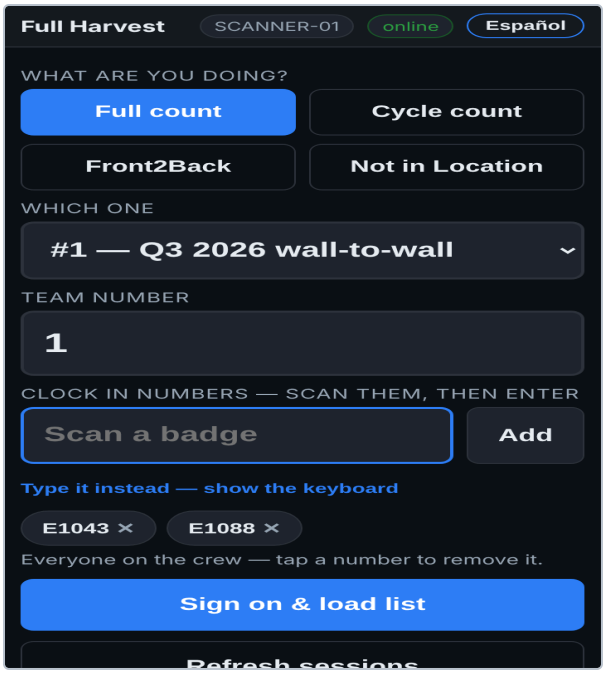
English or Spanish

The button at the top right of every screen switches the gun between English and **Español**. It stays that way on that scanner until somebody switches it back. Everything the counter reads is translated; what reaches the office — the lines, the SOS, the reasons — stays in English, so the office reads one language whoever counted.

Signing on

1. Open the app from the home screen. It shows the scanner's own name.
2. Pick the **count session** (it lands on the default one).
3. Enter the **team number**.
4. Scan or type every **clock-in number** on the crew, pressing Enter after each. Tap a number to remove it.
5. Press **Sign on & load list**. The handheld downloads the bin and pallet lists for that count and can work from then on **with no signal**.

A clock-in number that is not on the crew list is turned away: *"…is not on the crew list — see a supervisor"*, who adds the person under Teams & crew. On the jobs one person does alone — a **cycle count**, **Front2Back** and **Not in Location** — the bar at the top of the scanner shows **the person's name** from the crew list (*GUN-03 · Rosa Delgado*) rather than their number, so anyone glancing at the gun knows whose it is.



Signing on: the count, the team number, and every clock-in number on the crew.

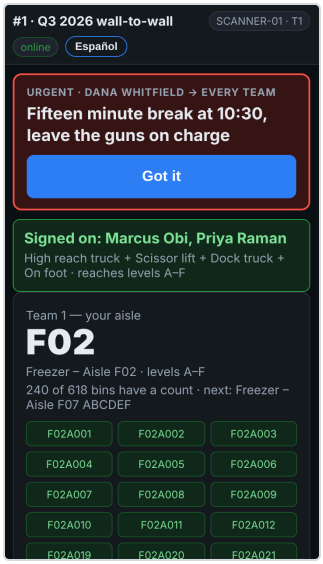
Counting a pallet

The gun asks one question per screen, in the order the site configured:

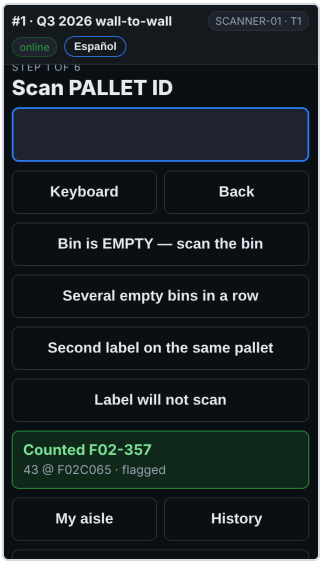
The gun checks that what was scanned is what the step asked for: a rack label at the pallet step, a pallet label at the bin or lot step, any label where the quantity is typed, is named for what it is and refused, and nothing is recorded (*"F01A001 is a bin, not a pallet — scan the label on the pallet"*).

1. **Scan PALLET ID** — the gun says which pallet, and nothing about what is on it. **This is a blind count**: the counter counts what is in front of them, not what the system expects.
2. **Enter QUANTITY** — big numbers may have to be typed twice.
3. **Scan LOT CODE** / **Enter EXPIRY** — only on counts that ask for them. A lot that disagrees with the report, or a date already past, is called out on the spot.
4. **Scan BIN LOCATION** — it says where that bin is and which face it is on.
5. **Comments** — optional. Tap a reason or type a note; leave it and the gun moves on by itself after a couple of seconds (Settings → Scanners sets how long). Typing or tapping a reason stops the countdown.

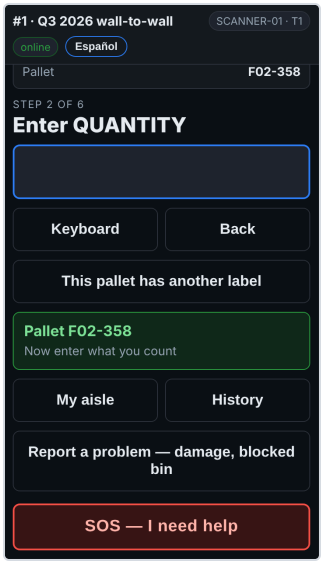
The line is saved on the handheld the moment the last question is answered, and pushed to the server whenever there is a signal. The header shows `online` / `OFFLINE` and how many lines are still queued.



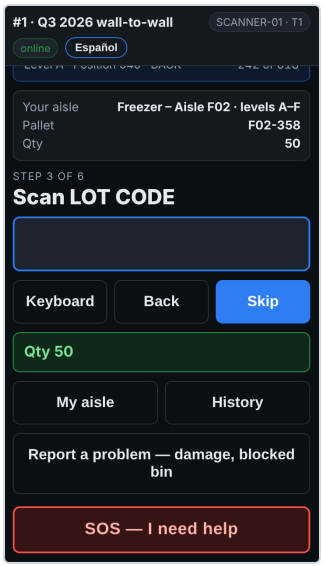
After sign-on — the aisle this team has been given, what is queued behind it, and any second counts waiting



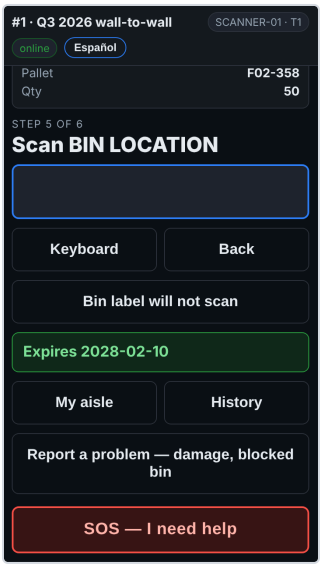
Question 1 — *Next bin* tells the team which bin comes next in the aisle, so nothing gets skipped



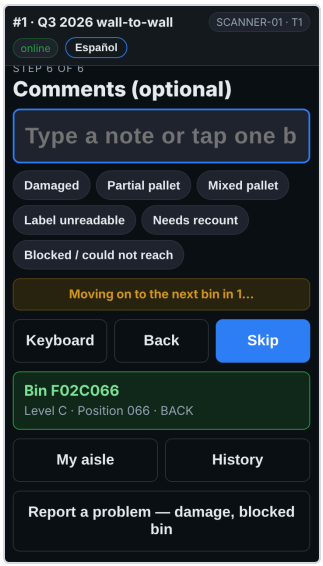
Question 2 — the pallet is recognized; the gun does not say what is on it, so the count is blind



Questions 3 and 4 — only on counts that ask for them. A lot that disagrees with the report is called out here, not a week later in a report



The last required question — the gun says where that bin is and which face it is on



Comments — tap a reason rather than type one. Left alone, the gun counts down and moves to the next bin by itself

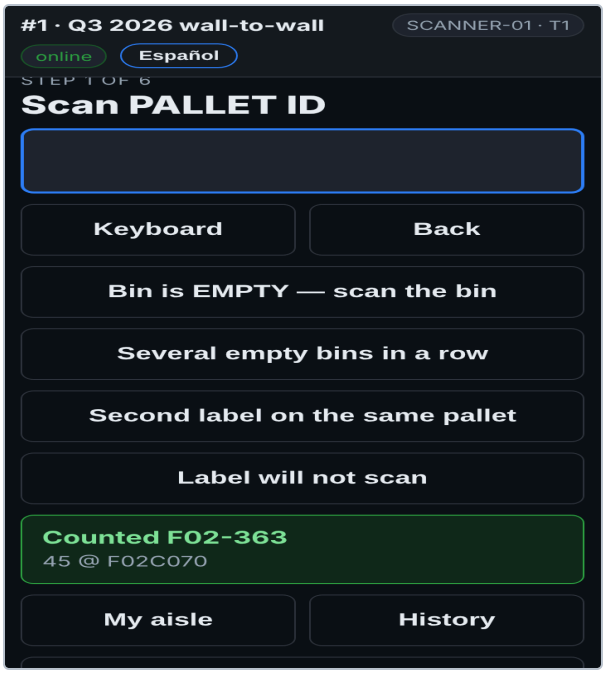
A bin with more than one pallet in it

Four pallet tags in one position is ordinary, and the guide stays put until they are all counted. After the first tag the banner turns amber and reads **Still in this bin** with a tally — *2 of 4 tags* — taken from the inventory report, so the team knows what is left to find before they walk on.

- Scan the next tag in the same bin. The tally goes up; when the report's pallets are all accounted for, the banner turns back to **Next bin** on its own.

- If the report is wrong and there is nothing more there, tap **Nothing more here →**. The bin is closed and the guide moves on.
- A bin the report lists nothing for stays open after a count, for the same reason — there may be a second pallet in it that the ERP has never heard of. Tap **Nothing more here →** when it is clear.
- **Bin is EMPTY** closes a bin outright. When the gun asks for the **bin first**, the bin is already scanned by then: the gun asks *Mark F01A004 EMPTY?* — **Yes** records it, **No** goes back — and never asks for the same bin twice. When the pallet comes first, it is **Bin is EMPTY — scan the bin**: one scan of the rack label.
- **This pallet has another label** — on the quantity screen, right after the pallet scan. A pallet wearing two tags: tap it, scan the other tag, and carry on with the quantity. The pallet is counted once; the second tag is saved beside it with no quantity, so neither label comes up as uncounted. (**Second label on the same pallet**, on the pallet screen, does the same after the line is saved.)
- **Report a problem** — damage (a pallet, the racking, the product), a bin **blocked** by a trailer or a forklift to come back to, or something else. Pick what, then why, add a word if it helps. Then **scan to confirm** which one it is: the **bin** for racking or a blocked bay, the pallet **or** its bin for a damaged, leaning or tipped pallet (its own label may be out of reach), the pallet for a problem with the pallet's label; *Something else* needs no scan. A pallet scanned where a bin is wanted (or the other way round) is refused with a word on what to scan; **Label will not scan — type it** takes it by hand and the fix list says *typed*. It goes on the office's **Fix list** against what was scanned, with no bar and no chime — it is not an emergency, that is what **SOS** is for. With no signal it is kept and sent later.
- **Several empty bins in a row** — for a stretch of an aisle with nothing in it. Scan the **first** empty bin and the **last**; the gun lists every bin between them in walking order, already ticked. Untick any that are not empty, then **Mark N bins EMPTY**. Each is recorded as its own empty bin, exactly as if they had been done one at a time; bins already counted are shown and left alone.

The tally counts what the whole team has done, not just this scanner, so two handhelds working the same aisle never double-walk a bay or leave one half counted.



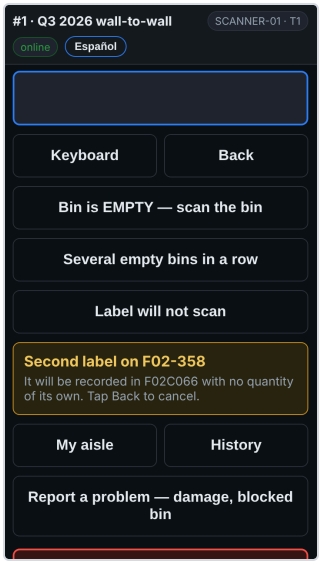
One tag into a position that holds three: the banner turns amber and stays on the bin until the other two are counted, or somebody says there is nothing more there.

A pallet with two labels on it

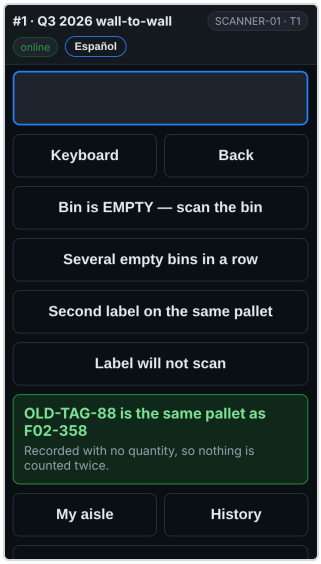
A re-tagged pallet that kept its old label, or one built from two, has two barcodes and is one pallet. Counting both as pallets doubles the stock; ignoring one leaves a live label for the next team to find.

1. Count the pallet once, as normal.
2. Tap **Second label on the same pallet**.
3. Scan the other label.

The tag is recorded against the pallet it is stuck to, with no quantity of its own. It shows on the pallet report as **SECOND LABEL**, the pallet it belongs to says *also tagged*, nobody is sent back for it, and it never reaches the ERP file as an adjustment. If the second tag is not on the inventory report, the gun offers the same answer in the "Pallet not on the list" screen.



The gun asks for the other label and names the pallet it will belong to



Recorded as the same pallet, with no quantity — so the stock is not counted twice

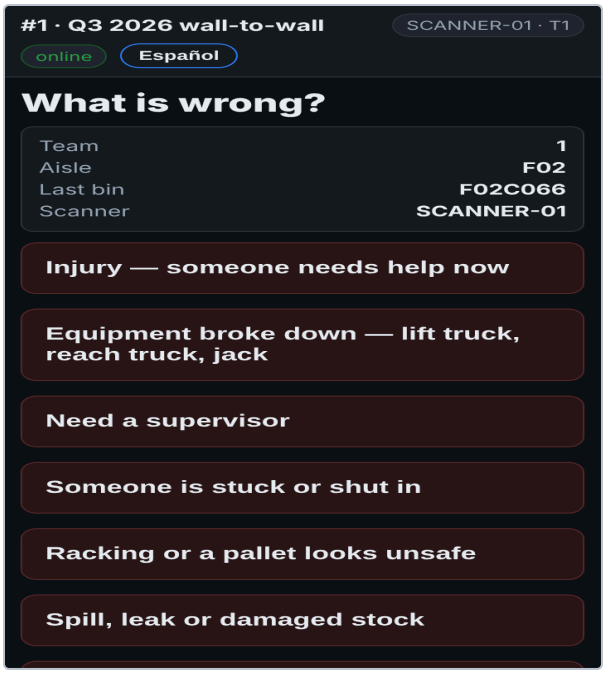
The SOS button

At the bottom of the counting screen, in red: **SOS — I need help**. A counter in the middle of a freezer aisle cannot radio the office through ear defenders, and walking out to find somebody is five minutes.

1. Tap **SOS**.
2. Pick what is wrong from the list — your site's own list, set in Settings.
3. Add a note if it helps, then it sends.

It carries the team, the aisle and the last bin counted, so nobody has to explain where they are. Within seconds it is on the dashboard, and in your Teams channel if one is set up. The bar at the top of the gun says **SOS sent** while it is waiting, and changes to **Dana has seen your SOS** the moment a supervisor picks it up — which is the thing the person who pressed it is waiting to know.

> **With no signal it says so.** *"No signal — this has NOT been sent."* It keeps trying and sends the moment there is a signal, but a counter is told the truth rather than left believing help is coming. If it cannot wait, walk to where there is signal or go and find somebody.



The SOS list — the site's own words, in buttons big enough for a gloved hand in a hurry.

The green "Update ready" bar

A new version of the app is on the server and this scanner has not got it yet. Nothing is wrong and nothing needs doing: it loads itself the moment the pallet in hand is finished and everything counted has reached the server. Tap the bar if you would rather take it now.

When a label will not scan

Freezer labels come off, ice over, get clipped by a forklift. The pallet is still there and still has to be counted, so the gun has a way out that keeps the count going and tells a supervisor where to send somebody with a label printer.

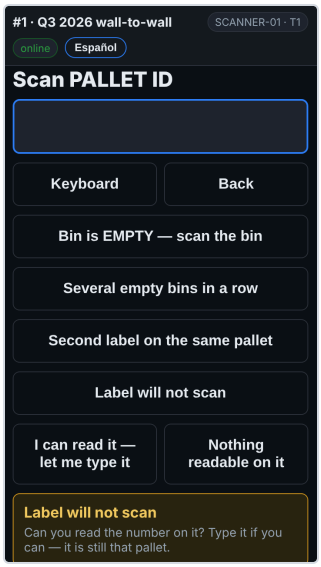
The same button is on both steps that read a label — **PALLET ID** and **BIN LOCATION** — and asks the same question: can you read it or not?

On the PALLET ID step — tap **Label will not scan**:

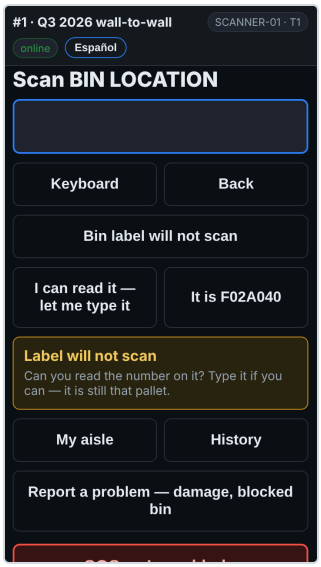
CHOICE	WHAT HAPPENS
I can read it — let me type it	The keyboard comes up, the counter types the number off the label, and counting carries on as normal. The line is flagged as a barcode to replace
Nothing readable on it	The pallet is counted anyway, under a name made from its bin — `NO-LABEL-F01A001-1`. Give the quantity and scan the bin as usual

On the BIN LOCATION step — tap **Bin label will not scan**:

CHOICE	WHAT HAPPENS
I can read it — let me type it	The keyboard comes up and the counter types the bin code. It is checked against the bin list, the team's aisle and its levels exactly like a scan
It is F01A005	The app names the bin it believes the counter is standing at — the one the guide is on, the one they are already counting out of, or the one the report puts this pallet in — and one tap takes it. Checked the same way
No readable bin label	Only when the app has nothing to offer. The line is recorded against the aisle rather than being lost, and flagged as an unknown bin for a supervisor to sort out



On the pallet step: can you read it, or not?



On the bin step the app offers the bin it believes you are at, so nobody types a code off a rack leg in a freezer

Either way the pallet still gets counted — walking away from it is the one thing that would put the count out.

The line is marked either way, and the bin turns up on **Dashboard → Reports → Labels to replace** with what was counted there. A pallet with nothing readable on it counts as a pallet not on the report, so it also shows on the pallet report as `NOT IN MASTER` — which is exactly what it is until somebody puts a label on it.

> **Print the rack labels first.** A pallet's bad label costs one counter one minute. A bin's costs every counter who walks up to that bay for the rest of the count — and the put-away driver afterwards. The relabel list marks them **RACK** and puts them at the top.

A message from the office

A supervisor can put a line on this team's scanners from the dashboard — *"come to the dock when you finish this aisle"*, *"leave F12, the forklift is in it"*. It arrives within about twenty seconds, buzzes,

and sits on the screen until somebody taps **Got it**. An urgent one is red. Tapping Got it tells the dashboard who read it.



A message waiting on the counting screen. Nothing else is blocked — the scan box is still live underneath it.

The other buttons

BUTTON	WHEN TO USE IT
Keyboard	Brings the on-screen keyboard up to type an entry by hand — a number pad on the quantity step. It never appears on its own: the scanner is the keyboard, and a keypad covers half the screen. Tap it again to put it away
Back	Wrong entry — steps back one question
Skip	Lot, expiry or comments the pallet does not have
Bin is EMPTY	The bin is genuinely empty. If the bin is already scanned (bin first), answer **Yes** to *Mark it EMPTY?*; otherwise scan the bin. It is recorded as counted and empty
My aisle	Back to the assignment screen
History	The last 50 lines from this scanner. **Void** removes a wrong line from the totals
Aisle complete — next aisle	Only when the aisle is finished. It frees the racking block and releases the team's next aisle

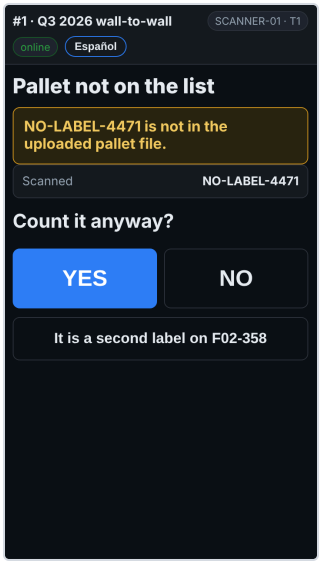


*History: the last 50 lines this scanner counted. **Void** takes a wrong one out of the totals.*

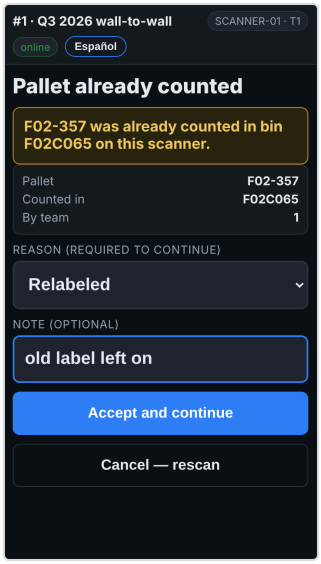
When the gun stops and asks

A pallet that is not on the list is one question: **Count it anyway? YES** counts it — flagged, and it shows up for a supervisor under **Adjustments** as a pallet to add (a positive adjustment). **NO** if it was simply the wrong barcode.

An unknown bin, a pallet already counted, a bin outside the team's aisle or level — the gun asks why before it will take the line. Pick a reason from the list (or **Other**), add a note if it helps, and press **Accept and continue**. The line is saved and flagged for a supervisor. **Cancel — rescan** if it was simply the wrong barcode.



A pallet that is not on the inventory report: **YES** to count it, **NO** to rescan



Everything else asks why. The reasons are the ones set in Settings — your site's words. **Other** is always offered

Not in Location (the pallets the system has lost track of)

Not in Location in the sidebar is the site's list of pallets that are not where the ERP says — upload it (Pallet, Item, Description, Qty, Lot, **Last known location**, Note) or type one in. Every scanner then watches for them: the moment any gun scans one, during any count or move, it is marked **found** with the bin, the team and the time, and the counter is told on the gun (*"F03-118 was on the Not in Location list — found!"*). **Found in…** marks one by hand; **Close** writes one off with the outcome. The list is the site's, not a count's: a pallet stays on it across counts until it turns up or is closed. It downloads as a CSV.

The find desk is at the bottom of the page: the next missing pallet, what it is and where it was last seen, ‹ › to step through, and the pallet system framed under it — type the bin and press **Found**, or **Close** it. The guns have the same desk: **Not in Location** at sign-on with a clock-in number alone, **pick the aisle** the pallets were last seen in (pallets with no known location are their own group), and the gun shows the pallet, *last seen in* its bin, what it is, and the pallet system under it. **Found — next** ticks one off and tells the office — badge, scanner, time; › leaves one; **Aisles** goes back to the list. Nothing is scanned in the app; the pallet system under the strip is where the pallet is booked.

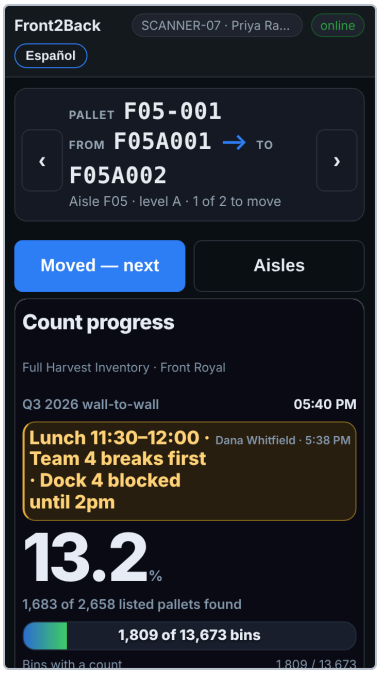
Moving pallets back (a job of its own)

Front bins → Pallets to move back builds the list — every pallet the report puts in a front bin with an empty bin behind it — or takes one you upload (Pallet, From bin, To bin). The guns then offer **Move pallets** at sign-on while any are waiting:

1. Pick **Front2Back** and scan **your clock-in number** — it is one person with a gun, so there is no team and no count to choose — then **Sign on**.
2. **Pick the aisle** to work; each shows how many pallets are waiting in it.

3. The gun opens the **move desk**, the same strip as the office's: the **pallet**, the bin it is in (**from**) and the bin it goes to (**to**), ‹ › to step through the aisle, and underneath, the **pallet system's own screen** (the address the office set under *Settings → Integrations → Pallet system*; the gun shows the screen, never the address). There is nothing to scan in the app: the move is booked, and scanned, in that screen. (Only when the office had to set the system to open *in its own window* does the gun show a button instead, **Open the pallet system for this pallet**; sign in there once, book the move, and come back with the recent-apps key.)
4. Move the pallet, book it in the system, tap **Moved — next**. The next one comes up; when the aisle is finished the gun says so and goes back to the aisle list (**Aisles** goes back any time). A pallet that cannot be moved is left with ›; the office skips it with the reason from *Front bins → Pallets to move back*.

Moves ticked off with no signal queue on the gun like count lines, so a dead spot loses nothing.



Front2Back on the gun: the office's strip — pallet, from → to — and the pallet system framed underneath.

The move desk. *Front bins → Move desk* is for the person doing the moves in the pallet system rather than on a gun: the next pallet, the bin it is in and the bin it goes to sit at the top, with ‹ › to step through the list, and the pallet system's own screen opens underneath, framed at the size of a Zebra's screen — the same slab as the Testing Suite's gun — so it shows what the crew see on a handheld; **Full width** opens it out across the page when a system's own screens need the room, and the choice is kept on that computer (the address is set once under *Settings → Integrations → Pallet system*). Make the move there, press **Mark moved**, and the next one comes up; **Skip** leaves one for a look, with the reason.

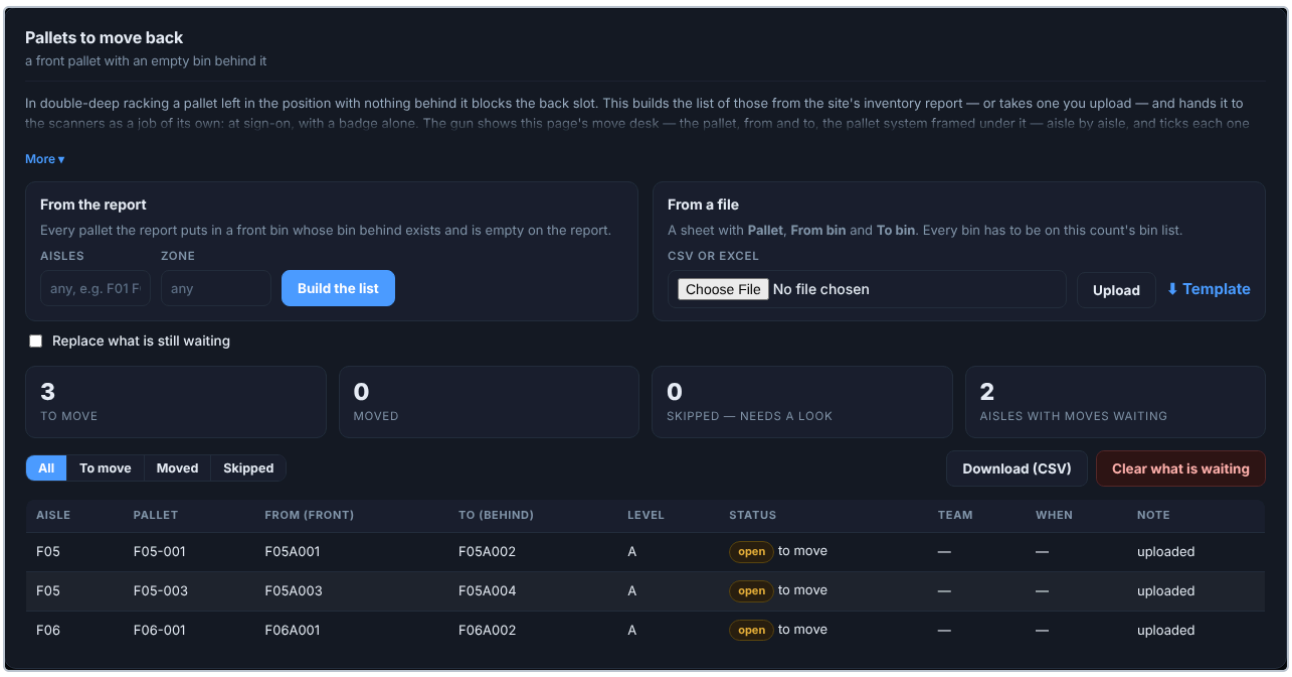
When the sign-in does not stick in the frame. A page shown inside another site's frame is "third-party" to the browser, and a system whose sign-in sets its cookies without `SameSite=None` —

classic **ASP.NET** sites (`.aspx` pages), most in-house systems — never stays signed in there: the form submits, the browser drops the cookie, the login page comes back. Many such sites also refuse to be framed at all (a blank box). This is the system's doing, not the app's. **Check the address** under *Settings → Integrations → Pallet system* reads the system's own headers and says which of the two it is; then **How it opens** has the answer:

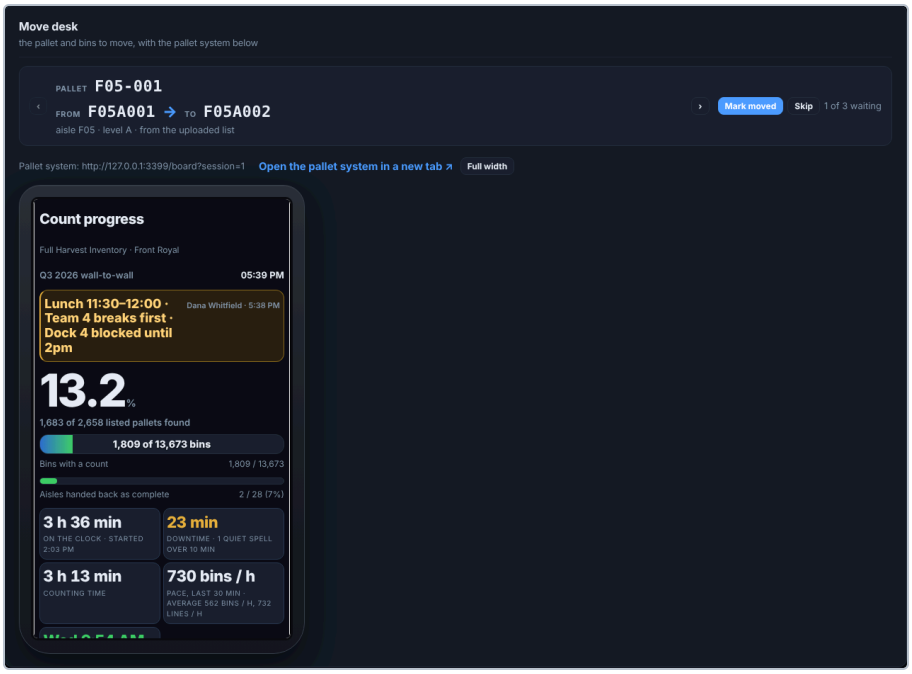
- **Through this app** keeps the frame under the strip and makes it work: this server fetches the system's pages and shows them as part of this site, so to the browser the frame is this site, the cookies are first-party and stick, and the "do not frame me" header never arrives. The move desk, the find desk and the guns all show it that way. Two conditions: the server has to be able to **reach** the system (a cloud system, or this app running on a PC in the building rather than on Railway — *Check the address* says whether it can), and the system's pages then run inside this site, so only a system you trust belongs here.
- **In its own window**, for a system the server cannot reach: one window beside the office desk that moves to each pallet; on a gun, a button that opens it on its own screen.

Put **{pallet}**, **{from}** and **{to}** (or **{bin}** and **{last}** on the find desk) into the address either way and the system opens straight on that record — `https://wms.example.com/Move.aspx?pallet={pallet}&to={to}`. The other way round, the system's owner can make a plain frame work by setting its cookies to `SameSite=None; Secure` in `web.config` (`<httpCookies sameSite="None" requireSSL="true"/>`, and `cookieSameSite="None"` on `<forms>` and `<sessionState>`) and letting this site frame it.

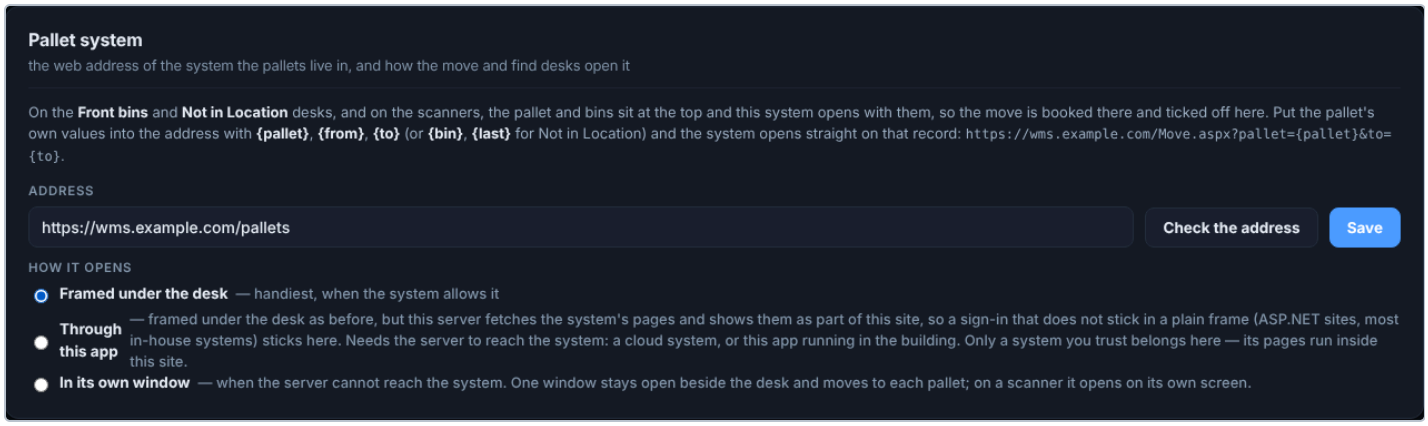
1. Done moves update the report, so the count that follows expects the pallet where it now is. With no signal they queue and send later, like count lines.



Front bins → Pallets to move back: the list built from the report, and how each one went.



The move desk: the next pallet and its two bins at the top, the pallet system underneath.



Settings → Integrations → Pallet system: the address the desk opens, set once.

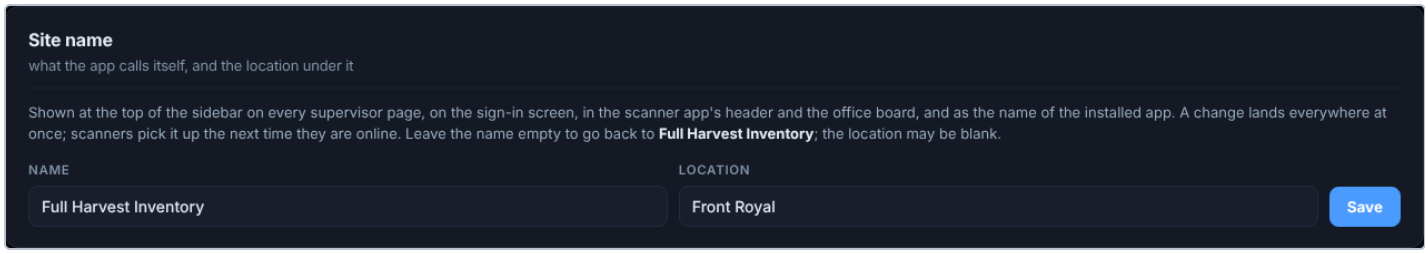
Step 16 — The supervisor's procedure (on the dashboard)

Dashboard, with the count picked in the header. It refreshes itself every 30 seconds, fetching only the tab on screen (plus the progress figures and the alerts, which are light), so three dashboards left open on a big count do not keep rebuilding the pallet report.

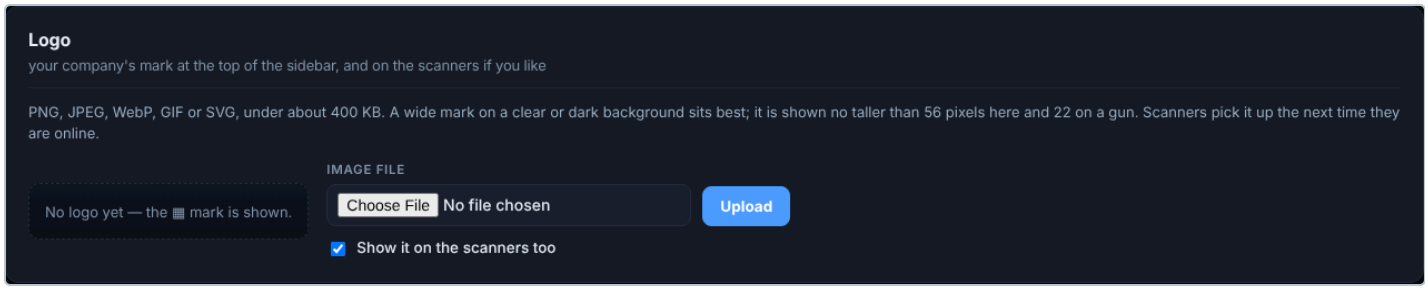
> **How it looks** is up to you: **Settings → Site & systems → Appearance** offers four themes — *Midnight* (dark blue), *Graphite* (dark gray), *Daylight* and *Frost* (light) — nine accent colors, the text size, how tightly the tables sit, the corners, the sidebar's width and the strength of the lines. It is kept per computer, so the office TV can be dark and a desk can be light. The same tab holds the **site name**: what the app calls itself (*Full Harvest Inventory* out of the box) and the location under it (*Front Royal*). Change either and it lands at once on every supervisor page's sidebar, the sign-in screen, the window titles, the scanner app's header, the office board and the name of the installed app; a blank name goes back to the default. The tab also takes a **logo**: your company's mark at the top of the sidebar and, if you tick it, in the header of every scanner.



Settings → Site & systems → Appearance: theme, accent, text size, density, corners, sidebar and contrast — per computer.

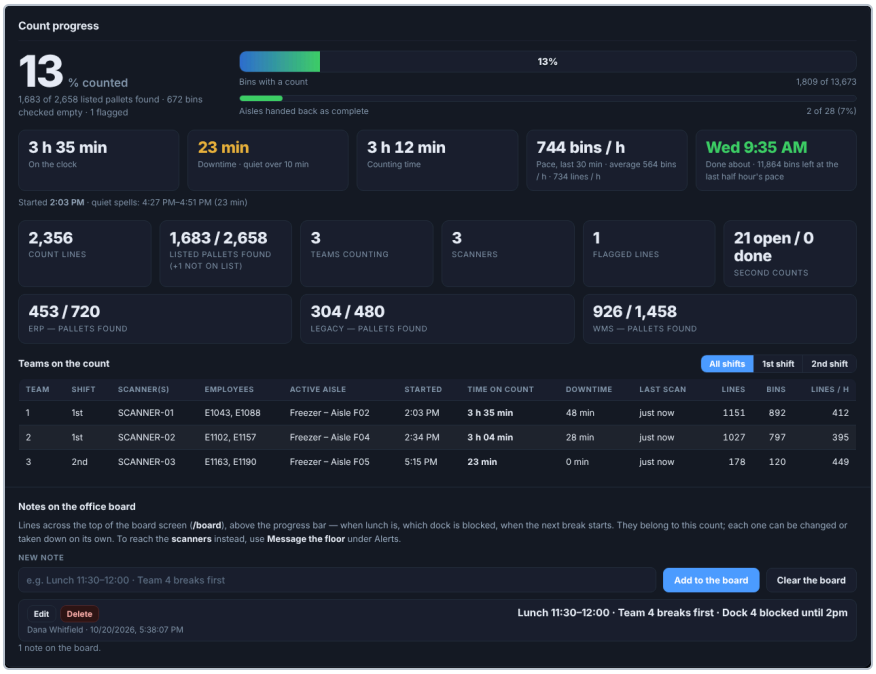


The site name card: the name and the location, changed here and shown everywhere.



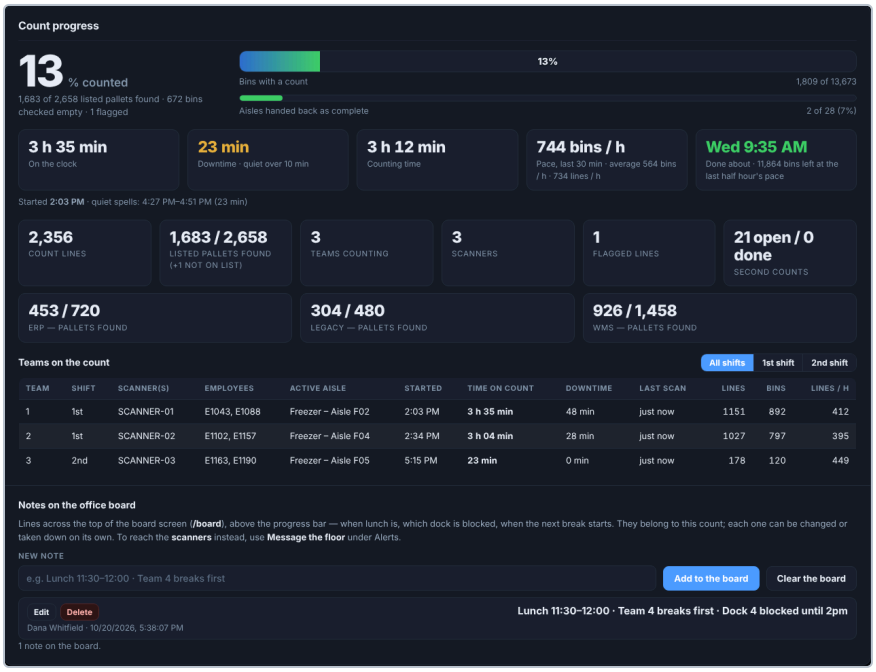
The logo card: one upload, shown on every supervisor page and, if ticked, on the scanners.

- **Progress** — lines, bins counted, pallets, exceptions; and a row per team with its **shift**, what it is counting right now, **when it started**, a **clock** of its time on the count, its **downtime**, when it last scanned and its **lines an hour**. *All shifts / 1st shift / 2nd shift* above the table narrows it. A team's clock stops when it signs off on the gun or finishes its last aisle. When the report carries a **System** column — three systems' stock in one warehouse — a tile per system shows how many of its pallets have been found, and the pallet report, adjustments and ERP files can each be narrowed to one system. The scanners never see the difference.



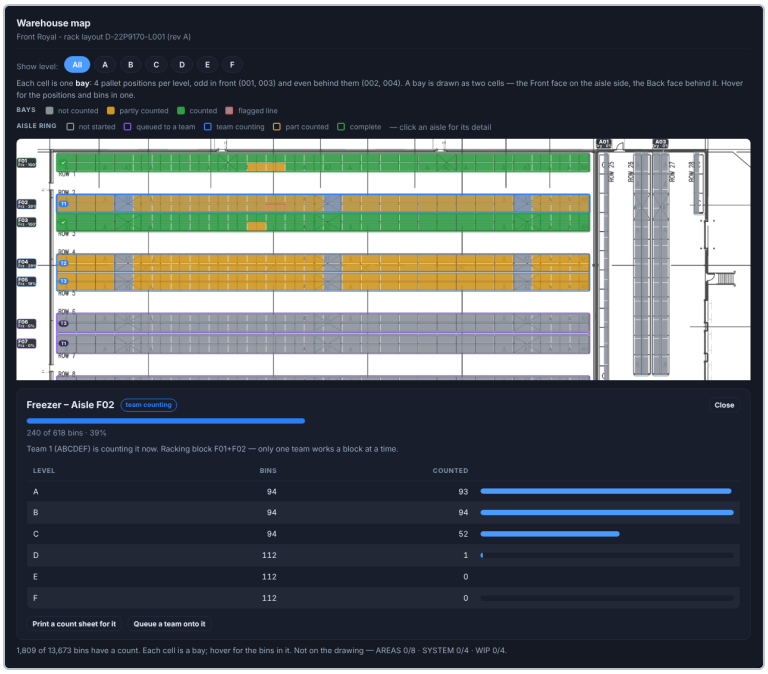
Progress: the count in numbers, then a row per team with its shift, aisle, clock and last scan.

- **The clock on the count** — the five figures under the progress bar, ticking. **On the clock** runs from the first sign-on (or the first line, whichever came first), not from when the count was created. **Downtime** is every stretch in which nobody on the count scanned, signed on or signed off for longer than the stopped-scanning limit (Settings → Scanners, 10 minutes to start) — lunch, a fire drill, the night between two days of counting; the line under the figures lists each spell, and says *quiet now for …* while one is running. **Counting time** is the rest. **Pace** is bins an hour over the last half hour, with the average over the counting time beside it — so a long lunch does not make the crews look slow. **Done about** is when the bins still to count will be done at that pace; it reads from the last half hour when there is one, else from the average. Once the count is closed the clock stops at the last thing anybody did on it, not at the click that closed it the next morning, and reads *Took*. All of it is in **Export everything**: the Summary sheet, a *Downtime* sheet of the spells, and the Teams sheet with each team's downtime, counting time and pace.



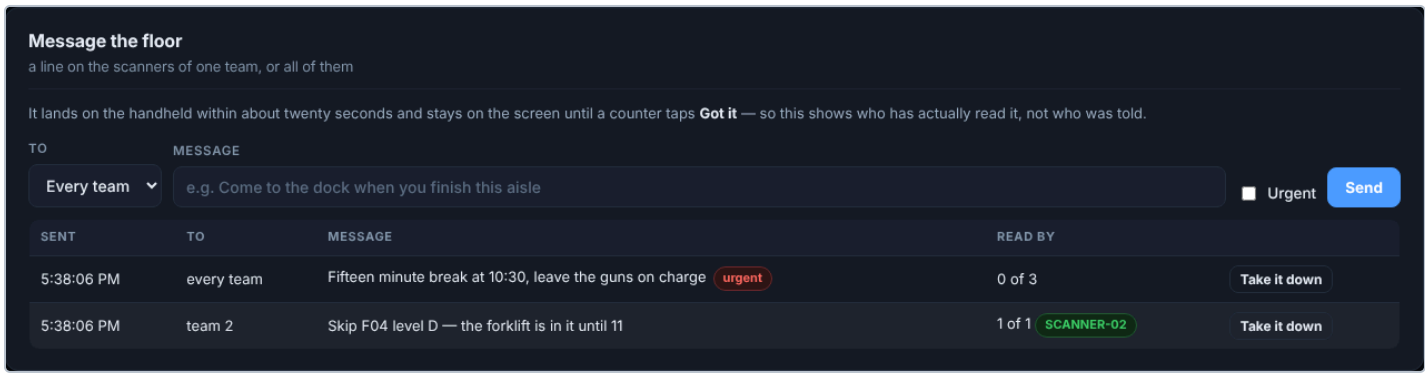
Pallets found, per source system.

- **Stopped scanning** — a team that is signed on and still has an aisle, but has not scanned for the site's limit (Settings → Scanners, 10 minutes to start), gets an **amber bar above every tab**, and its last scan turns amber in the table. **On break** (30 minutes) and **Lunch** (45) quiet it for that long; **Seen — I am on it** quiets it until the team scans again. If the site turned it on, the same alert goes to the Teams channel.
- **Map** — the warehouse from above. Aisles are outlined by state: not started, being counted now (with the team's number), done. Click an aisle for its bins, who counted it and what is flagged.



Dashboard → Map, with an aisle picked: 240 of 618 bins, level by level, and which team holds that racking block.

- **Team plan** — who is where, queue the next aisles, hand an aisle back. It also holds **Message the floor**: type a line, pick one team or all of them, tick **Urgent** if it cannot wait. The table underneath shows who has read each one — *2 of 3* means a scanner has not seen it yet, so do not assume. **Take it down** removes a message from the handhelds without losing what was said.



Message the floor, under Alerts. "1 of 1" with the scanner named is a message that landed; "0 of 3" is one nobody has looked at yet.

- **Second counts** — see step 18.
- **Adjustments** — **Positive and negative** at the top, on every count: stock to add (found more, or a pallet not on the report) and stock to take off (found less, or not found), with the units and the net. Click a side to list it, and download that list. Below it, on counts with approvals turned on, who signed for each one; see step 20.
- **Reports** — see step 19.

Watch for, during the count:

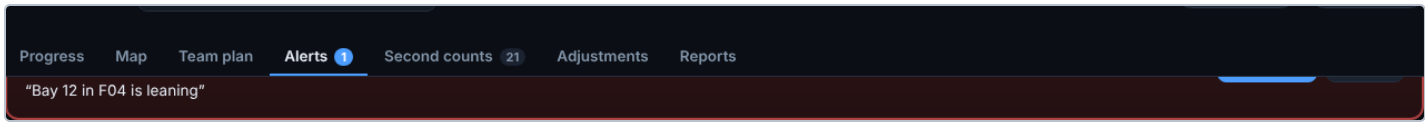
- Exceptions climbing on one team — usually a training problem, worth a radio call.
- A team stopped for a long time — dead battery, or stuck behind a trailer. The amber *stopped scanning* bar is there so you do not have to watch for it.
- An aisle nobody has started by mid-afternoon.

Step 16a — Answering an SOS

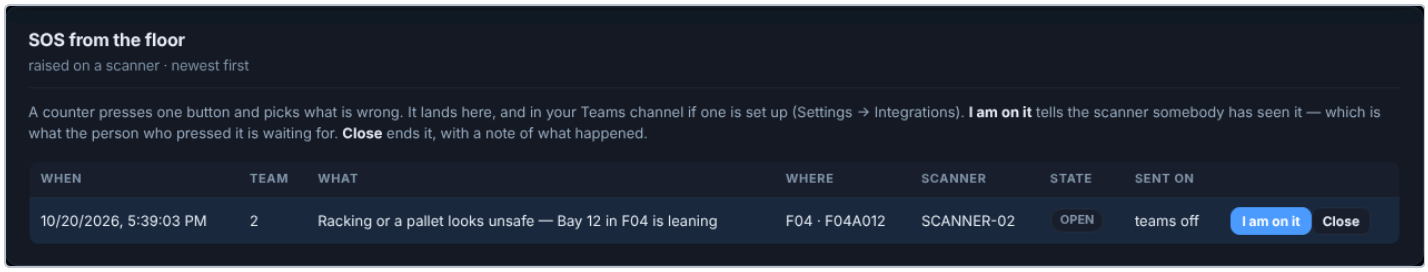
Anywhere on the dashboard. An SOS from a scanner appears as a red bar across the top of the page, whichever tab you are on, and makes a noise once. It names the team, what is wrong, the aisle and the last bin they counted, and whoever is on that gun.

- **I am on it** — tells the scanner somebody has seen it, by name. This is the one that matters: the counter is standing in an aisle waiting to know that anybody at all has picked it up.
- **Close** — ends it, and asks what happened. That goes in the log, and to Teams if the channel is set up, so nobody drives over for something sorted twenty minutes ago.

Every alert, open or closed, stays in **Alerts → SOS from the floor**, with who raised it, when, what happened and whether the Teams channel took it.



An SOS above the page: team, what is wrong, where they are, and the two things a supervisor can do about it.



Alerts → SOS from the floor: every call, open or closed, and what became of it. The tab carries a badge until the open ones are cleared.

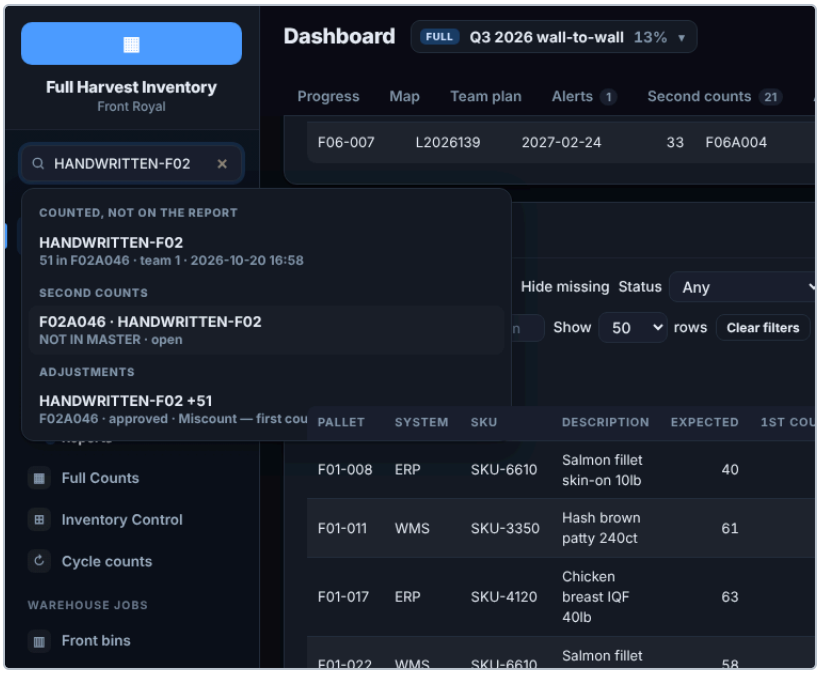
Step 16b — Finding anything: the search box

Every supervisor page, top of the sidebar. One box. Press **/** from anywhere on the page (or **Ctrl-K**) to jump into it.

It searches two things at once:

- **Your data** — a pallet ID, an item number, a word from a description, a bin code, a lot, an aisle, a name or clock-in number, a scanner, a count, a second count, an adjustment, something the office said to the floor, or a line in the log. Each hit carries the answer rather than just a link: a pallet shows what the report expected, what was counted, in which bin and by which team.
- **The app itself** — *"upload bin list"*, *"keyboard"*, *"approve"*, *"racking blocks"*, *"lunch note"*. These come back under **Go to** and take you to the card that does it, on whichever page it lives.

Choosing a hit takes you there: a tab on this page, or a page load that lands on the right tab. A lot code goes one better — it fills the lot box and runs it.



Typing a pallet ID: what the report expected and what was counted, the second count raised for it and the adjustment waiting to be signed — without opening any of those three cards.

Step 17 — The office board

Put `/board` on the office TV. It needs no sign-in and is read-only: the percentage counted, bins with a count, aisles handed back as complete, the **clock on the count** (time on the clock, downtime, counting time, pace and *done about* — the same five figures as the dashboard, big enough for the far wall), a row per team with what they are on, their lines an hour and their downtime, and every aisle as a tile colored by state. It deliberately shows no pallet IDs and no clock-in numbers.

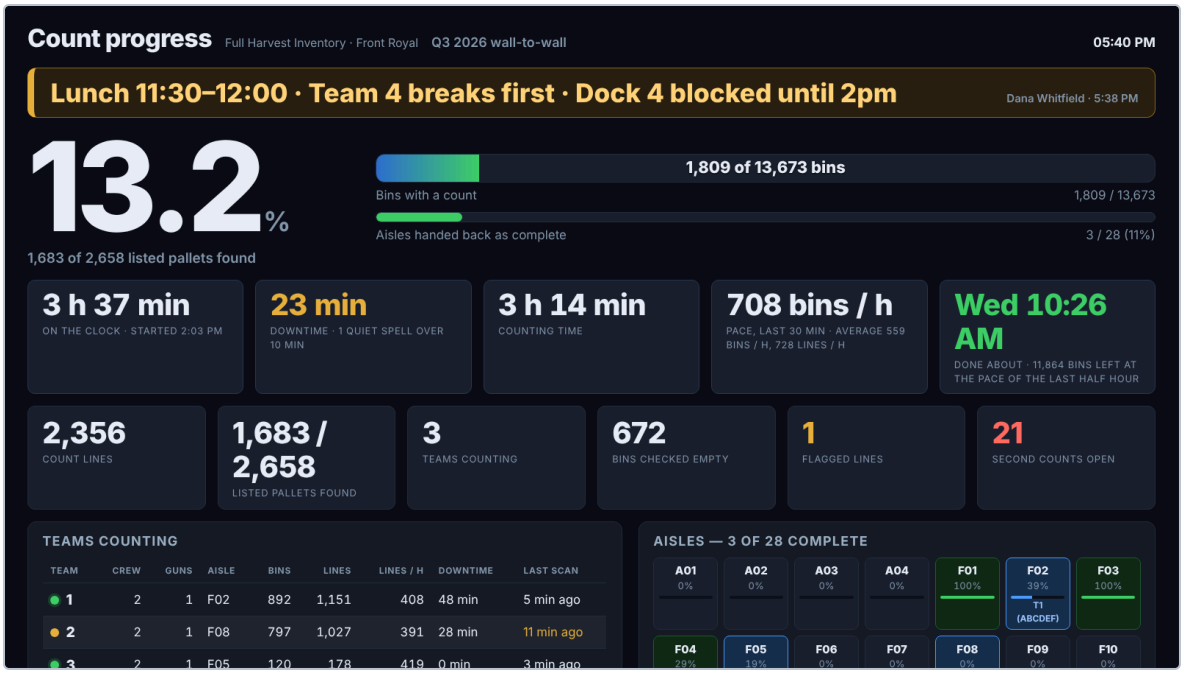
It follows the newest open count on its own. To pin it to a particular one, add the session to the address: `/board?session=12`.



The note across the top of the board, above everything else — because it is the one thing somebody walking past is looking for.

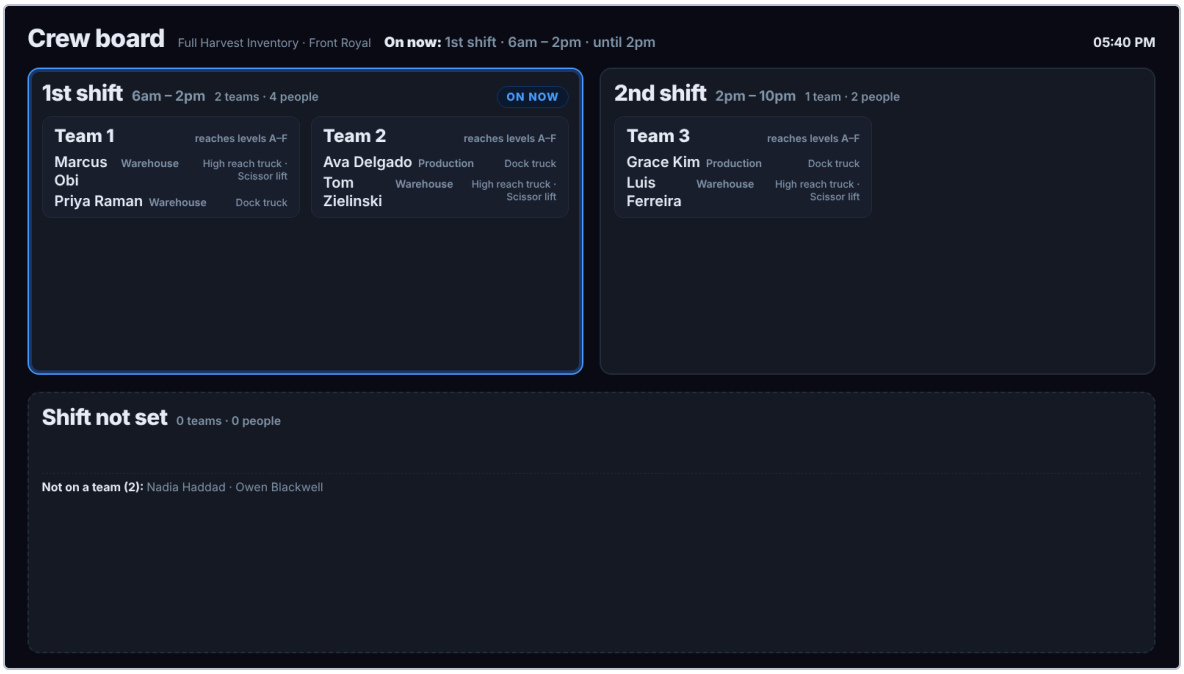
The notes across the top. *Dashboard → Progress → Notes on the office board* writes lines that appear above the progress bar on the board, big enough to read from across the room: *"Lunch 11:30–12:00 · Team 4 breaks first"*, *"Dock 4 blocked until 2pm"*. They belong to the count and say who wrote them and when; each has **Edit** and **Delete**, and **Clear the board** takes them all down.

Use the notes for anything the floor reads walking past. To reach the **scanners** instead — one team, or all of them, with a read receipt — use **Message the floor** under Alerts (step 16).



The office board, for the TV: percentage counted, aisles complete, every team, and every aisle colored by state.

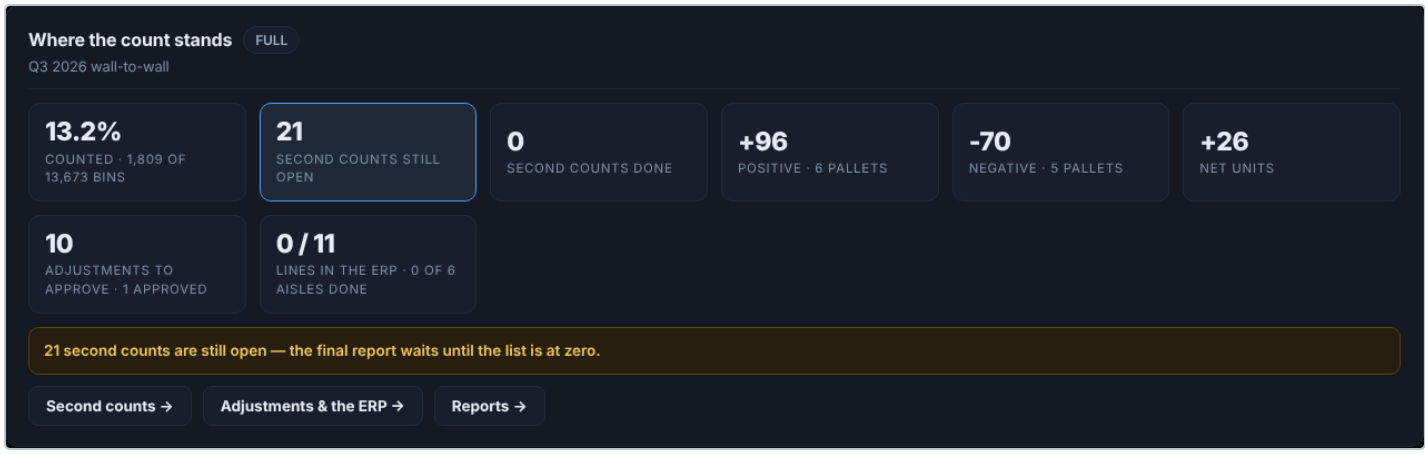
The crew board. `/crew` is the same idea for the time clock: every team and the people on it, by shift — **1st shift 6am–2pm** and **2nd shift 2pm–10pm** side by side, the one on now marked by the warehouse's clock (*On now · until 2pm*; between shifts it says when the next starts). Each team card has its people with their department and equipment, the levels the team reaches, its note, and any open seat from the team sheet; under each shift, who is on that shift but not on a team. A team's shift is the one set on Teams & crew; a team with none set takes its people's shift when they all share one (*shift from its people*). Teams with neither sit under *Shift not set*. It needs no sign-in, is read-only, refreshes every half minute, and shows no clock-in numbers. `/crew?shift=1` or `?shift=2` pins a screen to one shift.



The crew board: 1st and 2nd shift, each team's people, and the shift on now marked.

Part 5 — Finishing a count

Everything in this part is also under **Inventory Control** in the sidebar (`/inventory`): the dashboard's **Second counts**, **Adjustments** and **Reports** tabs on a page of their own, so the people turning the count into stock have nothing else in front of them. In front of the three sits an **Overview**: how much of the count is in, how many second (and third) counts are still open, the positive, negative and net units, how many lines and aisles are in the ERP, and whether the final report can be filed yet — each figure opens the tab it comes from. With cycle counts on, the open cycle counts are listed there too, with **Show it here** to bring one onto the same tabs; switched off for the site, they are gone from the page and its count picker. A login given the dashboard's Second counts, Adjustments or Reports has Inventory Control's too, and the other way round; *Inventory Control* is also a page of its own in a login's access list.



Inventory Control → Overview: where the count stands, in one read.

Step 18 — Work the second counts

Dashboard → Second counts.

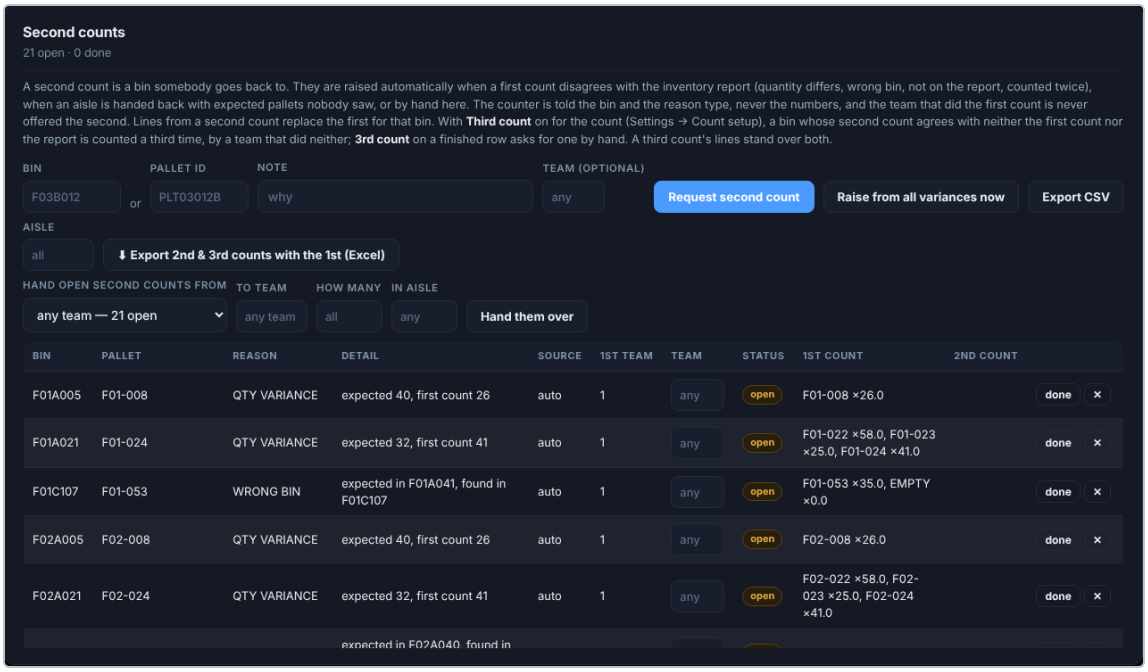
A second count is a bin somebody has to walk back to. They are raised automatically when a line disagrees with the inventory report (subject to your thresholds), or by a supervisor.

- **Raise from all variances now** goes through everything that currently disagrees and raises the tasks. It deliberately leaves alone pallets in aisles nobody has counted yet — mid-count those are not missing, just not reached — and tells you how many it skipped and why.
- The tasks appear on the handhelds: a team taps **Start second counts** on the assignment screen, counts the bin again, and presses **Bin done — nothing more here**. The list is **nearest first**: the aisle the team is in, then its racking block, then the aisles nearest by number, so a team finishing F11 is not sent to F01.

- **How a second count is judged.** The second count of a bin replaces its first count for every report — the bin's figures are the second count's from then on. What it means depends on what it agrees with: The line goes to Adjustments as counted;
  - **it agrees with the first count** — the difference is real: two teams found the same thing.
  - **it agrees with the report** — the first count was wrong, and the line now matches;
  - **it agrees with neither** — nobody knows which number is right. That is what a third count is for.
- **One team for the second counts.** If one team does them all — inventory control's, say — tick **Send them all to team** in the count's settings (*Settings → Count setup → Count session → Second counts*) and type its team number. Every new second count is given to that team, and the open ones nobody had yet go to it at once. It stays blind: a bin that team counted first is never its own second count — that one stays open for any other team — and a third count only goes to it if it did neither earlier count. The Second counts card says *new second counts go to team ….* **When it is swamped**, the **Hand open second counts from … to team …** bar on the card moves some or all of a team's open ones to another team (or back to *any team*): leave *How many* blank for all of them, or give a number; *In aisle* for one aisle. A bin a team has already started stays with it, and a bin the other team counted first stays where it is — the message says which. The **Team** box on each row still moves one at a time. Untick it, and new second counts are for any team again; the ones already handed out stay where they are.
- **Third count.** Tick **Third count when the second settles nothing** in the count's settings (*Settings → Count setup → Count session → Second counts*; off on a new count). From then on a bin whose second count agrees with neither its first count nor the report is sent back a **third time**, by a team that did **neither** count — on the gun it reads *THIRD COUNT · bin …* — and its count stands over both. Judged when the second count's lines reach the server, so a gun that finished the bin out of signal is judged when it sends. On the dashboard a third count is marked *3rd count*, with both earlier teams, and a **3rd count** column shows what it found. **3rd count** on any finished second count asks for one by hand, option or not. A supervisor who assigns a third count to a team by hand has made the call; one left open goes only to a third team.
- **Export 2nd & 3rd counts with the 1st (Excel)** — every bin counted again, one row a pallet: the report's bin and quantity, the **1st, 2nd and 3rd counts** with the team that did each, the **final count** and which count it is, the final against the report, and *which number stands* (*the second count agrees with the first*, *…with the report*, *the third count stands*, or *waiting for…*). A *Summary* sheet counts how each bin was settled. Type an **Aisle** first for one aisle alone. The same sheet is in **Export everything** as *Counts side by side*.
- **A pallet found in another bin than the report's** brings back the bin it was found in, as always — and also **the report's own bin**, once that bin has been counted (or its aisle handed back): reason *LABEL CHECK*, told on the gun as *Check for a pallet with a duplicate label*. It is counted blind by a team other than the one that counted it first; a second pallet there wearing the same label shows up as *Counted twice*. A bin already second-counted for another reason is not sent back a second time; the one count covers both.

Work these down before you close the count. A count closed with open second counts is a count with known-wrong numbers in it.

Practice it first. Testing Suite → What to practice → **Second counts** sets up another crew's aisle on your practice count, four bins that come back to the gun and one to raise by hand from the dashboard, with the tips following each step (Step 8c).



Dashboard → Second counts. The counter is told the bin and the reason type, never the numbers — and never the team that counted it first.

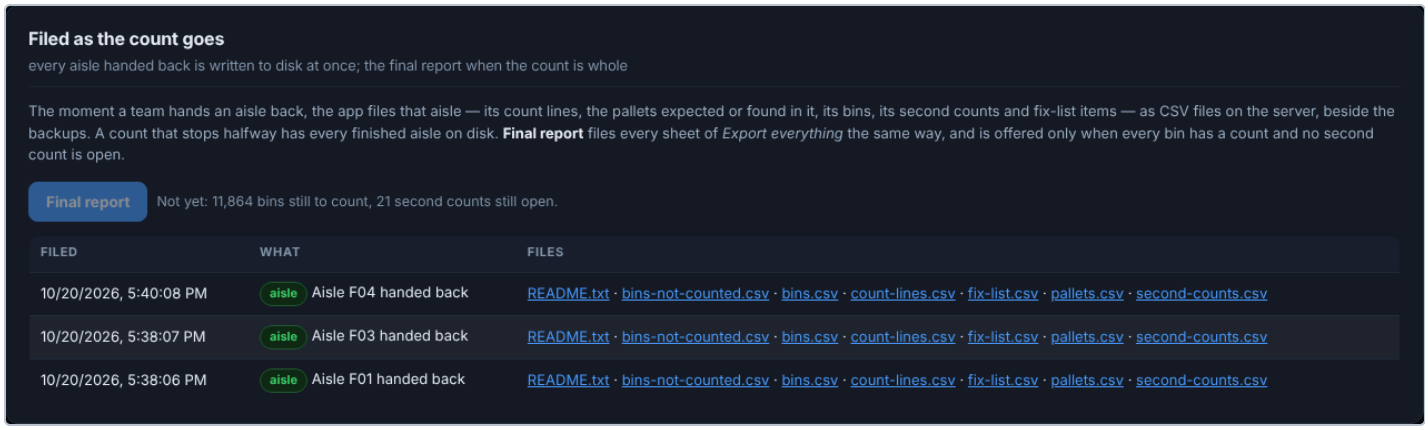
Step 19 — Read the reports

Dashboard → Reports.

Fix list — what the floor found that somebody has to put right, on one list: **labels that would not scan** (pallet or rack — rack first, every counter after walks up to that bin), **damage** reported from the gun's *Report a problem* button (a pallet, the racking, the product, each with a reason and a note), and **bins blocked** by a trailer or a forklift that a team said it would come back to. Each shows the bin, the pallet (marked ✓ **scanned**, or ✓ **typed on the gun**, when that is what the counter confirmed it with), who saw it and when. **Fixed** ticks it off with a word on what was done; *Still open* is the walk-round list. It downloads as a CSV and has its own sheet in the export.

Filed as the count goes. The moment a team hands an aisle back, the app writes that aisle to disk — its count lines, the pallets expected or found in it, its bins, its second counts and fix-list items — as CSV files in an `exports` folder beside the backups, one folder per hand-back. A count that stops halfway has every finished aisle on disk already. The card lists what has been filed, with a download link for each file. **Final report** files every sheet of *Export everything* the same way, and the button

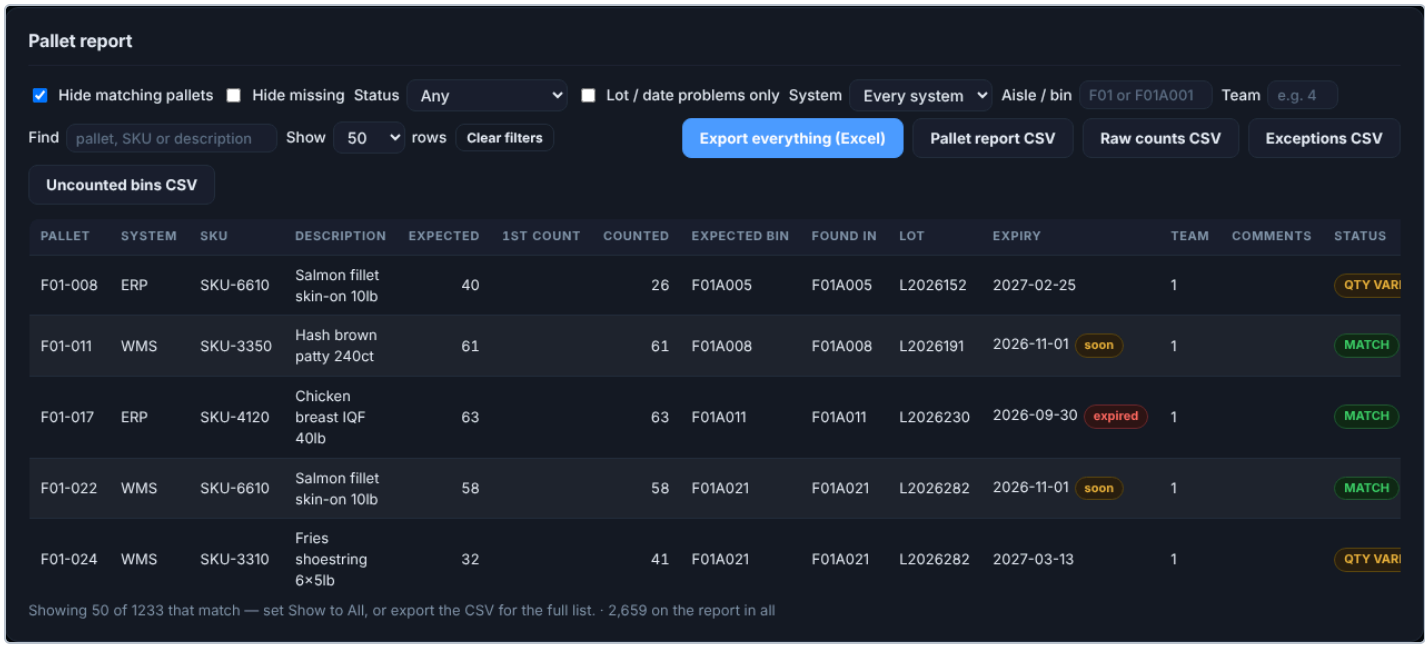
is on only when the count is whole: every bin has a count and no second count is open; until then the card says what is still missing. The log records every filing.



Filed as the count goes: one row per aisle handed back, the final report once the count is whole.

Export everything (Excel) — one workbook with every table the app keeps for this count, in a form a person can read: a **Summary**, every **count line**, the **pallet** report, the **adjustments** (positive and negative), the pallets **not on the report** (sold or shipped in the system, found on the floor — with where each was found, where its approval and ERP entry stand, and an *Investigation* column to fill in before it is adjusted back into stock), the **bins** and the ones **not counted**, the **teams** and their times, **sign-ons**, **SOS**, **stopped scanning**, **second counts** and the **log**. Headings are plain words, flags are *Yes* / *No*, and times are on the warehouse's clock. It is the file for an auditor, a manager or the shared drive. (The CSV buttons beside it are for machines — the ERP, a reconciliation sheet.)

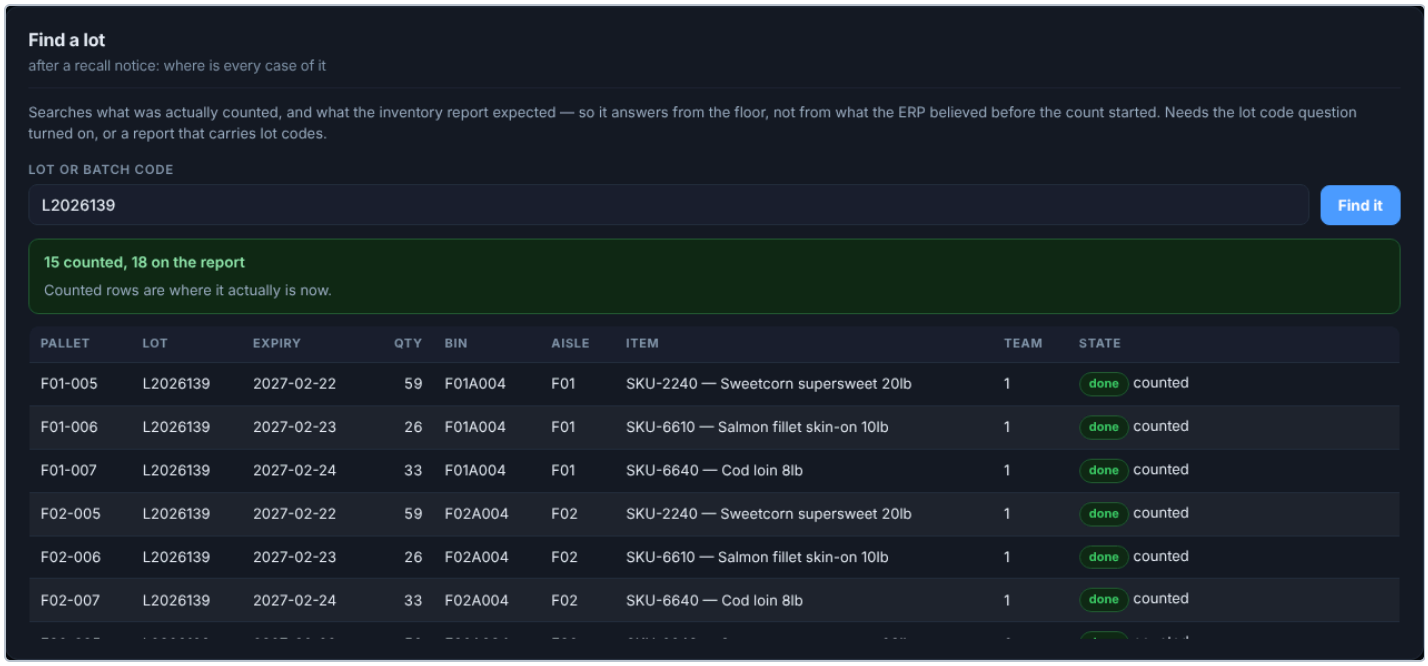
- **Pallet report** — one row per pallet: expected vs counted, which bin the ERP expected and where it was actually found, and a status you can act on: `MATCH`, `QTY VARIANCE`, `WRONG BIN`, `MISSING`, `NOT IN MASTER`, `COUNTED TWICE`. Lot and expiry get their own columns, because the right count of the wrong lot is still wrong. **Only exceptions** narrows it to what needs attention — including a wrong lot or a date about to run out, even when the quantity is right.



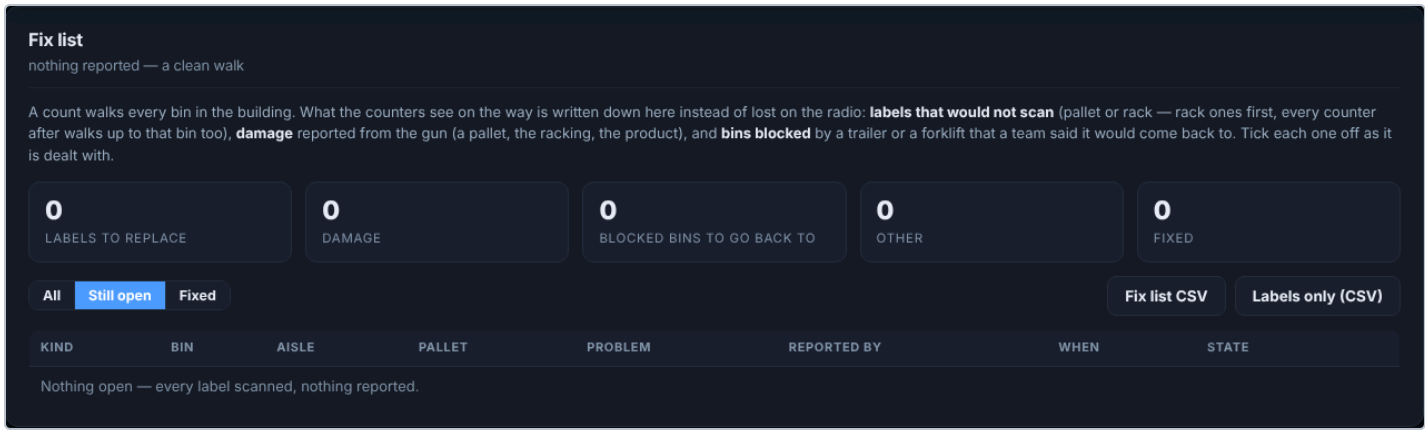
*The pallet report with **Hide matching pallets** ticked: a short pallet, an expired date, one expiring soon. This is the list to work down.*

The filter row reads the report without exporting it: **Hide missing** drops the pallets nobody has found or counted yet, **Status** picks one kind of difference, **Lot / date problems only** keeps the wrong-lot and out-of-date rows, **System** one of the site's systems, and **Aisle / bin**, **Team** and **Find** narrow by place, crew, pallet, SKU or description. The line under the table says how many rows match out of the whole report; **Clear filters** is back to the usual view.

- **Find a lot** — after a recall notice: type part of a lot code and get every case of it, both where it was actually counted and where the report expected it, so a pallet nobody found still shows up.



A recall: five cases of the lot found on the floor, and a sixth the report still expects in an aisle nobody has reached.



The fix list: labels that would not scan, damage and blocked bins from the gun, each with the bin, the pallet and who saw it. Fixed ticks it off.

Step 20 — Approve the adjustments

Dashboard → Adjustments. Only on counts with **Adjustments need approval** turned on (step 13). With it off this tab says so, and every difference goes into the ERP file as it always did.

A variance is not an adjustment until somebody owns it. This is where that happens.

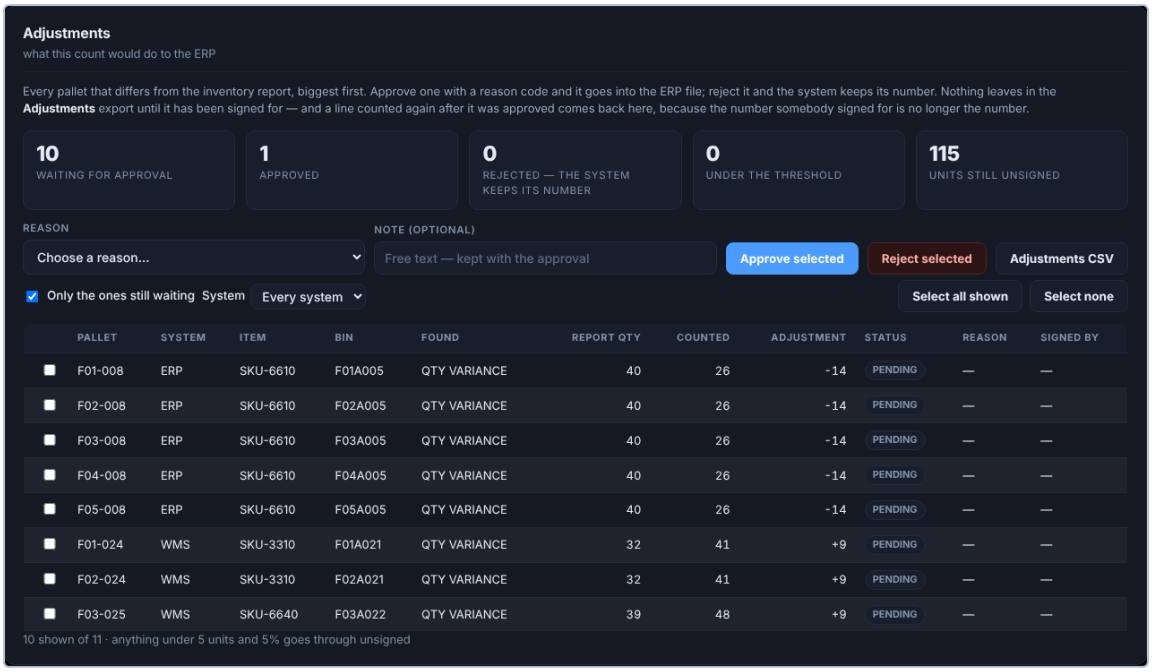
1. The list is every pallet that differs from the report, biggest first: what the system says, what was counted, and what the ERP would move.
2. Tick the ones you have satisfied yourself about. **Select all shown** takes the screenful.
3. Pick a **reason** — your site's list, set in Settings → Integrations → Adjustment reasons — and add a note if it helps.

4. **Approve selected**, or **Reject selected** if the count is wrong and the system should keep its number.

What the feature is actually for:

- **Nothing unsigned reaches the ERP.** The Adjustments export holds back anything still waiting, and the preview says how many it left out.
- **Anything under the threshold goes through unsigned.** A two-case difference on a pallet of 600 is noise; a count where every line needs a signature gets rubber-stamped, not read.
- **Counted again means signed again.** If a second count changes a pallet after it was approved, it comes back to this list with the approval cleared — the number somebody signed for is no longer the number.
- **A variance a second count clears disappears.** There is nothing left to adjust.
- **It is all in the log**: who approved what, with which reason, and when. That is the answer to "who authorized writing off 400 cases of chicken".

A pallet nobody has reached yet is not on this list while the count is running. Once the count is **closed**, a pallet the report expected and nobody found becomes an adjustment like any other.



The adjustments waiting for a signature, biggest first. AUTO is under the threshold and goes through without anybody being asked; APPROVED carries the reason and the name.

Step 20a — Mark what has been keyed into the ERP

Dashboard → Adjustments → Entered in the ERP. For a site that keys the adjustments into the ERP by hand, from the worksheet, over a day or two — and must not key a line twice.

1. **Download the ERP worksheet (CSV)**: every adjustment line — side, pallet, item, bin, why, report and counted quantities, the adjustment, its approval — with an **In ERP** column.
2. Work the lines into the ERP from the worksheet. As each goes in, fill its **In ERP** cell: a *yes*, an *x*, a date or your initials all count as entered; *no* takes a mark off; a blank leaves the line as it is.
3. **Upload the worksheet back.** The lines you filled are marked, with who and when, and the message says how many were marked, how many unmarked and how many left alone. A pallet that is not on this count is named and ignored.
4. Or mark a few at a time from the table: tick the lines, **Mark as entered**; **Take the mark off** undoes it. **Hide lines already in the ERP** leaves only what is still to key.

An aisle at a time: Dashboard → Adjustments → ERP upload, an aisle at a time. For putting a wall-to-wall into the ERP aisle by aisle, as the aisles are finished:

1. The table lists every aisle: its pallets, how many are counted, how many differ from the report, how many are **uploaded to the ERP** (*not yet*, *2 of 40*, *all in the ERP*), and what is **still being counted** there — open 2nd/3rd counts and bins nobody has counted. Wait for an aisle's second and third counts before you put it in.
2. **⬇ Excel** on an aisle downloads its sheet: **every pallet in the aisle**, matches included — bin, pallet, item, lot, the report's bin and quantity, the **counted quantity** (the latest count: a 2nd or 3rd count stands over the 1st), the **adjustment**, the result, which count it is — and an **Uploaded to ERP** column. A *How to use* sheet goes with it.
3. Put the lines into the ERP. As each goes in, write *yes* (or an *x*, a date or your initials) in its **Uploaded to ERP** cell; *no* takes a mark off; blank leaves it.
4. **⬆ Upload a filled-in sheet…** — the Excel file as it is (or a CSV). Those lines are marked as in the ERP, with who and when; the aisle's row moves on, and the next download of that aisle shows the marks. A workbook with several aisles' sheets in it is read sheet by sheet.
5. **Only aisles not fully uploaded** hides the aisles that are done.



Dashboard → Adjustments → ERP upload, an aisle at a time: each aisle's state, and its Excel sheet to fill in and bring back.

These are the same marks as the worksheet's *In ERP* above, so the two never disagree. Only the **Uploaded to ERP** column is read back — the quantities in the file are not: a count changes only by being counted. A line second- or third-counted after it was uploaded is flagged **changed since uploaded**, with the number the ERP was given. Put it right in the ERP, write **again** (or *updated*, *re-entered*) in its cell and upload the sheet: it is marked at today's numbers. A plain *yes* on a line already marked keeps its first mark, so an old sheet uploaded twice cannot hide a change. The full routine is in the ERP upload guide.

The **In ERP** column shows the mark with who and when; the chip in the heading counts *entered of total*. A line **counted again after it was entered** is flagged **changed**, with the number the ERP was given, because the ERP now holds the old figure. The mark also appears in the adjustments CSV and in the *Adjustments* sheet of **Export everything**. Marking is for logins that may approve adjustments — inventory control, admins, supervisors.

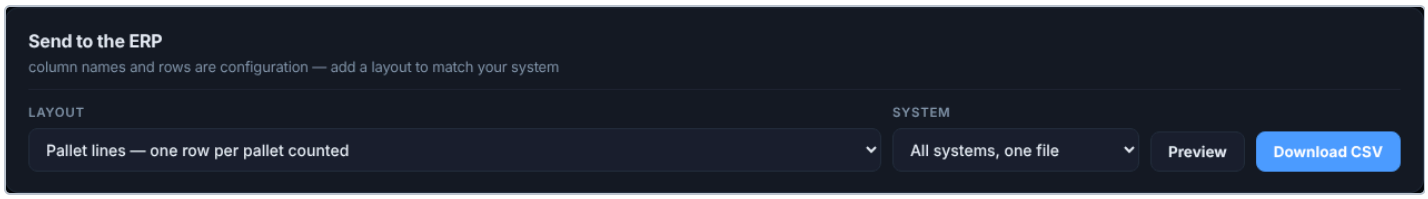
Step 21 — Send it to the ERP

Settings → Integrations → Send to the ERP.

1. Choose the **layout** that matches your ERP. Column names and which rows are included are configuration — add a layout rather than editing the file by hand afterwards. Where three systems share the warehouse, pick the **System** too: one file per system, named for it, with only that system's pallets in it. (Each system's report is named when it is uploaded — Settings → Lists & racking → Inventory report → **System** — or carries a *System* column; the pallet report, adjustments and Export everything show it, and the scanners never do.)
2. **Preview** and read the first rows. On a count with approvals on, the preview says how many adjustments it left out because nobody has signed for them yet.

3. **Download CSV** and import it into the ERP.

The **Adjustments** layout carries the reason code and the name of whoever approved it, so the file the ERP gets is the same story the log tells.



Settings → Integrations. Preview before you download; the layout decides the column names and which rows are included.

Step 22 — Close the count

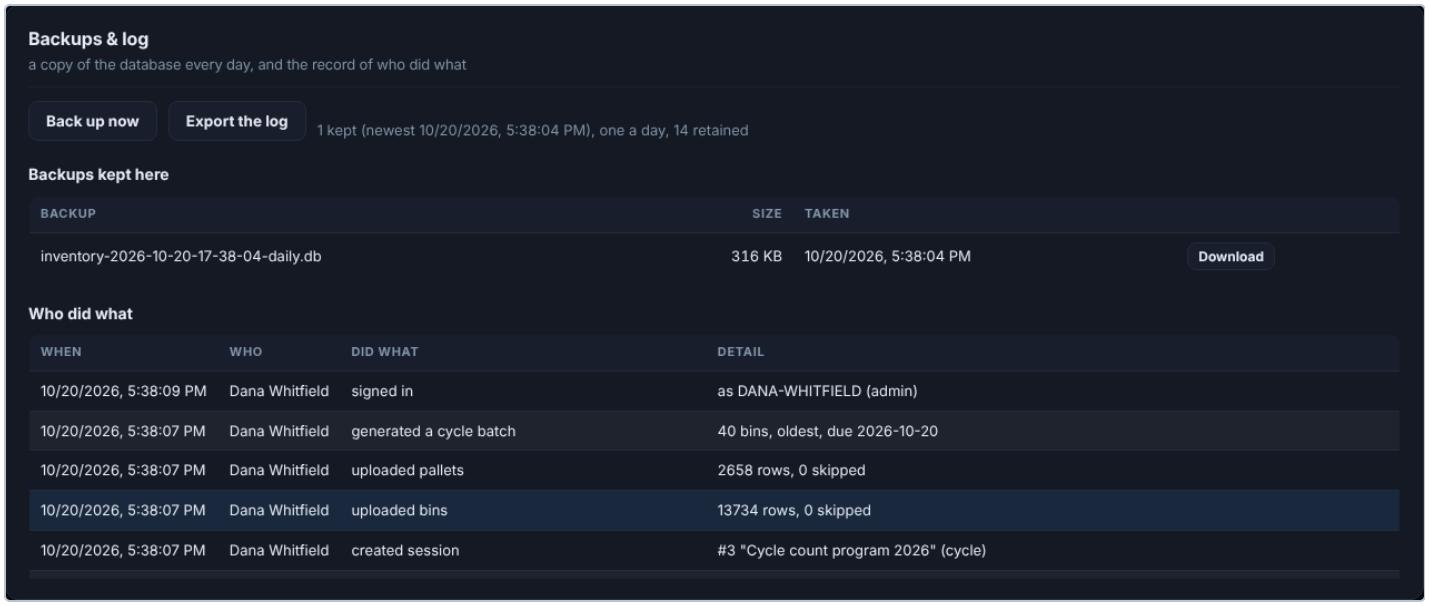
Settings → Count setup → Count session → Close session. A closed count is read-only: scanners can no longer post lines to it, and its reports stay available for ever.

Delete… is separate and deliberately harder: the count must be closed first, you have to type its name, and a backup is taken before anything is removed. Everything counted against it goes with it. Close counts; delete only the ones created by mistake.

Step 23 — Backups and the log

Settings → Backups & log.

- **Back up now** before and after anything large — an ERP import, a bulk re-upload, a deletion. The app also backs itself up daily and keeps the last 14.
- **The log** records every supervisor action: who signed in, who uploaded what, who overrode what, who raised second counts, who searched for a lot. **Export the log** for an auditor.



The backups kept on the server, and the log under them. Both export.

Off-site copy: Google Drive — nothing for IT to do. The same, to a folder in a Google Drive, through a small **Apps Script** the admin deploys in that Google account: no Google Cloud project, no app registration, no IT ticket. *Settings → Backups & log → Off-site copy: Google Drive.*

One-time set-up (about ten minutes):

1. Sign in to Google with the account whose Drive should hold the backups, and open **script.google.com → New project**.
2. On the card, press **Copy the script**. In the editor, select everything, paste over it, and **Save**. (The script carries a key only this server knows; a request without it is turned away.)
3. **Deploy → New deployment →** gear by *Select type* **→ Web app**. **Execute as: Me**, **Who has access: Anyone**. **Deploy**.
4. **Authorize access**, pick the account. Google warns *Google hasn't verified this app* — it is your own script: **Advanced → Go to Full Harvest backups (unsafe) → Allow**.
5. Copy the **Web app URL** (it ends in **/exec**) into the card and press **Save and check the connection**. The card names the account, links the folder, and the newest backup goes at once.

Each copy is the backup **gzipped** — a database shrinks five to ten times — named *inventory-… db.gz*; unzip it (7-Zip, or double-click on a Mac) to get the database back. **Send a copy** and **Keep the newest** work as for OneDrive. A failed copy shows on the card in Google's words and is tried again every half hour. **Disconnect** stops the copies; **Make a new key** also stops the old script (paste the new one and deploy again). Treat the web app address like a password. A Google Workspace account whose admin blocks *Anyone* web apps cannot do this — use a personal Google account, or OneDrive. One copy can be up to 30 MB squeezed, far more than a site's database; the card says so if it is ever passed.

Either off-site copy will do, and both can run at once.

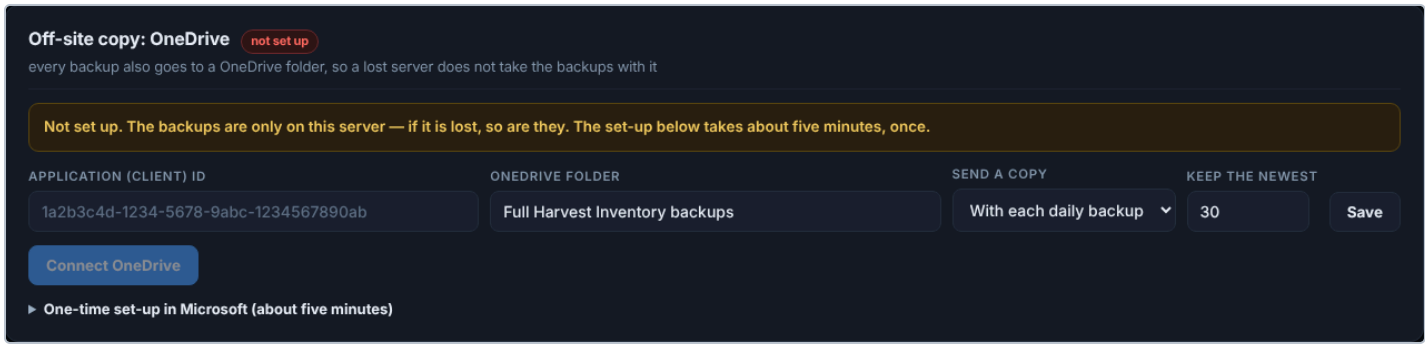
Off-site copy: OneDrive. The daily backups live on the same Railway volume as the database, so a lost volume takes them with it. Connected once, the app copies every backup to a folder in a OneDrive as well and keeps the newest there (30 unless you change it).

One-time set-up (about five minutes):

1. At **portal.azure.com**, sign in with any Microsoft account and open **App registrations → New registration**. Name it *Full Harvest backups*; supported account types: **Accounts in any organizational directory and personal Microsoft accounts**; no redirect URI. **Register.**
2. On the app's **Authentication** page turn on **Allow public client flows**, and save.
3. Copy the **Application (client) ID** from its **Overview** into the card, choose the folder, and **Save**. (It can instead be set as the `ONEDRIVE_CLIENT_ID` variable in Railway.)
4. Press **Connect OneDrive**. A code appears: open **microsoft.com/devicelogin** on any computer or phone, type it, and sign in with the account whose OneDrive should hold the backups. The card notices by itself and turns green, naming the account.

How often. It is a copy at a moment, not a live sync. **Send a copy** chooses when: *with each daily backup* (and any taken by hand), or **every 12, 6, 2 hours or every hour** — those extra copies are taken for OneDrive alone and not kept on the server, so the server's dailies are not crowded out. During a wall-to-wall, every hour or two is sensible.

Back up to OneDrive now takes a fresh backup and sends it while you watch. A failed copy shows on the card and is tried again every half hour; a sign-in Microsoft has withdrawn says to connect again. **Disconnect** stops the copies (those already in OneDrive stay). No password is kept — only a sign-in token that lets the app write to that account's files.

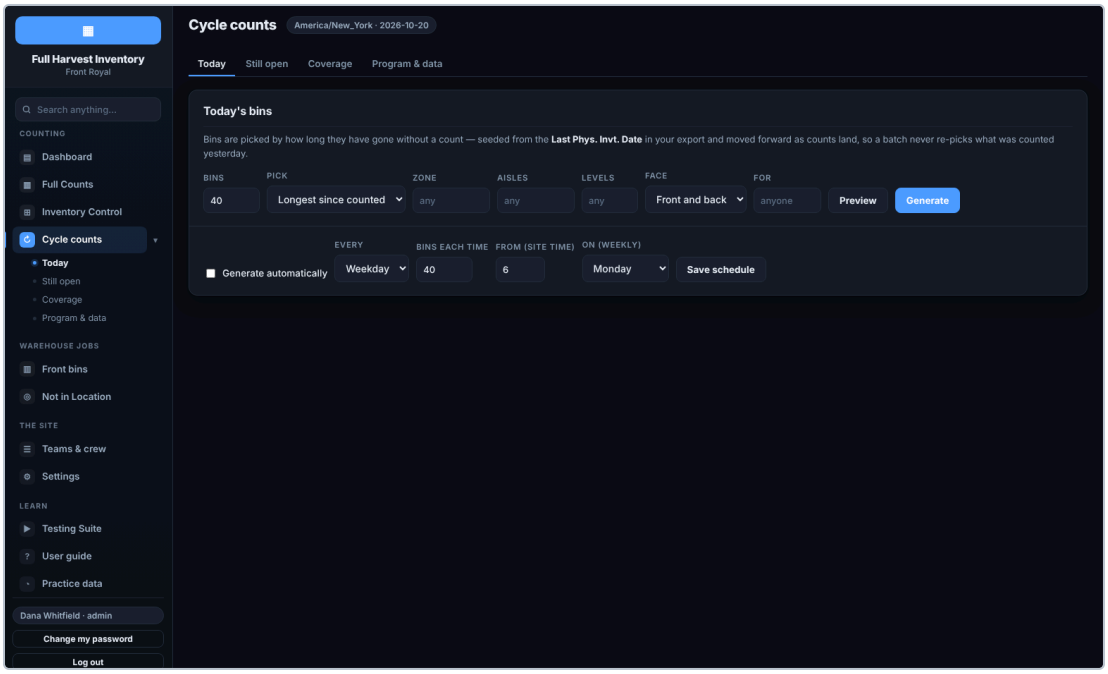


Off-site copy: OneDrive — connected, how often, and the last copy sent.

Part 6 — The cycle-count program

A cycle count is the same app with a different rhythm: instead of stopping the warehouse, a handful of bins are counted every day.

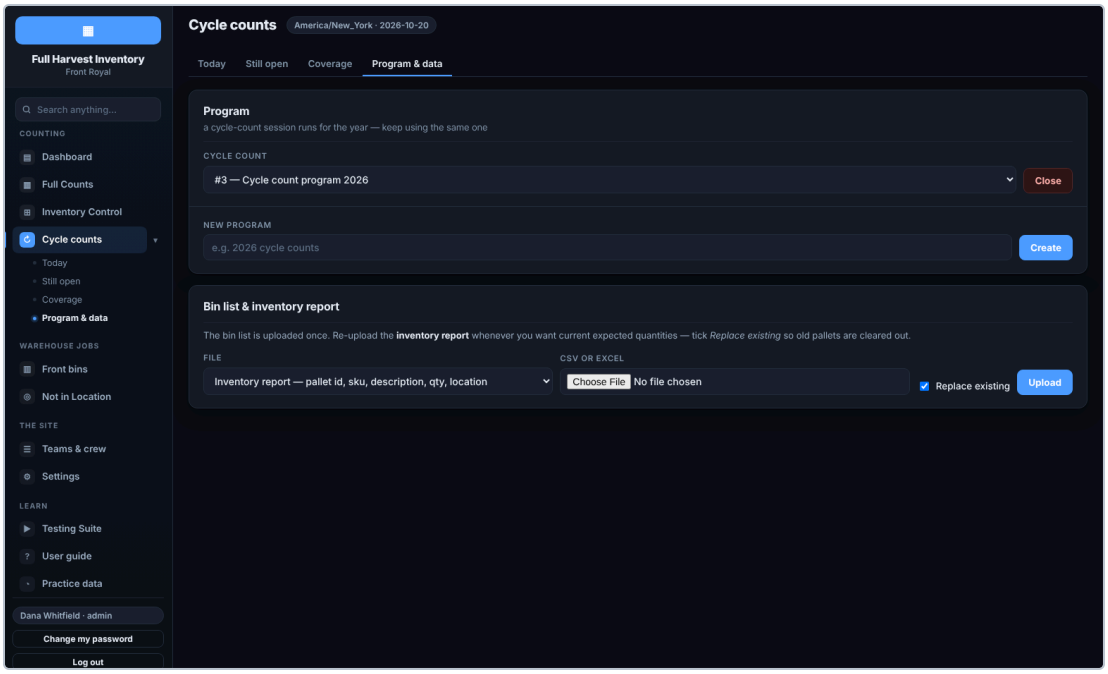
1. **Create one count session of type "Cycle count"** and keep using it. It runs for the year — do not create a new one each week, or you lose the history of when each bin was last counted.
2. Upload the bin list and the inventory report. If your ERP export has a *last physical inventory date* column it is read automatically, and the program picks up where the ERP left off.
3. Go to `/cycle` → **Today's bins**. Choose:
   - **how many bins** in the batch,
   - **which bins**: *longest since it was counted*, *never counted*, or a *random sample*,
   - optionally a **zone, aisle or levels** to stay inside.
   - optionally only **front** or only **back** bins (*Face*).
4. **Preview**, then **Generate**. The bins appear on the handhelds as tasks.



`/cycle` → *Today's bins: how many, which ones, and optionally the zone or aisles to stay inside.*

Front-placed bins. The **Front bins** tab lists the bins on the aisle face of the racking — narrow it by zone, aisle or level — to look at, **download**, or send to the scanners in one go with **Count these — make a batch**. Which face a bin is on comes from the bin list's own description (*… Position # 001 – Front*), or, where the list does not say, from the site drawing (odd positions front, even back).

1. Counters pick **Cycle count** at sign-on. A cycle count is one person with a gun, so the gun asks for **their clock-in number only — no team**. Several people can be on the same program at once: each signs on as themselves and takes their own bins; a bin one person has taken is not offered to the next. The dashboard shows each by clock-in number and name. They then work the list. Each bin is counted as it stands — it is *the* count for that bin, not a second opinion.
2. **Coverage** shows how much of the warehouse has been counted in the period, and what has not been touched. Export it for the auditors.
3. A **schedule** generates the batch automatically each day or week so nobody has to remember.



The batches that have been generated, and the program's own bin list and inventory report.

Part 7 — When something goes wrong

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
Gun: *"this scanner is no longer authorized"*	Its link was reset or the scanner was removed	Settings → Scanners → **Reset link**, open the new link on the device
A page is not in the sidebar, or a bookmark to it says *"... is switched off"*	An admin switched that part of the app off for the site (Step 4a)	Settings → Site & systems → **What is switched on**: tick it back, or **Everything on**
The jobs under Settings → Scanners are grayed out	Their part of the app is switched off for the site; the ticks are kept	Switch the part back on under **What is switched on** if the scanners should offer it
Gun: *"Offline and no list cached for this session"*	The handheld has never downloaded this count	Carry it into Wi-Fi once and sign on again
Gun says `OFFLINE` with lines queued	Normal in a dead spot	Nothing. They upload when it gets a signal. Do not wipe the device
A scan does nothing	DataWedge is not sending a suffix	DataWedge → Basic data formatting → send **ENTER** (or TAB)
A keypad covers the screen	Somebody left the **Keyboard** button on	Tap **Keyboard** again. It is off by default and never comes up by itself
The Adjustments tab is empty	Approvals are off for this count, or nothing differs from the report yet	Turn on **Adjustments need approval** under Count session. A pallet nobody has counted yet is not an adjustment until the count is closed
The ERP file is shorter than the variance list	Adjustments are waiting for approval and are held back deliberately	Work the Adjustments tab, then export again. The preview says how many were left out
the file has no "In ERP" column when uploading the worksheet	The file uploaded is not the ERP worksheet	Download the ERP worksheet under Adjustments → Entered in the ERP, fill its In ERP column, upload that

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
A line under Adjustments is tagged **CHANGED**	It was counted again after it was keyed into the ERP	Correct it in the ERP to the adjustment shown; the tag says the number the ERP was given
A pallet is on the report as `NOT IN MASTER` with a name like `NO-LABEL-F01A001-1`	Its label would not scan and nothing on it was readable, so it was counted under the bin's name	Send somebody to relabel it — **Reports → Labels to replace** has the list
A line is in a bin called `NO-LABEL-BIN-F01-1`	The rack label would not scan, and the app had nothing to suggest — an unguided count, with the pallet not on the report either	The quantity is safe. Relabel the bay, then correct the bin on the line if it matters; the relabel list says which aisle it was in
The note is not on the board	The board polls every 15 seconds by default, or the note is on a different count	Wait a few seconds; check the board is pinned to the same count (`/board?session=12`)
A change you made does not appear on the scanners	The gun is still running the app it loaded before the change	Site settings (questions, reasons, text size) reach a running gun within about half a minute and need nothing. A new **version of the app** needs the page to reload: with **Update itself after a deploy** on it does that by itself between pallets, within a couple of minutes of somebody picking the gun up. To force it: tap the green **Update ready** bar, or close the app fully (not just the home button) and reopen it. Reinstalling is never necessary
An SOS did not reach Teams	The channel address is wrong, expired, or Teams was down	The alert is still on the dashboard — the row says what Teams answered. Settings → Integrations → **Send a test card** to check the address. Microsoft is retiring the old *Incoming Webhook* connectors; if yours was one, make a **Workflows** one instead
A counter says they pressed SOS and nothing happened	The gun had no signal	The gun tells them: *"No signal — this has NOT been sent."* It sends itself the moment there is signal. In a dead zone, walking ten meters usually fixes it
The printed barcodes will not scan	The page was printed “fit to page”, or on glossy paper	Print again at **100%** on plain white paper. If a gun still refuses them, print with **Per row: 1** — the bars are wider on a bigger label

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
A code in the spreadsheet will not print	Code 128 carries plain ASCII only — an accented letter or a smart quote cannot be drawn	The book says which character it was; retype that code with plain letters and numbers
The app still shows the old name somewhere	The name was changed under Settings → Site & systems → Site name	Supervisor pages change at once. A scanner picks it up when it is next online; the board on its next refresh. The name under a home-screen icon comes from the app's install and updates when the app is next reinstalled
The strip on the gun shows the pallet but no system under it	No pallet system address is set (the gun says so), or the system refuses to be framed	Settings → Integrations → Pallet system: **Check the address**, then **How it opens → Through this app** (the frame works again) or, if the server cannot reach the system, **In its own window**
The pallet system under the desk shows its login page, and signing in only brings the login page back	The system's sign-in cookies are set without SameSite=None (an ASP.NET site, most in-house systems), so the browser drops them inside a plain frame	Settings → Integrations → Pallet system → **Through this app**: the frame stays and the sign-in sticks. Or the system's owner sets its cookies to SameSite=None; Secure
Through this app, the frame says "The pallet system did not answer"	This server cannot reach the system — it is on the warehouse's own network and the app runs on Railway	Run the app on a PC in the building (README: *On a PC in the warehouse*), ask IT to expose the system to the server, or use *In its own window*
The site's address opens the supervisor sign-in, not the scanner	That is how it works: the bare address is the office's	A scanner is opened from its own link under Settings → Scanners (`/?d=…`); after that its home-screen icon lands on the app. If a handheld lost that (site data cleared), open its link again
A gun does not offer the cycle count, Front2Back or Not in Location	That job is unticked under Settings → Scanners → What the scanners offer	Deliberate during a wall-to-wall; tick it back on when the count is over
A gun offers Front2Back but says *Nothing waiting to move right now*	The job is ticked and the move list is empty	Normal: build a move list under Front bins, or untick the job

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
A count is missing from the picker	The login may not work that kind of count	A login without the Cycle counts page sees no cycle count; a cycle counter sees no full count. An admin adds the page under Supervisor logins
Gun: *Not signed on — see a supervisor* (*E9999 is not on the crew list*)	That clock-in number is not on the crew list	Add the person under Teams & crew → Employees, then they sign on again. Once a crew list is loaded, nobody signs on with a number the site does not know (the Testing Suite's practice gun is exempt)
You do not know what a message means	The gun or the dashboard said something unfamiliar	Paste it into **User guide → Ask the guide**: it knows every message the gun and the server can show, with the fix. A new starter should begin at the guide's *Your first day*
You cannot find where something is set	The app has four pages and twenty-odd cards	Type what you would call it into the search box at the top of the sidebar — *"upload bin list"*, *"keyboard"*, *"approve"* — and the **Go to** hits take you straight there. **/** puts the cursor in it
Not sure whether a gun has the latest version	—	The sign-on screen's bottom line reads *"App build a1b2c3d4e5f6 on the server — this scanner is up to date"*. Compare it across two guns, or against a fresh reload
A gun keeps saying "update did not take"	It reloaded and is still on the old version — something between it and the server is serving stale files	Close the app fully and reopen it. If it persists, clear the site data for the app on that device (Android → Settings → Apps → the app → Storage), then open its link again
The screen keeps rotating, or reads upside down	Auto-rotate is on, and a browser tab is never allowed to lock the orientation	The app turns itself back upright on its own, sideways or end over end. If it is not, check **Keep it upright** is ticked under Settings → Scanners, then read the gray line at the bottom of the sign-on screen — it says the screen size the handheld gave the app, how far the device says it has turned, and what the app did about it. To stop the device turning at all, install the app (it asks for one way up) or turn auto-rotate off on the handheld
A team says they never got a message	Look at **Read by** on the dashboard	It shows which scanners have tapped Got it. A scanner that is offline gets it on its next sync

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
A scan seems to land on a button instead of the box	Something else took the focus	Nothing — the app puts the keystrokes in the box and carries on. Tell us if it still happens
The guide is a bin or two ahead of the team	The bins hold several pallets each	Fixed: the guide now stays on a bin until its tags are counted. Check **Show the next bin in the aisle** is on
Gun: *F01A001 is a bin, not a pallet* / *P-100 is a pallet label, not a bin* / *That is a bin, not a quantity*	The wrong label was scanned into that step	Scan what the prompt asks for: the pallet's label at PALLET, the rack label at BIN, a typed number at QUANTITY, the lot code at LOT. Nothing was recorded. For an empty bin tap **Bin is EMPTY** first
A scan opens the address bar and the text goes into it	The page has lost the keyboard — either the scanner sends a TAB that used to move focus out of the page, or somebody tapped the browser's own bar	Fixed: TAB now ends a scan like ENTER and the focus never leaves the box. If it ever happens again the app shows a red **Tap here to scan** bar — one tap puts it right
A bar with the web address appears on every scan	The app is being run as a page in Chrome rather than installed	Install it: Chrome menu → **Install app**, or the **Install on this scanner** button on the sign-on screen, then open it from the home-screen icon
The browser's address bar is in the way	Same thing — the app is not installed	As above. For a device that should run nothing else, use Zebra's Enterprise Home Screen
Gun: *"Team N is counting aisle X"*	Another team holds that racking block	Wait, or hand the other aisle back first
Second-count list is enormous	Thresholds are at 0	Set **Recount over** and **or over %**, and a **cap** (Part 3, step 13)
The map is a schematic, not your drawing	No rack drawing on this count	Settings → Count setup → Count session → **Map drawing**

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
A reason code edit "did not reach" a gun	It takes up to about half a minute, and only lands between pallets; a gun on an old build does not take it at all	Wait for the counter to finish the pallet they are on; check the build line on the sign-on screen matches the server
Can nobody sign in?	All logins lost	The superadmin is made from the Railway variables the first time the app starts; if it was deleted or demoted later, set `SUPERADMIN_USER` to a new username and restart, which makes a fresh admin login
Numbers look wrong after an ERP import	The inventory report moved on	Re-upload it with **Replace what is there**, then re-read the pallet report
A scanner's scan times are hours out	The handheld's clock is wrong	Nothing to do: every line carries the gun's own send time, and the server corrects the scan time by the clock's error. Set the clock when you can, all the same
Everyone was signed out of the dashboards	The server restarted (a deploy, a Railway restart)	Sign-ins now survive a restart; if it still happens, the server lost its database volume — check Railway

Part 8 — Capacity: what the system will take

Two measurements, both against a count bigger than the warehouse.

A whole count, played end to end. Twenty teams with two scanners each counted 20,000 bins over three shifts, with the three shifts compressed into 6.0 hours of real time (`node tools/sim-floor.mjs`): the sign-on rush at six, the breaks, a lunch that trips the stopped-scanning alert, SOS calls, second counts raised behind the variances, and a supervisor approving adjustments over the top of it all. Every ten minutes the dashboard and the office board were photographed; `docs/load-test/index.html` is the result, to scrub through.

Bins counted	20,000 of 20,000, 22,655 count lines, 0 failed requests out of 27,059
Nothing lost, nothing doubled	0 pallets counted twice; every aisle job handed back (71)
A scan, with the whole floor working	9 ms typical, 747 ms at the 95th percentile
A dashboard refresh over the top of it	279 ms typical, 1595 ms at the 95th percentile
The office board	350 ms typical, 1468 ms at the 95th percentile
Forty scanners pulling the whole list at once	11 ms typical, 2574 ms worst
The office: approvals, SOS, messages	296 approved, 62 rejected, 25 SOS raised and 25 closed, 657 second counts walked

The same floor at full tilt. The rehearsal of that run plays the three shifts in eighteen minutes — twenty-eight times real speed, every scanner posting a line every second or two — and the server still answered: 20,000 bins in, a count line at 8 ms typical and 536 ms at the 95th percentile, dashboards at 150 ms typical. (`node tests/run-all.mjs load-15-teams` keeps the older fifteen-team burst test in the suite: thirty scanners posting flat out, nothing lost, nothing doubled, the racking-block rule honored.)

What the simulation found and fixed. On a count this size the pallet report took six seconds to build, and the adjustments list, its summary and *Export everything* twelve to eighteen; the server answers one request at a time, so while it built them nothing else moved. It is now five indexed passes joined in memory: the report in about a tenth of a second, the export in under one.

The constraints that matter in practice are warehouse Wi-Fi coverage and battery life, not the server. Real counters scan a pallet every ten to twenty seconds; the whole floor at that pace is a small fraction of what the rehearsal threw at it.

Appendix A — File column reference

Files may be CSV, TSV or Excel (`.xlsx`/`.xlsm`). Column names are matched loosely — case, spaces and punctuation are ignored — so most ERP exports load unchanged. Each upload card shows exactly which columns it recognized.

Bin list — the only required column is the location.

WHAT	ACCEPTED COLUMN NAMES
Bin (required)	location, loc, bin, bin location, location code, slot, code, warehouse location
Aisle	aisle, row, aisle no, aisle number — *derived from the bin code if absent*
Zone	zone, area, section, region, warehouse
Level	level, levels, tier, shelf — *derived from the bin code if absent*
Description	description, desc, name
Last counted	last phys invt date, last counted, last count date, last inventory date

Inventory report — upload it exactly as the site's systems export it. The columns are:

COLUMN	WHAT IT IS	REQUIRED
Bin Code	the position (F18D064); lower case is fine	yes
Container No.	the pallet's number as printed on its label (FR12345678, F61409-006, 4748025); a leading `$` from the ERP is dropped	yes
Item No.	the item	
Description	the item's description	
Variant Code	DIST, REWORK, DONATE, ALLERGEN, DESTROY, REJECTED, RND — kept on the pallet and in the exports	
Quantity	the system's quantity; zero and negative quantities are kept as the report says, and counted in the upload summary	
Unit of Measure Code	CS, CS30LB, C10KG, CASE, BAG, BOX …	
Entry No.	the ERP's item ledger entry number, carried through to the pallet report and the ERP file so adjustments post against the right entry	

COLUMN	WHAT IT IS	REQUIRED
Lot No.	the lot (F61409, 50479-1, PO123456)	
System	which system the pallet belongs to (JustFood, SGI, NTFF); `OPEN` with no container is an empty position	

Other names for these columns are understood too (pallet id, LPN, item, SKU, qty, location, bin, lot code, unit, source …), as is a **Best Before** or **Expiry** column when the file has one.

Three things in a real export, and what the upload does with them:

- **Empty positions.** A row with a bin and no container (System `OPEN`) is a position the count expects to find empty. It is not a pallet; the summary says how many there were.
- **A container on two rows.** In the same bin it is two ledger entries for one pallet (a −134 and a +180 are a pallet of 46): the quantities are added up. In two different bins the first row stands and the summary lists the rest, with both bins, so somebody can look. Nothing is overwritten quietly.
- **Odd quantities.** Zero and negative quantities are kept exactly as the report has them, and the summary counts them.

Barcode test book — one row per practice line.

WHAT	ACCEPTED COLUMN NAMES
Bin	bin, bin location, location, loc
Pallet	pallet, pallet id, tag, LPN
Qty	qty, quantity, count, cases, units
Note	note, notes, label, description

A row needs a bin or a pallet; everything else is optional. Sheets made with the first version of the template (Type / Code) are still read.

Counting plan — one row per aisle, in the order the team counts it.

WHAT	ACCEPTED COLUMN NAMES
Team	team, team number, crew, group
Aisle	aisle, row, aisle no
Levels	level, levels, tier, shelf — e.g. `A-C`, `D-F`, `A-F`

Crew list — one row per person.

WHAT	ACCEPTED COLUMN NAMES
Clock in number (required)	clock in number, badge, badge id, employee id, emp id, employee number, id
Name	name, employee name, full name, employee
Department	dept, department, area
Shift	shift, shift no, shift number — 1 or 2, however it is written (*1st*, *first*, *day*; *2nd*, *second*, *night*)
Equipment	equipment, certifications, certified, licenses, can operate, machines — a list, e.g. `Scissor lift; High reach`
Jacket	jacket, jacket size, coat, coat size
Bibs	bibs, bib size, overalls, overall size, pants
Boots	boots, boot size, shoe, shoe size

Dates are read in either order — `2027-03-15` and `03/15/2027` both work.

Appendix B — Server settings reference

SETTING	DEFAULT	WHAT IT DOES
`SUPERADMIN_PASSWORD`	—	The superadmin's password. Set it
`SHARED_PASSWORD_LOGIN`	`on`	`off` requires named logins — ignored if no admin account exists, so it cannot lock you out
`SUPERADMIN_USER`	—	Username of the permanent admin, created at start-up
`SUPERADMIN_NAME`	—	That person's name, as it appears in the log
`SUPERADMIN_PASSWORD`	falls back to `ADMIN_PASSWORD`	Must be 8+ characters and not `changeme`
`SCANNER_AUTH`	on	`off` lets anything on the network post counts. Leave it on
`DB_PATH`	`data/inventory.db`	Put this on a persistent volume
`BACKUP_DIR`	`data/backups`	Where the daily backups go
`BACKUP_KEEP`	14	How many backups to keep
`SITE_TIMEZONE`	`America/New_York`	Dates on reports and cycle-count due dates
`PORT` / `HOST`	3000 / 0.0.0.0	Where the server listens
`MAX_UPLOAD_MB`	64	Largest file an upload will accept